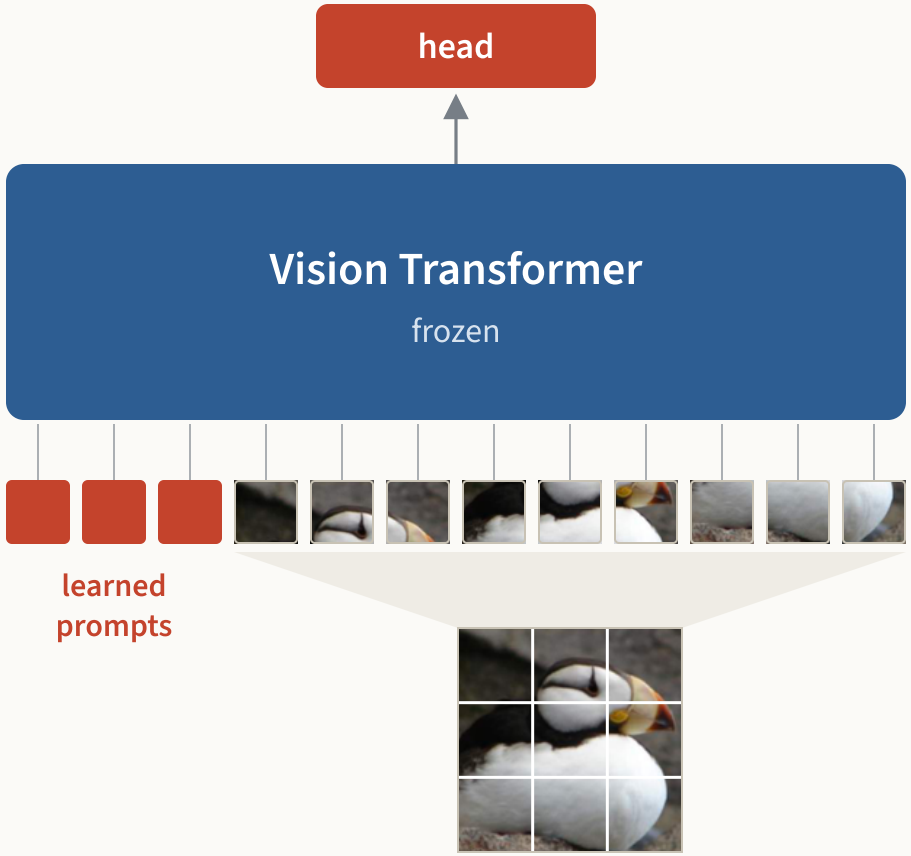
Visual Prompt Tuning

Teach a frozen Vision Transformer a new task by learning only a few extra input vectors.



Menglin Jia, Luming Tang, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, Ser-Nam Lim
Cornell University, Meta AI and University of Copenhagen. ECCV 2022.

Presented by **Ertugrul Senturk**
October 2026

Pretrain one big model once, then adapt it to each task

ImageNet-21k

14 million photos in 21,000 everyday categories. Pretraining takes weeks and is done once, by Google.

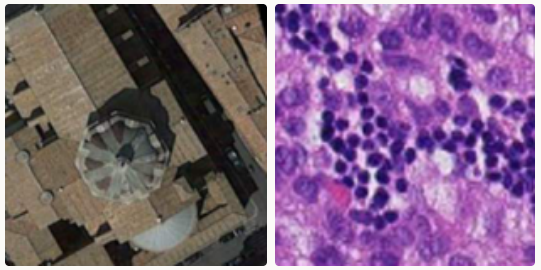
→

ViT-B/16

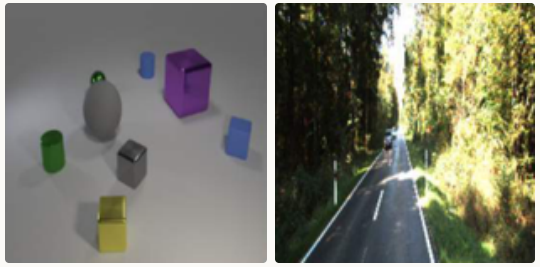
A Vision Transformer with 85.8 million weights that has learned general visual features.

→

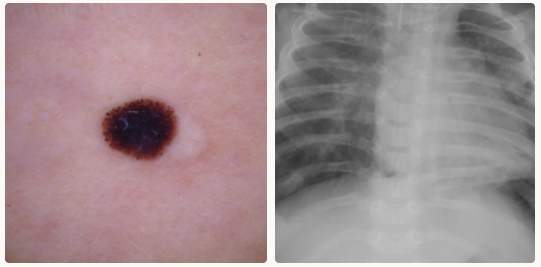
Fine-grained categories
200 bird species, 196 car models



Satellite and microscope
land use from above, tissue slides



Counting and distance
how many objects, how far away



Medical images
my own experiments, shown later

Each new task is small: most tasks in the paper have only **1,000 training images**, far too few to train from scratch. So we start from the pretrained model and **adapt** it.

The paper asks how to adapt it best: accurately, and cheaply enough to store for every task. Next, the standard answer and what it costs, then the cheaper methods tried before VPT.

Dataset examples from Jia et al. 2022, Fig. 10. ViT-B/16: Dosovitskiy et al. 2021. The pretrained checkpoint used here and in the paper is Google's ImageNet-21k ViT-B/16.

Full fine-tuning stores a new copy of the model per task

Full fine-tuning retrains every weight of the pretrained model on the new task, and it is usually the most accurate option. The cost is that each task keeps its own retrained copy, so storage grows with every task and with the size of the model.

Shown for **1 task** on **ViT-B, 86 million weights**. The next pages show 24 tasks, then the largest model.

Full fine-tuning: 1 retrained ViT-B copy

343 MB

1 copy at 343 MB each

VPT: one shared ViT-B plus 1 small file

each red speck is one task file, to scale

345 MB

343 MB once, plus 2.2 MB per task

32-bit weights: ViT-B/16 85.8M, ViT-L/16 303M, ViT-H/14 632M. A VPT file is 50 prompts at every layer plus a 102-class head. Squares are sized by area.

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Shown for **24 tasks** on **ViT-B, 86 million weights**.

Full fine-tuning: 24 retrained ViT-B copies

8.2 GB

24 copies at 343 MB each

VPT: one shared ViT-B plus 24 small files

395 MB

343 MB once, plus 2.2 MB per task

32-bit weights: ViT-B/16 85.8M, ViT-L/16 303M, ViT-H/14 632M. A VPT file is 50 prompts at every layer plus a 102-class head. Squares are sized by area.

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Full fine-tuning retrains every weight of the pretrained model on the new task, and it is usually the most accurate option. The cost is that each task keeps its own retrained copy, so storage grows with every task and with the size of the model.

Shown for **24 tasks** on **ViT-H, 632 million weights**.

Full fine-tuning: 24 retrained ViT-H copies

61 GB

24 copies at 2.5 GB each

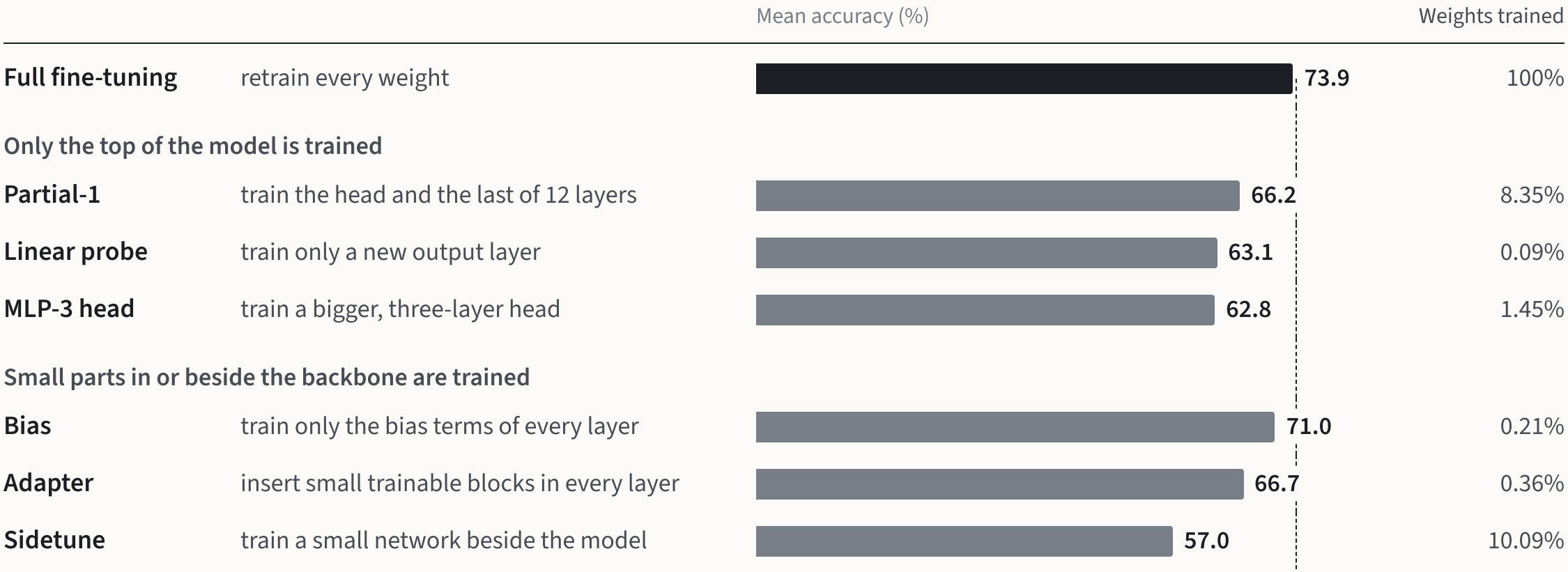
VPT: one shared ViT-H plus 24 small files

2.7 GB

2.5 GB once, plus 8.7 MB per task

32-bit weights: ViT-B/16 85.8M, ViT-L/16 303M, ViT-H/14 632M. A VPT file is 50 prompts at every layer plus a 102-class head. Squares are sized by area.

Cheaper methods existed, but none matched full fine-tuning



Bias came closest and even beat full fine-tuning on 8 of the 24 tasks, but on average no cheaper method reached it. VPT takes a third route: it leaves the model alone and changes the input.

Mean test accuracy over the paper's 24 tasks, ViT-B/16 (Jia et al. 2022, Table 1, mean of the four group averages). The right column is the share of weights trained for each task.

Where the word "prompt" comes from

In 2020, a prompt was text. GPT-3, a frozen model with 175 billion weights, was only trained to continue text, yet it does a new task when you type a description of the task in front of the input. Not a single weight changes. But words are discrete, so the only way to improve a prompt is trial and error.

In 2021, prompt tuning (Lester et al.) replaced the words with **learned vectors** that correspond to no word. They are trained by gradient descent on the task's examples while the model stays frozen, and they are all you store per task. Prefix-tuning (Li & Liang) does the same at every layer.

The sentiment example is illustrative. Brown et al. 2020 (GPT-3); Lester, Al-Rfou, Constant 2021 (prompt tuning, T5); Li & Liang 2021 (prefix-tuning).

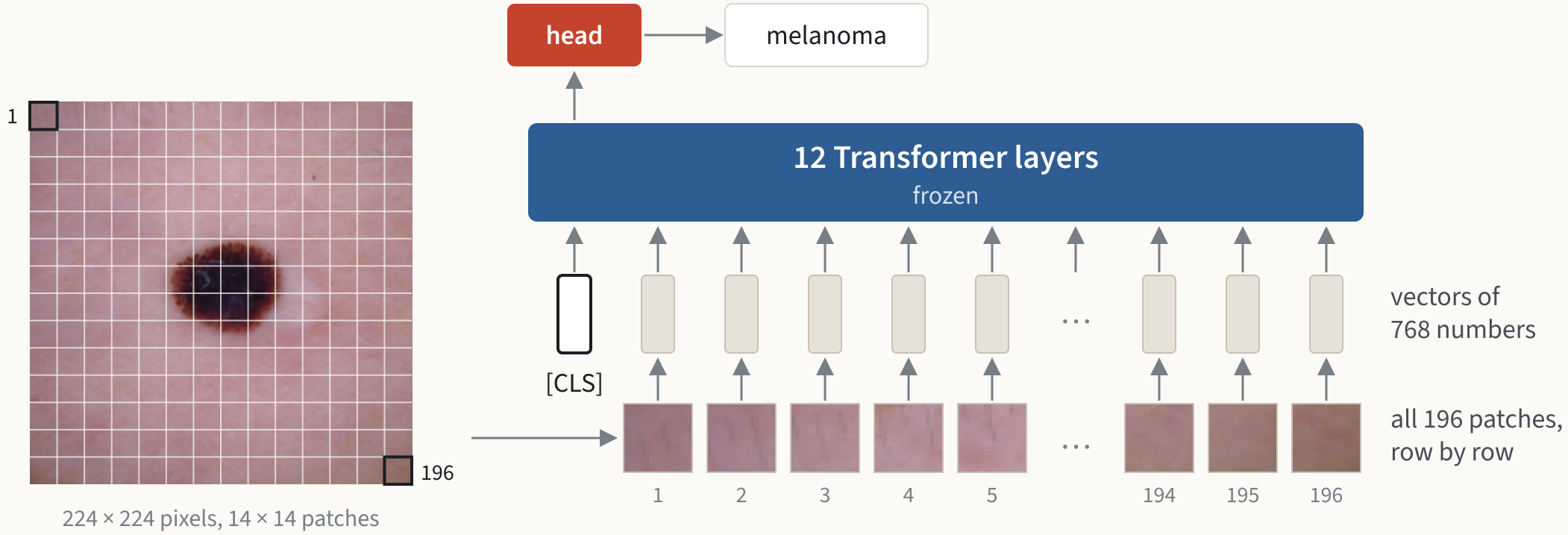
A visual prompt is the same idea applied to image patches

The grey vectors come from the input, as before. The red ones do not: they start as random numbers, are trained on the task's examples while the model stays frozen, and are then put in front of every input of that task. A visual prompt is therefore not text and not a pattern painted on the pixels, only learned numbers in the model's input space.

> In language, prompt tuning matched full fine-tuning only once the model had about 10 billion weights. VPT beats full fine-tuning on 20 of 24 tasks with an 86-million-weight ViT-B.

Lester et al. 2021 (T5); Jia et al. 2022, Table 1. Not to be confused with CoOp (Zhou et al. 2021, text prompts for CLIP) or Bahng et al. 2022 (a pixel pattern).

How a Vision Transformer reads an image

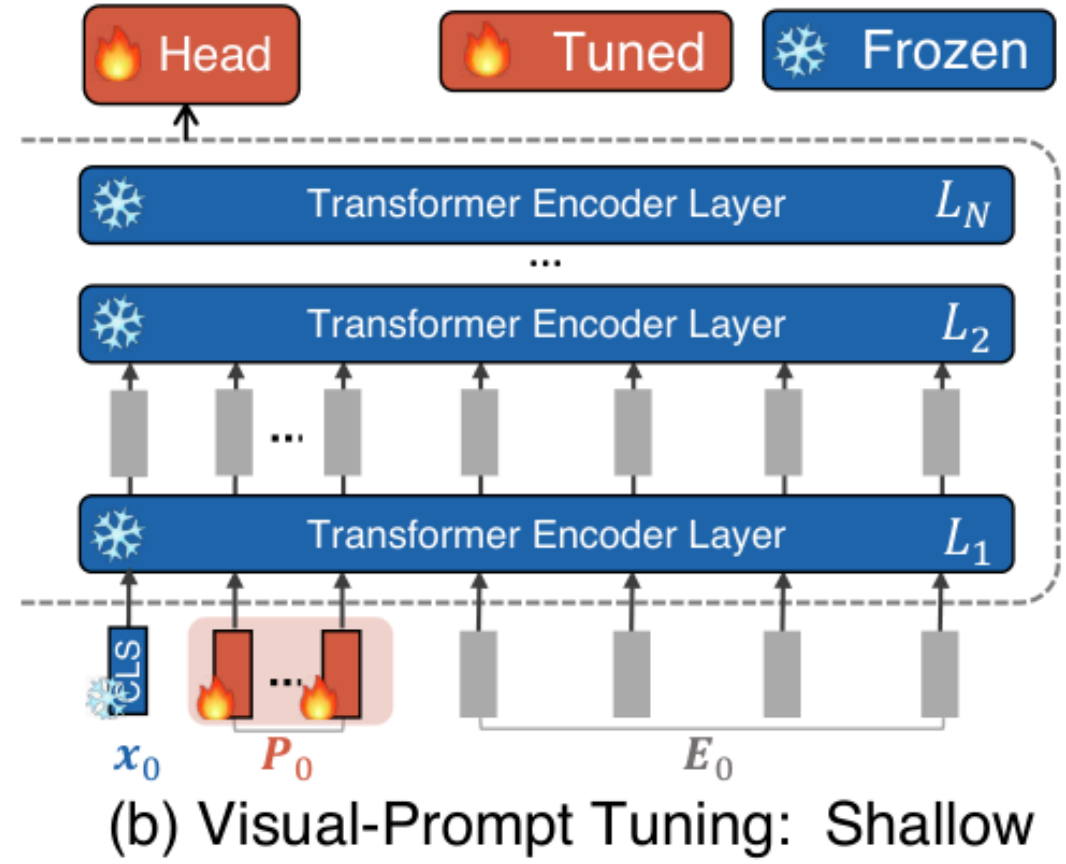


A fixed grid cuts the image into 196 patches of 16 × 16 pixels, and each patch, read row by row, becomes a vector of 768 numbers. One extra vector, [CLS], goes in front to collect a summary of the whole image: the 12 layers mix all 197 vectors, and the head reads only [CLS] to name the class.

> The layers only see a sequence of vectors, so we can add a few more.

ViT-B/16: 12 layers, vectors of 768 numbers, 85.8 million weights. Dosovitskiy et al., ICLR 2021. Image: a melanoma from DermaMNIST, 224 × 224 pixels.

VPT trains a few extra input vectors and the head



VPT-shallow | VPT-deep

The p prompt vectors P are added once, before the first layer. After that they pass through the 12 frozen layers like any other token.

$$[x_1, Z_1, E_1] = L_1([x_0, P, E_0])$$ prompts enter once

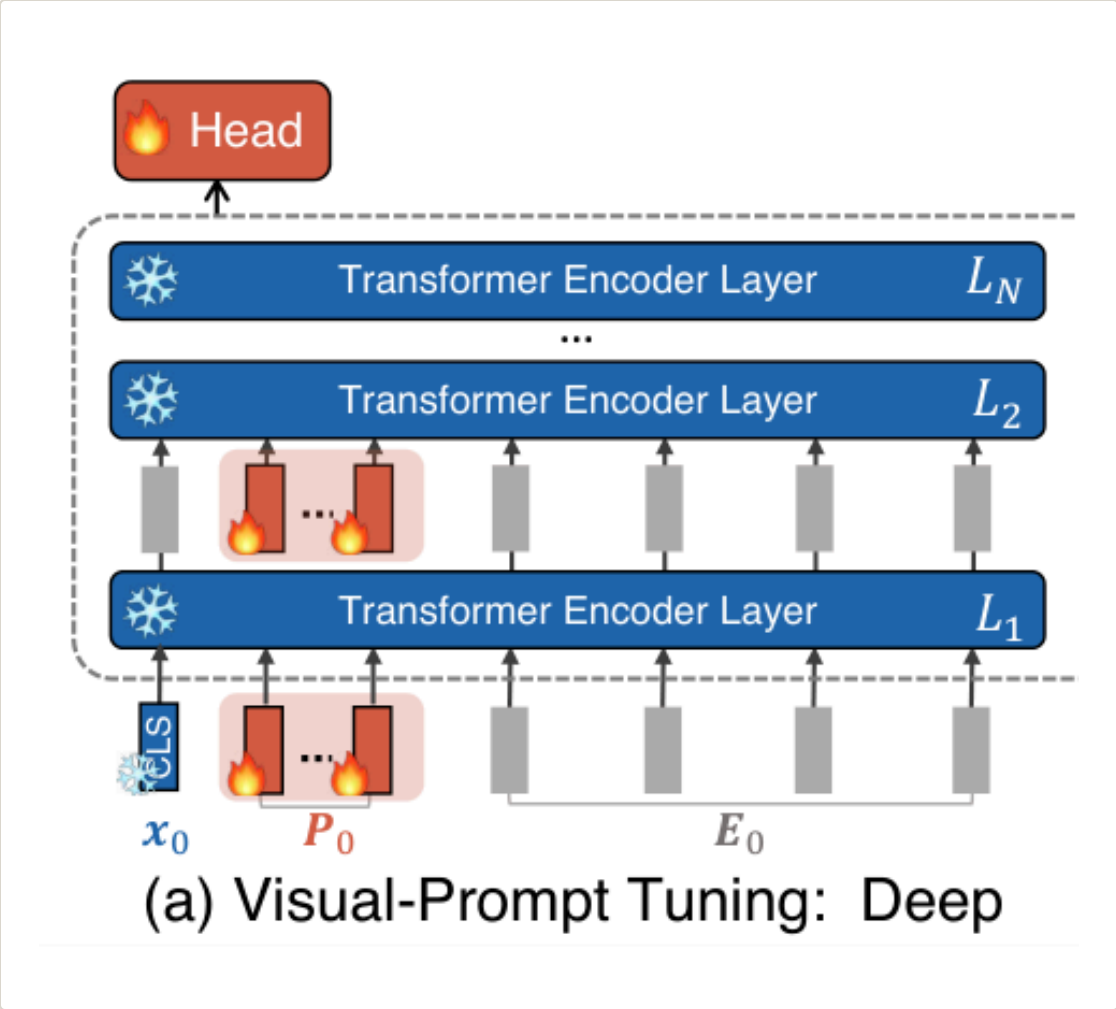
$$[x_i, Z_i, E_i] = L_i([x_{i-1}, Z_{i-1}, E_{i-1}])$$ i = 2 … 12

$$y = \text{Head}(x_{12})$$

Per task, only the red parts are trained and saved: 50 prompt vectors and the head, **0.14%** of the model.

Figures: Jia et al. 2022, Fig. 2. Equations (6) and (8) of the paper: x class token, E patch vectors, P prompts, Z the prompt positions after a layer, L_i layer i.

VPT trains a few extra input vectors and the head



VPT-shallow | **VPT-deep**

Every layer gets its own fresh set of p prompt vectors. What the previous layer produced at those positions is thrown away: the blank in the equation.

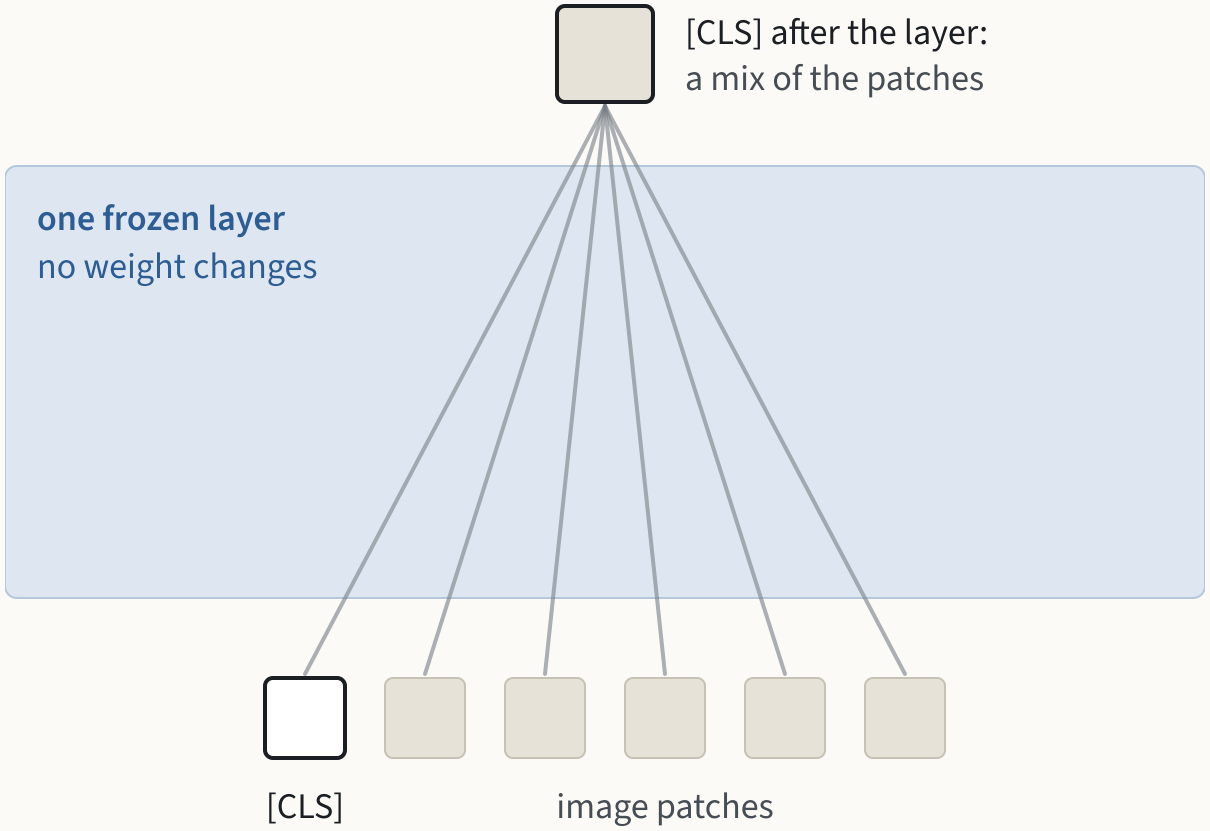
$$[x_i, __, E_i] = L_i([x_{i-1}, P_{i-1}, E_{i-1}]) \quad i = 1 \dots 12$$

$$y = \text{Head}(x_{12})$$

Per task, only the red parts are trained and saved: 50 prompt vectors for each layer and the head, **0.63%** of the model.

Figures: Jia et al. 2022, Fig. 2. Equations (6) and (8) of the paper: x class token, E patch vectors, P prompts, Z the prompt positions after a layer, L_i layer i.

Why a few extra tokens can steer a frozen network



Without prompts | With prompts

In every layer, each token takes in information from all the other tokens; this is attention. With prompts in the sequence, [CLS] and every patch also take in information from the prompts.

> So training the prompts changes what the frozen network computes, without changing a single weight.

Mechanism: Sec. 3 of Jia et al. 2022; Appendix B relates it to prefix-tuning ("VPT-prefix"). Line widths and the red share of [CLS] are illustrative, not measured.

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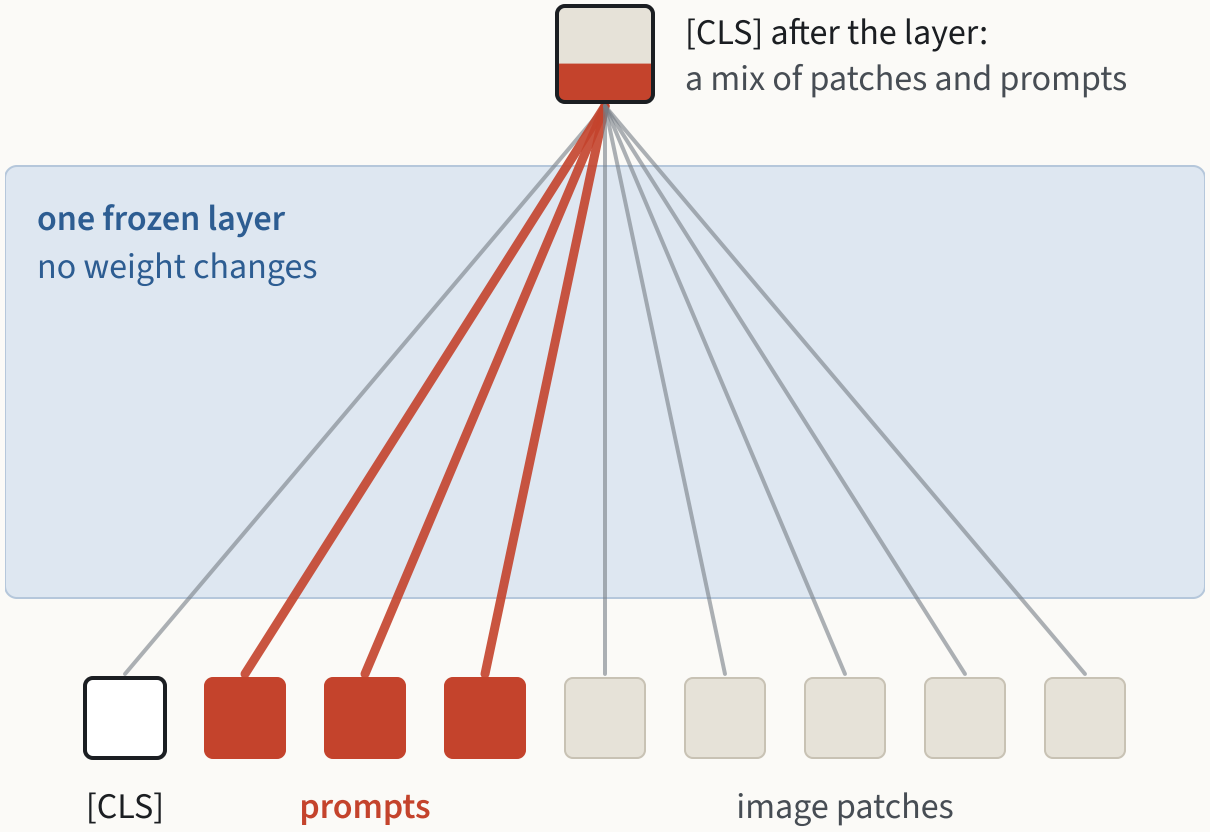

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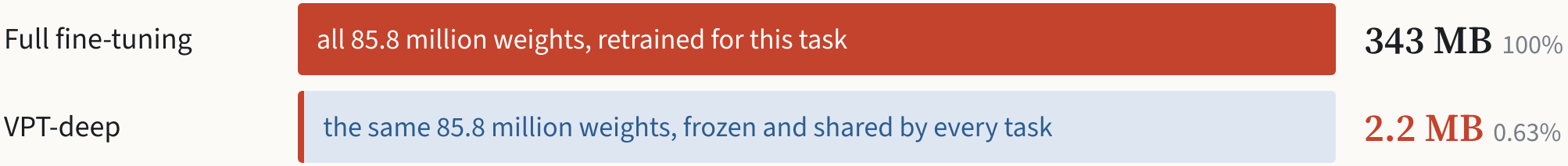
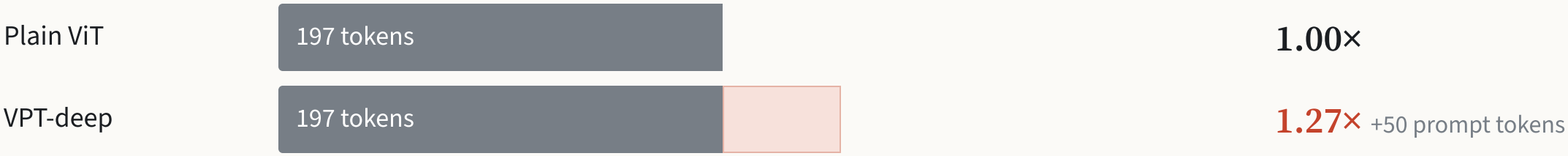
How much VPT trains and stores per task

$N \cdot p \cdot d = 12 \times 50 \times 768 =$ **460,800** prompt numbers, plus 78,438 for a 102-class head

Numbers trained and stored per task



Compute per image



Shown for p = 50; at p = 200 the file is 7.7 MB and compute is about 2.1×. More prompts keep the file small, but every image carries them through all 12 layers.

ViT-B/16, $N = 12$ layers, $d = 768$, 32-bit numbers. Compute counts multiply-adds per layer for n tokens ($12nd^2 + 2n^2d$). The paper measures about 2× test latency at $p = 200$ (Jia et al. 2022, Table 12).

VPT in code: a few lines and one new hyperparameter

```
# VPT-deep on a frozen ViT-B/16, simplified from demo/vpt/vpt_model.py
vit = timm.create_model(BACKBONE, pretrained=True)              # ImageNet-21k ViT-B/16
vit.requires_grad_(False)                                       # backbone frozen
prompts = nn.Parameter(torch.empty(12, p, 768).uniform_(-b, b))   # trained: 12 layers × p × 768
head = nn.Linear(768, num_classes)                              # trained: the classifier

def forward(img):
    x = vit._pos_embed(vit.patch_embed(img))                    # [CLS] + 196 patch vectors
    for i, layer in enumerate(vit.blocks):
        rest = x[:, 1:] if i == 0 else x[:, 1 + p:]             # drop last layer's prompts
        P = prompts[i].expand(len(x), -1, -1)                   # this layer's prompts
        x = layer(torch.cat([x[:, :1], P, rest], dim=1))        # frozen layer on [CLS, P, rest]
    return head(vit.norm(x)[:, 0])                              # classify from [CLS]
```

The prompts start random, with Xavier-uniform values as in the official code. The paper tried smarter starts, such as class prototypes, and random did as well or better.

Training is plain SGD with momentum for 100 epochs on a cosine schedule. Learning rate and weight decay are picked on a validation split, for every method alike.

The one new hyperparameter is the prompt length p, chosen per task from {1, 5, 10, 50, 100, 200} on the validation split.

Recipe: Jia et al. 2022, Sec. 4.1 and Appendix A. Code is my re-implementation, which reproduces the paper's Oxford Flowers result (99.1% vs the paper's 99.0%).

The paper tests VPT on 24 very different tasks

CUB-200

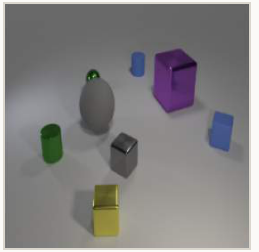
Oxford Flowers

Caltech101

DTD

EuroSAT

Resisc45


CLEVR count

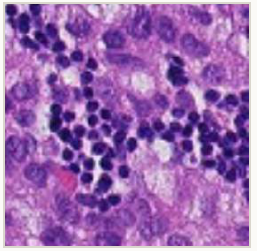
dSprites location

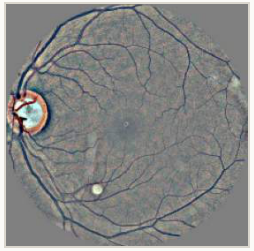
Stanford Cars

Stanford Dogs

Oxford Pets

SUN397


Patch Camelyon


Retinopathy

KITTI distance

SmallNORB azimuth

FGVC, 5 tasks

Fine-grained categories, such as 200 bird species, with thousands of training images.

VTAB Natural, 7 tasks

Ordinary photos of objects, textures, animals and scenes, close to the pretraining data.

VTAB Specialized, 4 tasks

Images from special sensors: satellites, tissue slides under a microscope, and retina cameras.

VTAB Structured, 8 tasks

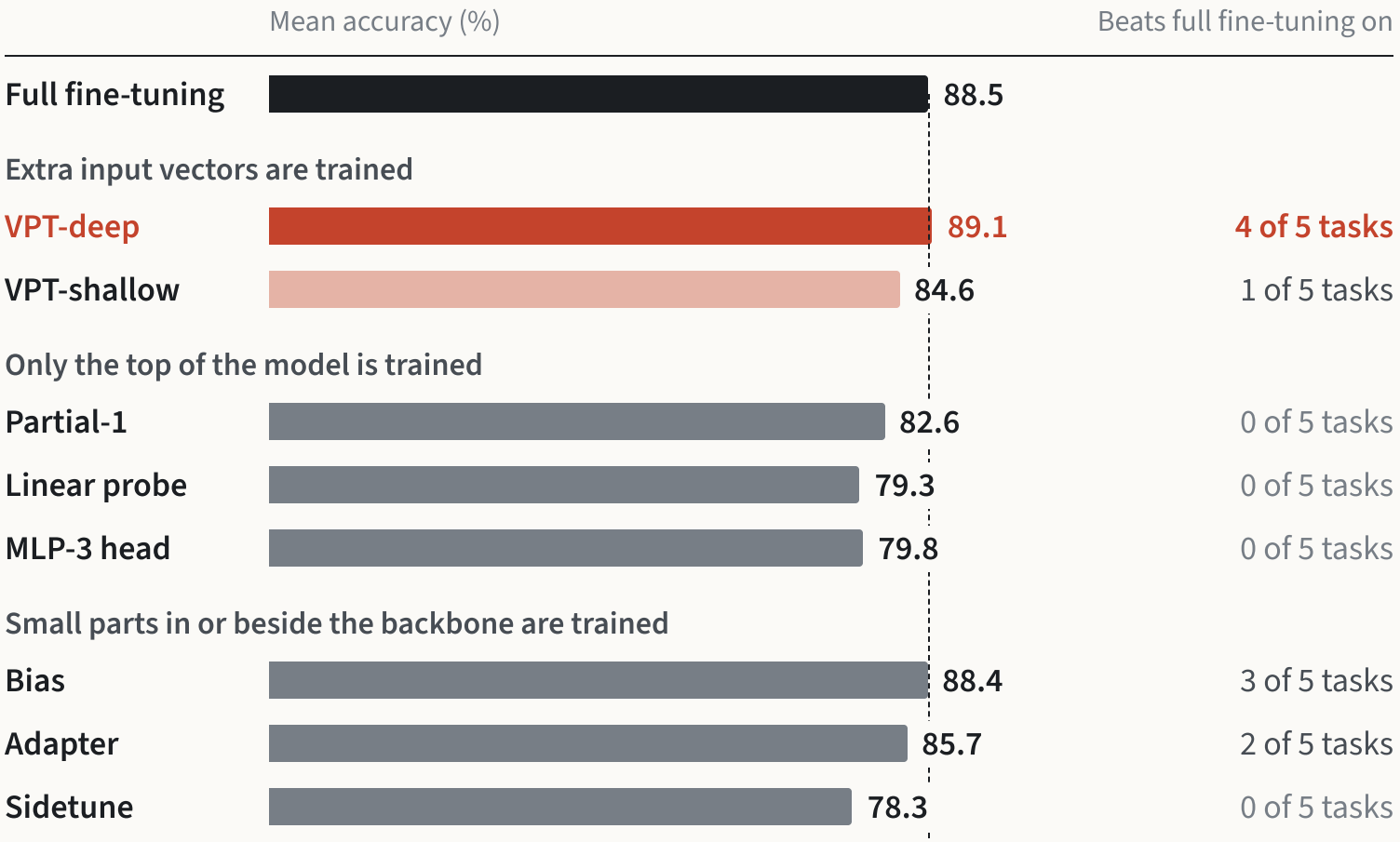
Geometry instead of categories, such as counting objects or judging distance.

All 24 tasks use the same frozen ViT-B/16, pretrained on ImageNet-21k. The 19 VTAB tasks have only 1,000 training images each.

Images: Jia et al. 2022, Fig. 10, four example tasks per group. VTAB-1k: Zhai et al. 2019.

VPT-deep beats full fine-tuning on 20 of 24 tasks

FGVC | Natural | Specialized | Structured | All 24 tasks



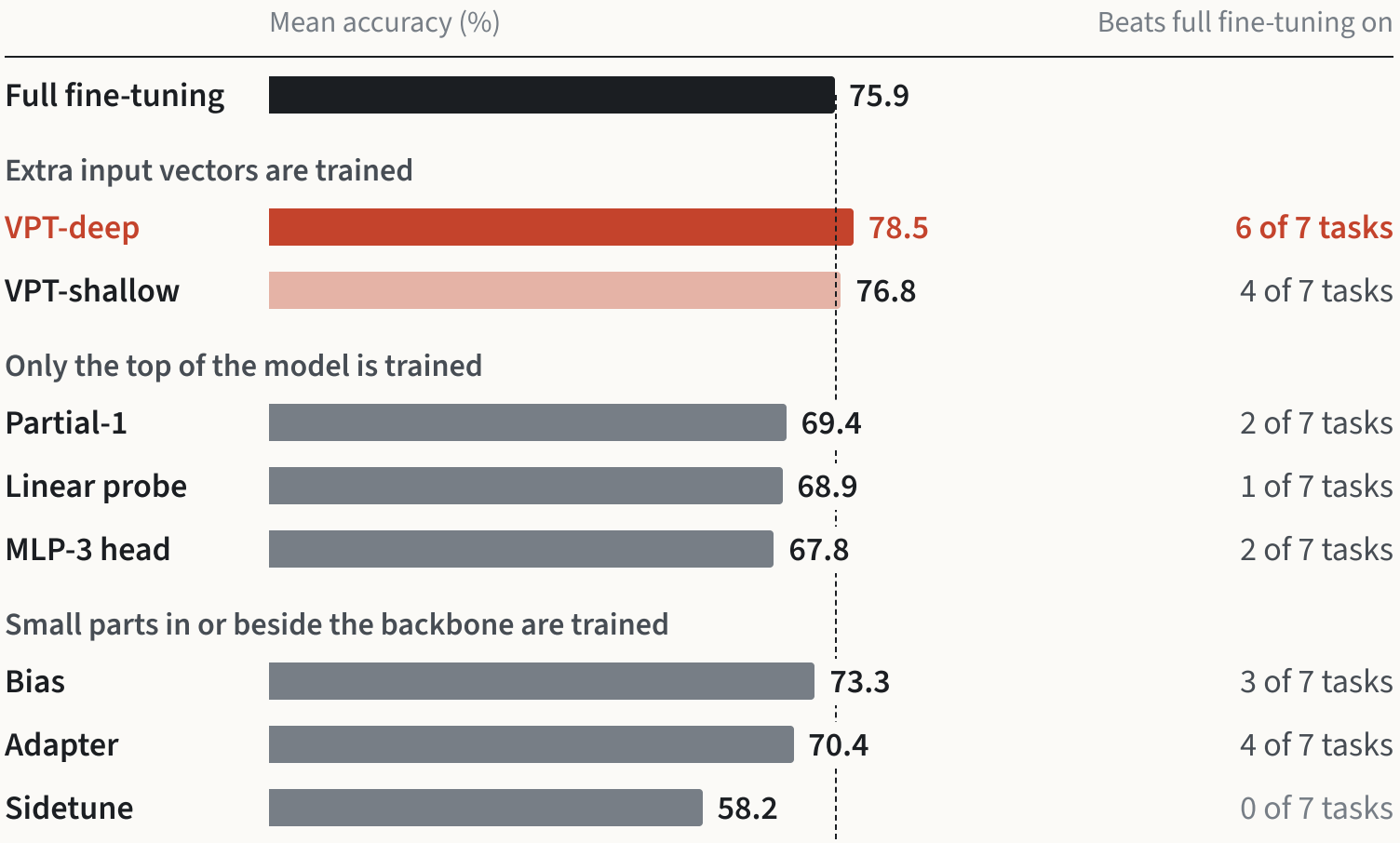
These five tasks ask the model to tell apart 200 bird species, 196 car models, 120 dog breeds or 102 kinds of flower, with thousands of training images per task.

Every method does well on such photos, and VPT-deep only edges out full fine-tuning, 89.1 against 88.5.

Jia et al. 2022, Table 1 and Tables 13–14: ViT-B/16 pretrained on ImageNet-21k, mean test accuracy within each group. For all 24 tasks, the mean of the four group averages.

VPT-deep beats full fine-tuning on 20 of 24 tasks

FGVC | **Natural** | Specialized | Structured | All 24 tasks



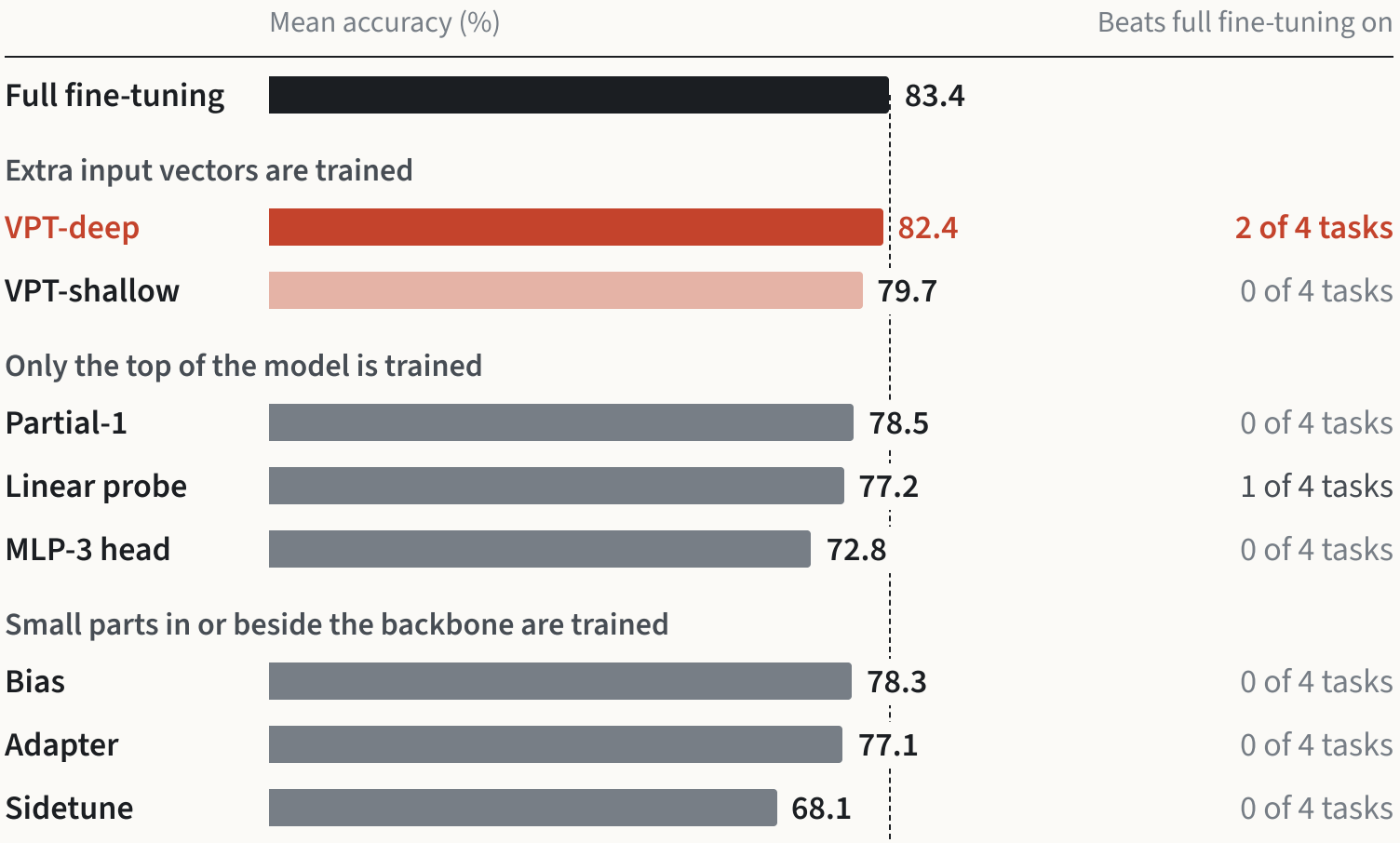
These seven tasks use ordinary photos of objects, pets, textures, scenes and house numbers, close to what the model saw in pretraining, but with only 1,000 training images each.

With so little data, retraining all 86 million weights overfits, so even VPT-shallow beats full fine-tuning here.

Jia et al. 2022, Table 1 and Tables 13–14: ViT-B/16 pretrained on ImageNet-21k, mean test accuracy within each group. For all 24 tasks, the mean of the four group averages.

VPT-deep beats full fine-tuning on 20 of 24 tasks

FGVC | Natural | **Specialized** | Structured | All 24 tasks



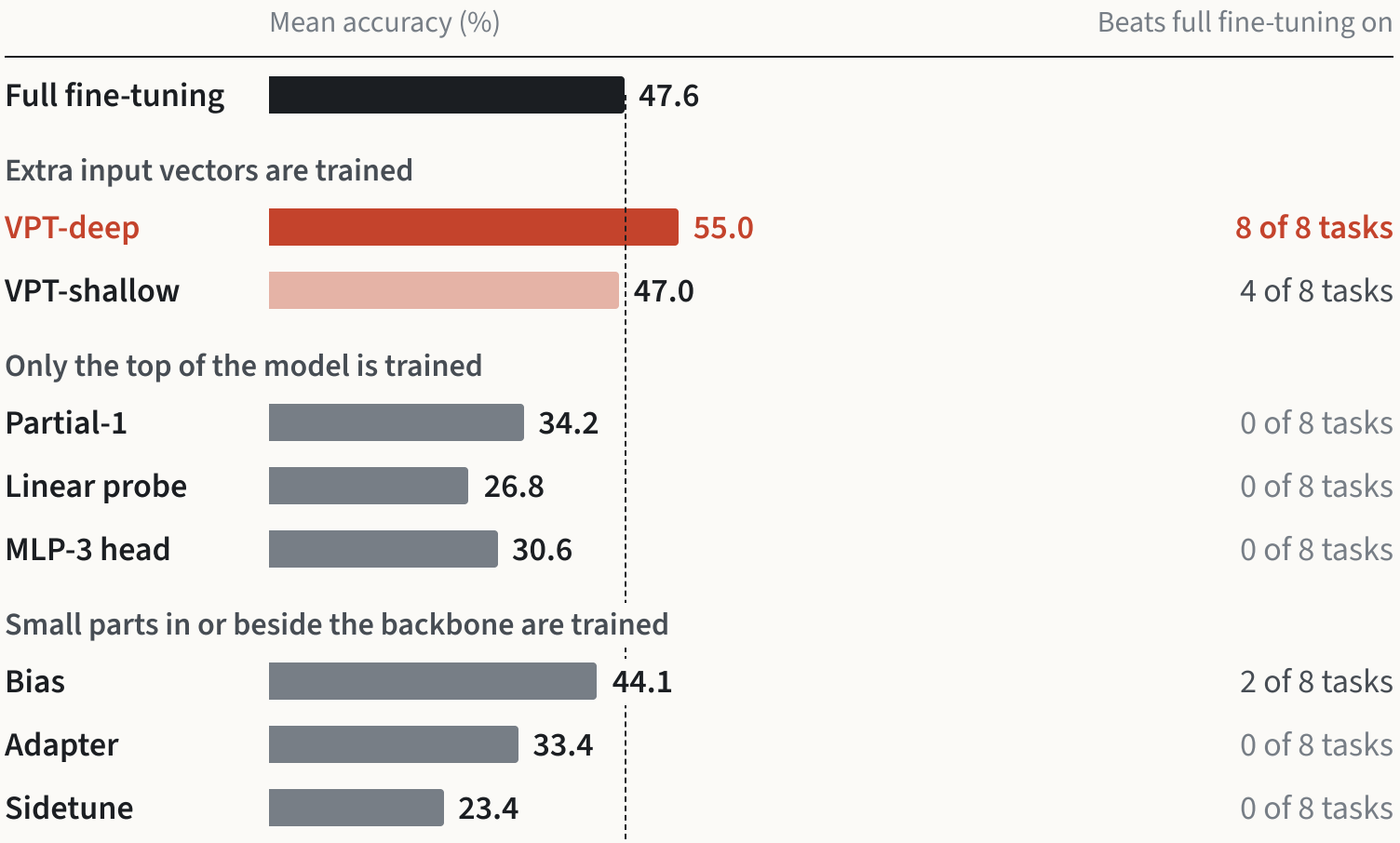
These four tasks come from special cameras: satellite images of land use, microscope slides of tumour tissue and photos of the retina, with 1,000 training images each.

This is the one group where full fine-tuning stays ahead, by one point, but VPT-deep is still the best of the cheap methods.

Jia et al. 2022, Table 1 and Tables 13–14: ViT-B/16 pretrained on ImageNet-21k, mean test accuracy within each group. For all 24 tasks, the mean of the four group averages.

VPT-deep beats full fine-tuning on 20 of 24 tasks

FGVC | Natural | Specialized | **Structured** | All 24 tasks



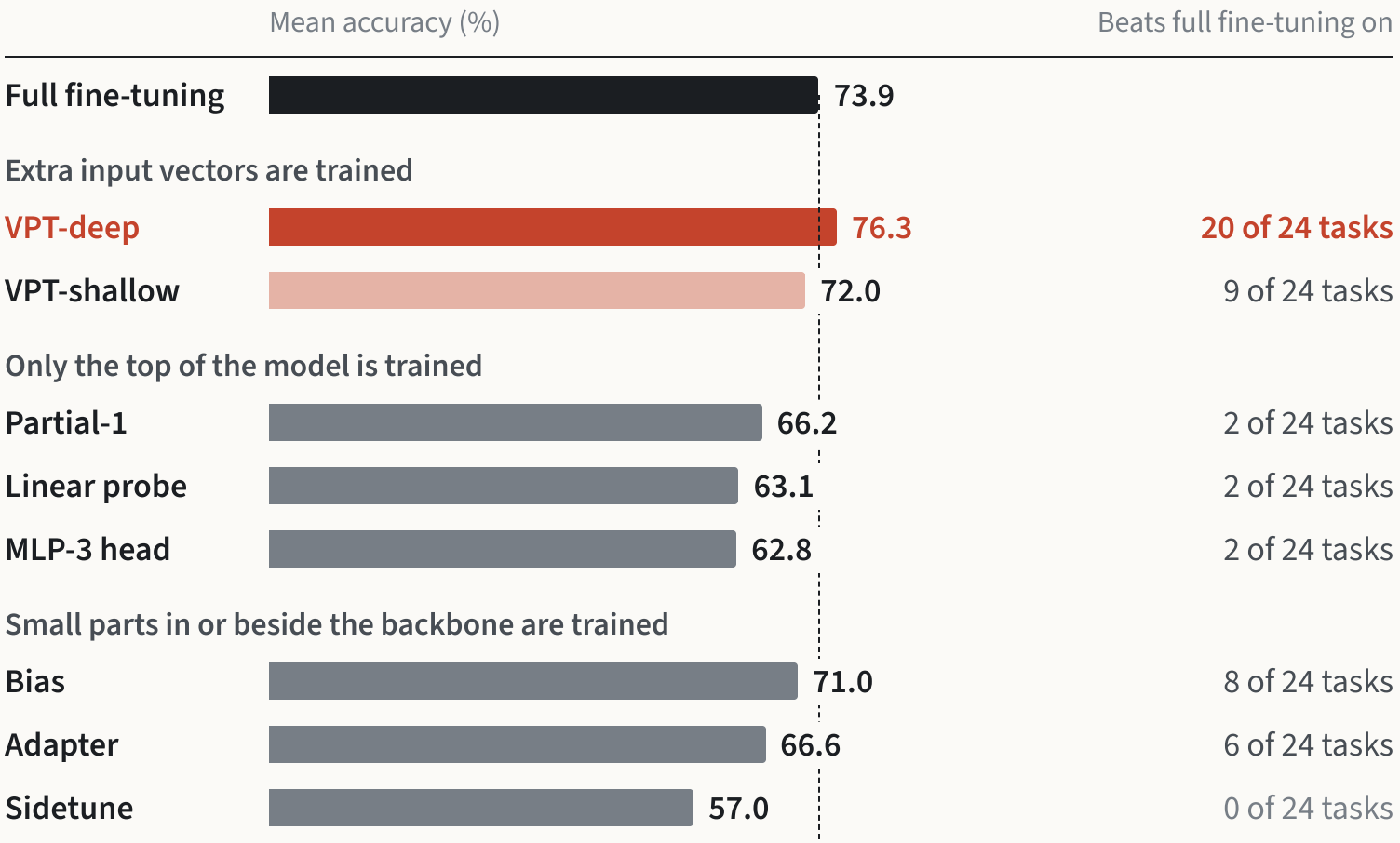
These eight tasks ask about geometry rather than objects: how many things there are, how far away, at what angle. Most are synthetic scenes, far from ImageNet.

Here VPT-deep wins all eight, 55.0 against 47.6, while the linear probe falls to 26.8, so the frozen features alone do not hold this information.

Jia et al. 2022, Table 1 and Tables 13–14: ViT-B/16 pretrained on ImageNet-21k, mean test accuracy within each group. For all 24 tasks, the mean of the four group averages.

VPT-deep beats full fine-tuning on 20 of 24 tasks

FGVC | Natural | Specialized | Structured | **All 24 tasks**



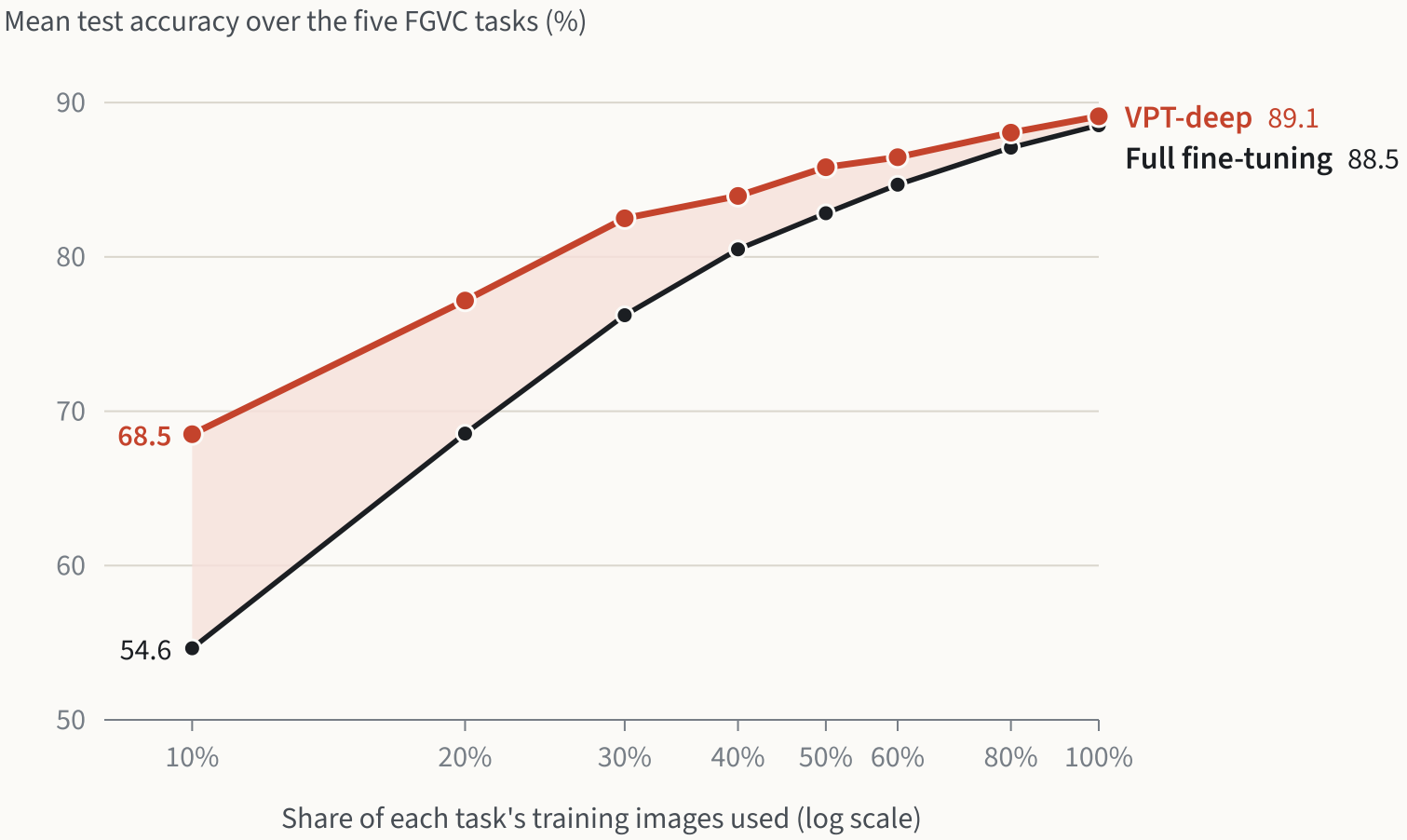
Averaged over all 24 tasks, VPT-deep is the only cheap method above full fine-tuning, 76.3 against 73.9.

It gets there while storing 1.18 backbones' worth of weights for the 24 tasks, where full fine-tuning stores 24.

Jia et al. 2022, Table 1 and Tables 13–14: ViT-B/16 pretrained on ImageNet-21k, mean test accuracy within each group. For all 24 tasks, the mean of the four group averages.

VPT leads with little data and stays ahead with more

VPT-deep against | Full fine-tuning | Linear probe | Adapter | Bias



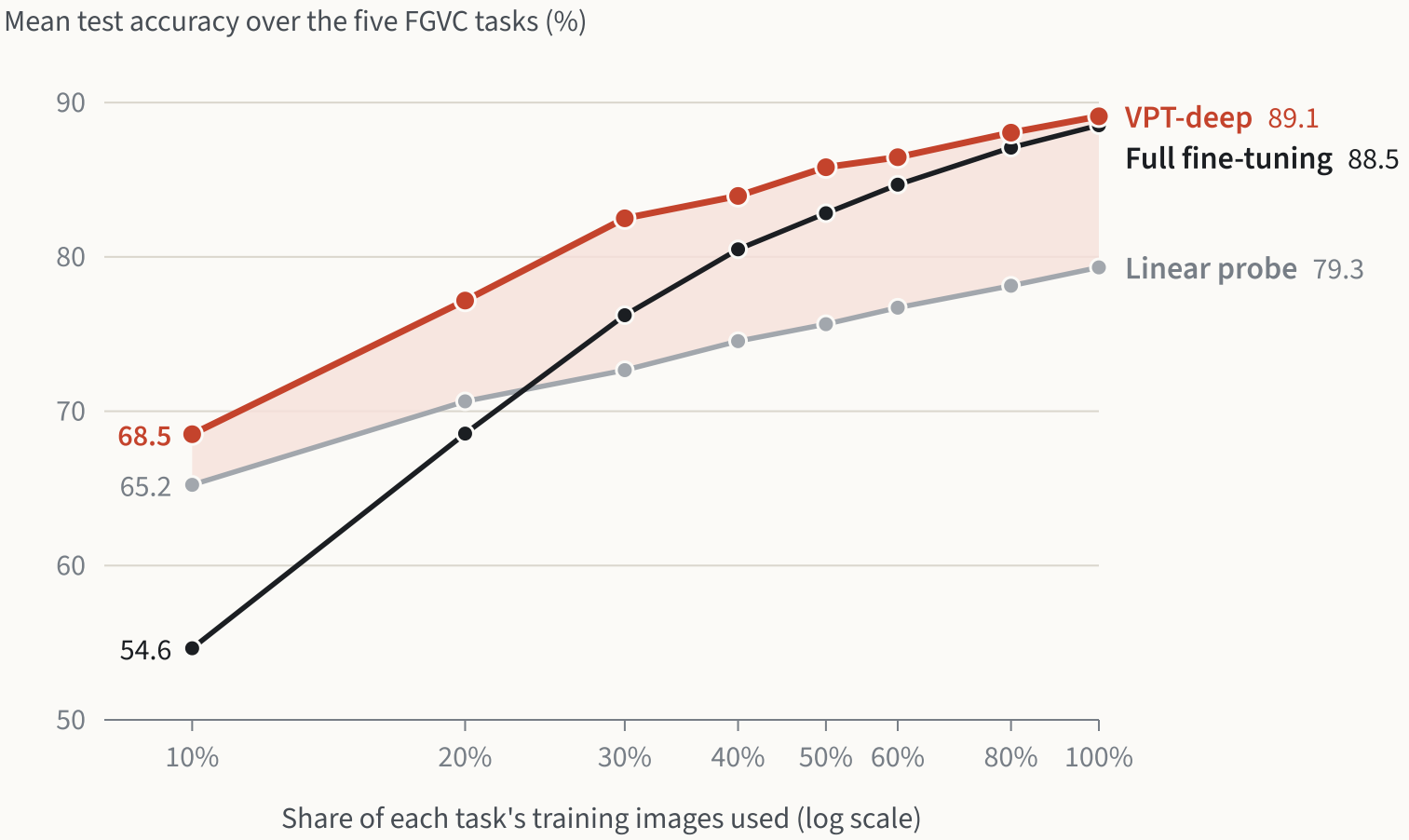
With a tenth of the training images, full fine-tuning has too many weights to fit and falls 14 points behind VPT-deep.

The gap shrinks as the data grows, but VPT-deep stays ahead all the way to the full training set.

Jia et al. 2022, Fig. 3: ViT-B/16, mean test accuracy over the five FGVC tasks, each method trained on a share of every task's training images. The paper gives these values only as a figure; they were read from its vector drawing in the PDF, and the 100% points match Table 1.

VPT leads with little data and stays ahead with more

VPT-deep against | Full fine-tuning | **Linear probe** | Adapter | Bias



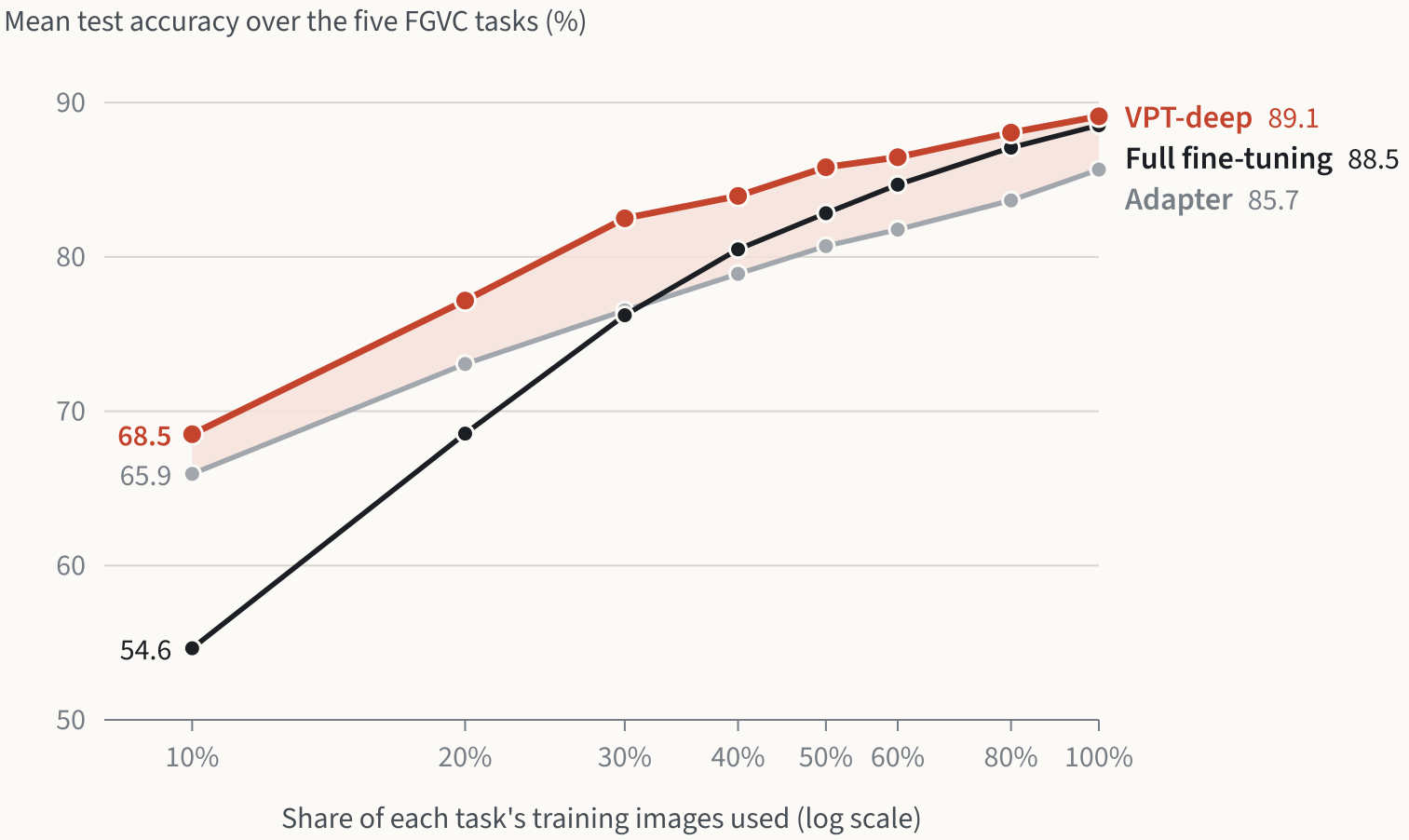
The linear probe trains only a new output layer, so it does well when data is scarce.

From 30% of the data on, full fine-tuning overtakes it, and with all the data it trails VPT-deep by almost 10 points.

Jia et al. 2022, Fig. 3: ViT-B/16, mean test accuracy over the five FGVC tasks, each method trained on a share of every task's training images. The paper gives these values only as a figure; they were read from its vector drawing in the PDF, and the 100% points match Table 1.

VPT leads with little data and stays ahead with more

VPT-deep against | Full fine-tuning | Linear probe | **Adapter** | Bias



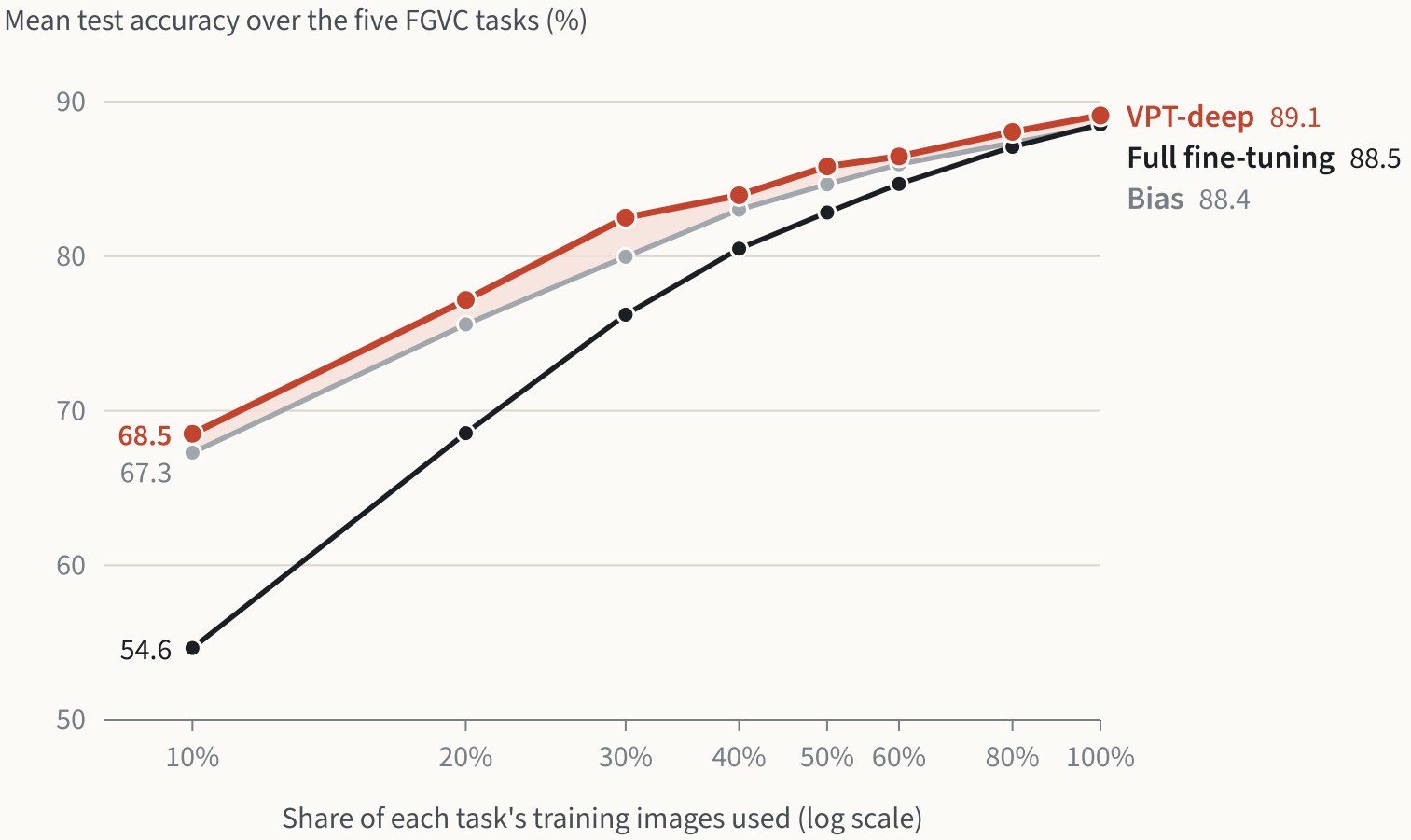
The adapter inserts small trained blocks into every layer, and it too beats full fine-tuning on little data.

Full fine-tuning passes it at 40% of the data, while VPT-deep stays 3 to 6 points above it throughout.

Jia et al. 2022, Fig. 3: ViT-B/16, mean test accuracy over the five FGVC tasks, each method trained on a share of every task's training images. The paper gives these values only as a figure; they were read from its vector drawing in the PDF, and the 100% points match Table 1.

VPT leads with little data and stays ahead with more

VPT-deep against | Full fine-tuning | Linear probe | Adapter | **Bias**



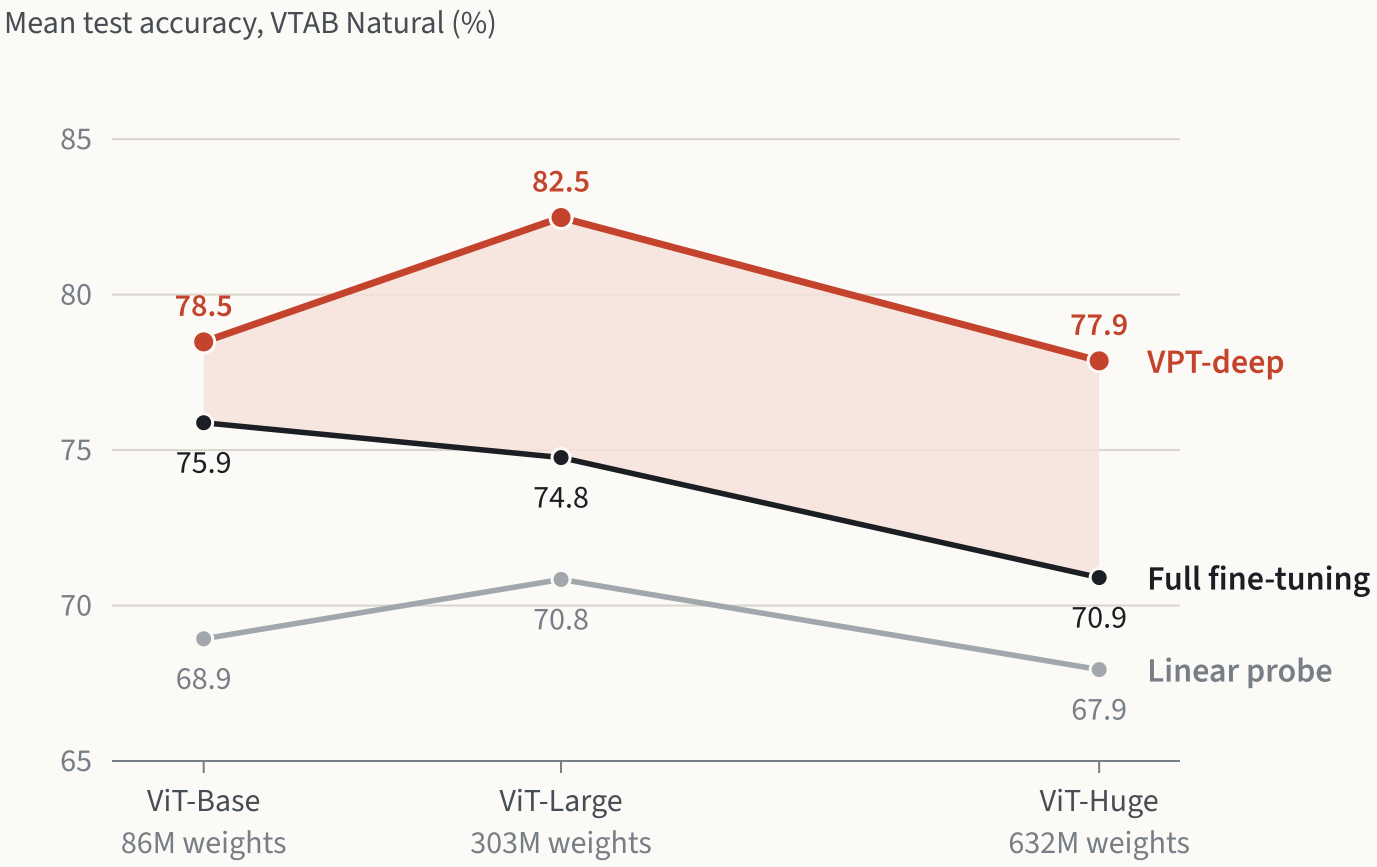
Bias trains only the bias terms and is the closest rival: it beats full fine-tuning up to 80% of the data.

VPT-deep is still ahead at every point, by half a point to 2.5 points.

Jia et al. 2022, Fig. 3: ViT-B/16, mean test accuracy over the five FGVC tasks, each method trained on a share of every task's training images. The paper gives these values only as a figure; they were read from its vector drawing in the PDF, and the 100% points match Table 1.

The result holds for larger backbones too

VTAB group: **Natural (7 tasks)** | Specialized (4 tasks) | Structured (8 tasks)



On ordinary photos, VPT-deep beats full fine-tuning with all three backbones: by 2.6 points on ViT-Base and by 7 to 8 points on the larger two.

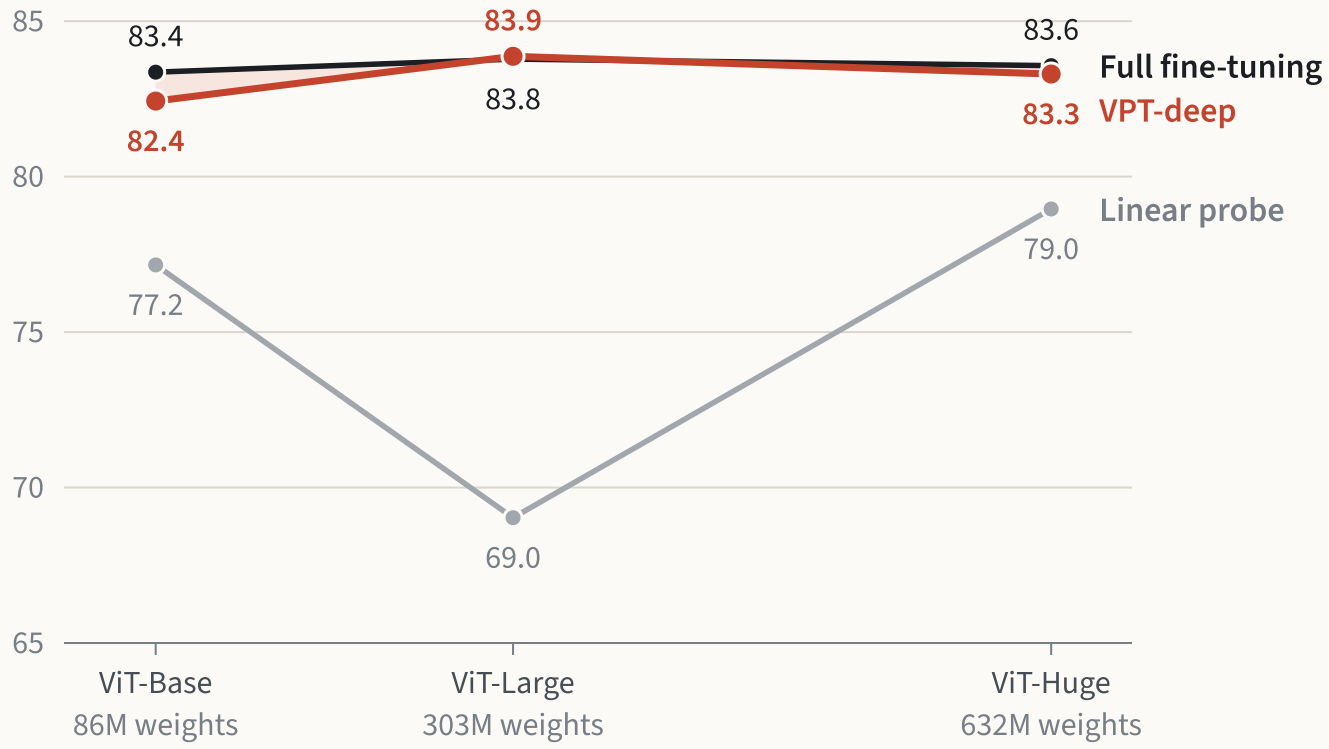
Full fine-tuning even gets worse as the model grows, from 75.9 to 70.9.

Jia et al. 2022, Fig. 4: VTAB-1k, mean test accuracy per group, backbones pretrained on ImageNet-21k. The paper gives these values only as a figure; they were read from its vector drawing in the PDF, and the ViT-B points match Table 1.

The result holds for larger backbones too

VTAB group: Natural (7 tasks) | **Specialized (4 tasks)** | Structured (8 tasks)

Mean test accuracy, VTAB Specialized (%)

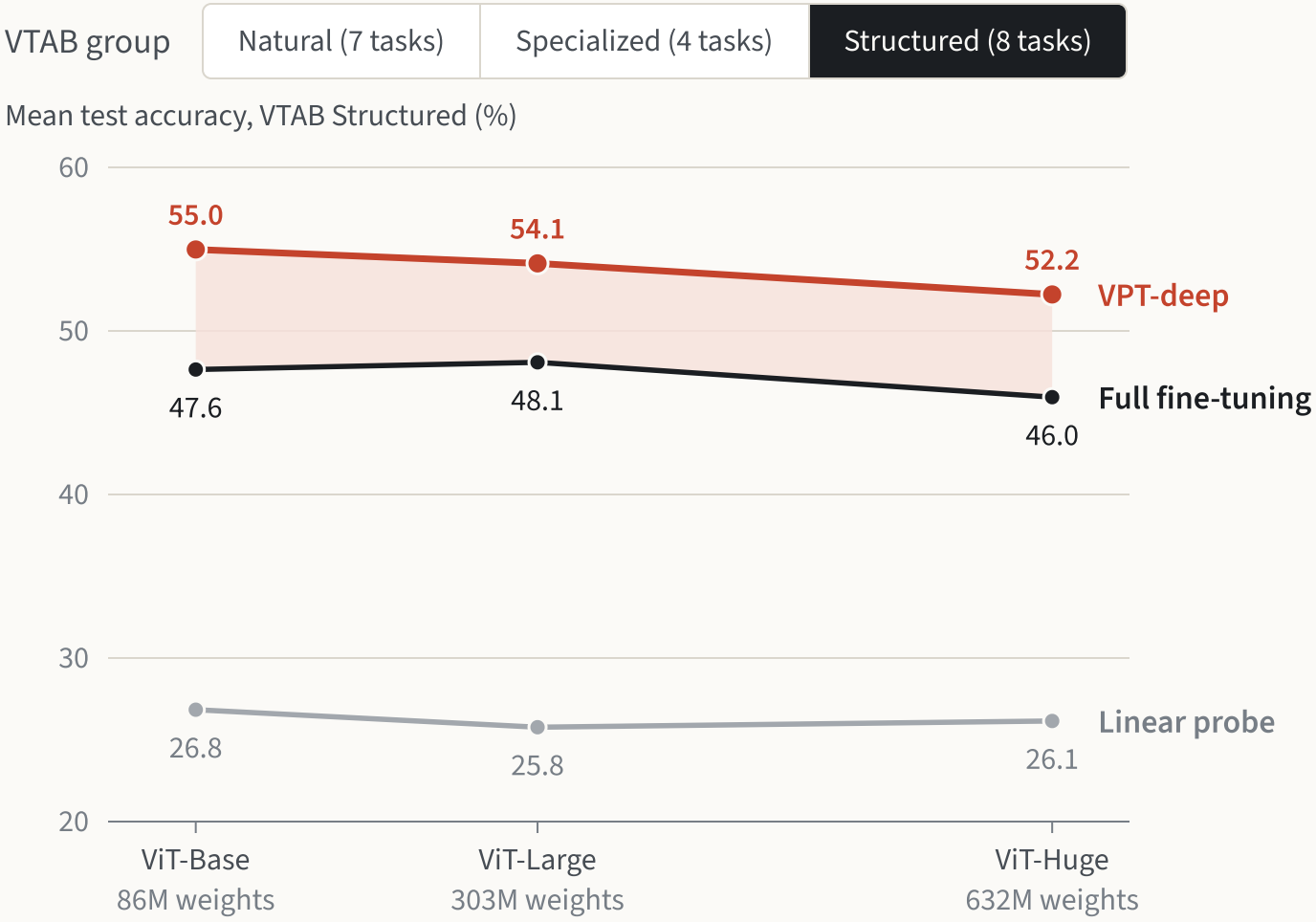


On medical and satellite images, the two are level with every backbone and never differ by more than one point.

This was already the one group where full fine-tuning kept a small lead on ViT-Base, and larger backbones do not change that.

Jia et al. 2022, Fig. 4: VTAB-1k, mean test accuracy per group, backbones pretrained on ImageNet-21k. The paper gives these values only as a figure; they were read from its vector drawing in the PDF, and the ViT-B points match Table 1.

The result holds for larger backbones too



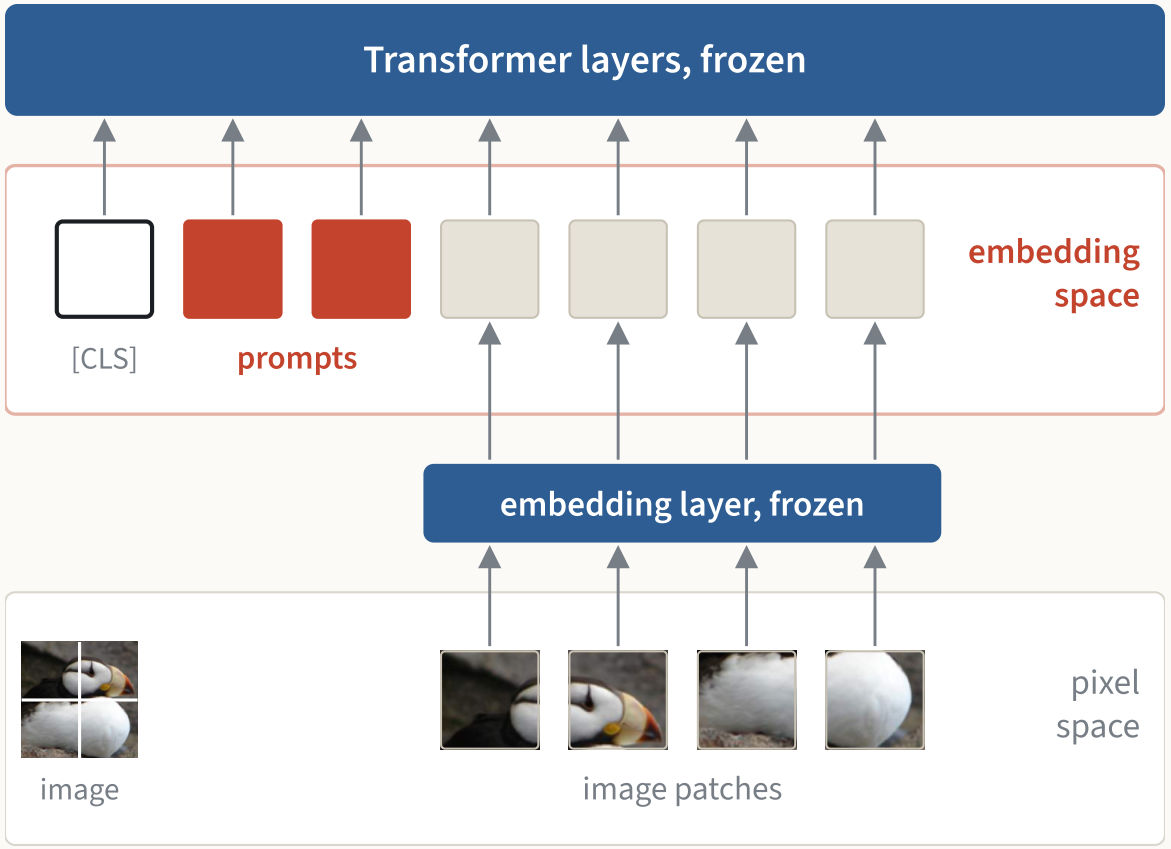
On counting and distance tasks, VPT-deep leads by 6 to 7 points with every backbone.

The linear probe stays near 26%, so the frozen features alone are not enough here, at any size.

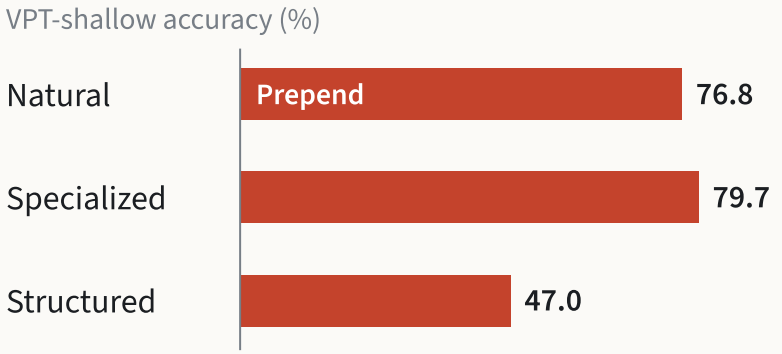
Jia et al. 2022, Fig. 4: VTAB-1k, mean test accuracy per group, backbones pretrained on ImageNet-21k. The paper gives these values only as a figure; they were read from its vector drawing in the PDF, and the ViT-B points match Table 1.

Four places to put the prompt

Prepend | Add | Prepend-pixel | Concat-channel | All four



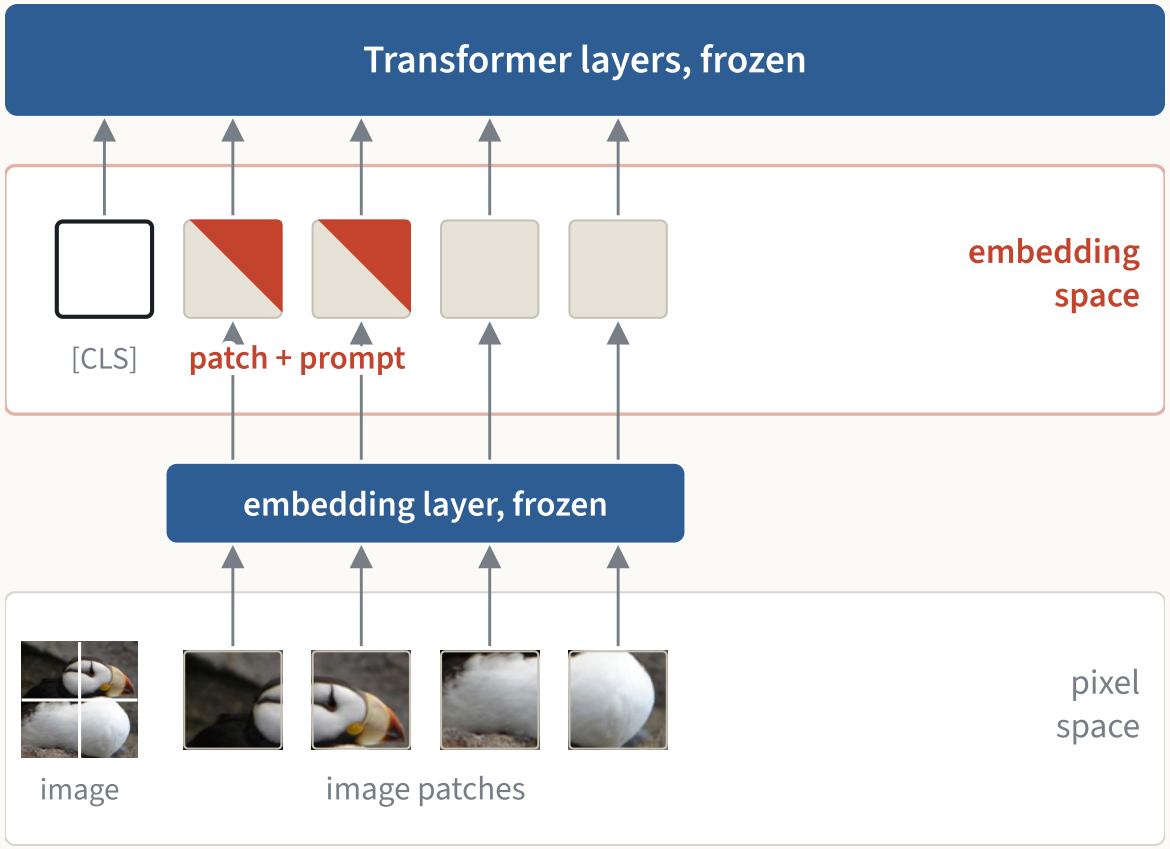
This is the default. The prompts are extra vectors placed next to the patch vectors, after the embedding layer, so every layer can attend to them as tokens of their own.



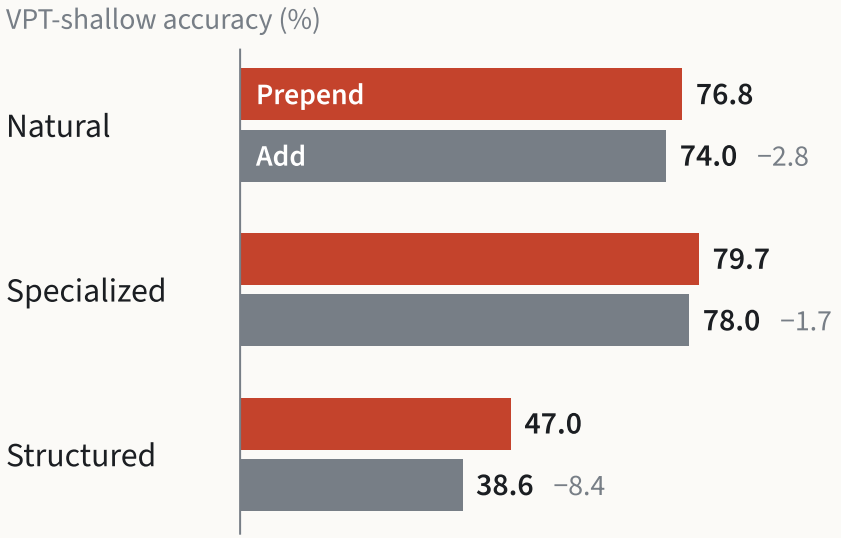
Numbers from Jia et al. 2022, Fig. 5 (ViT-B/16, prompts in the first layer only); "All four" is the mean of the three groups. Diagrams redrawn.

Four places to put the prompt

Prepend | **Add** | Prepend-pixel | Concat-channel | All four



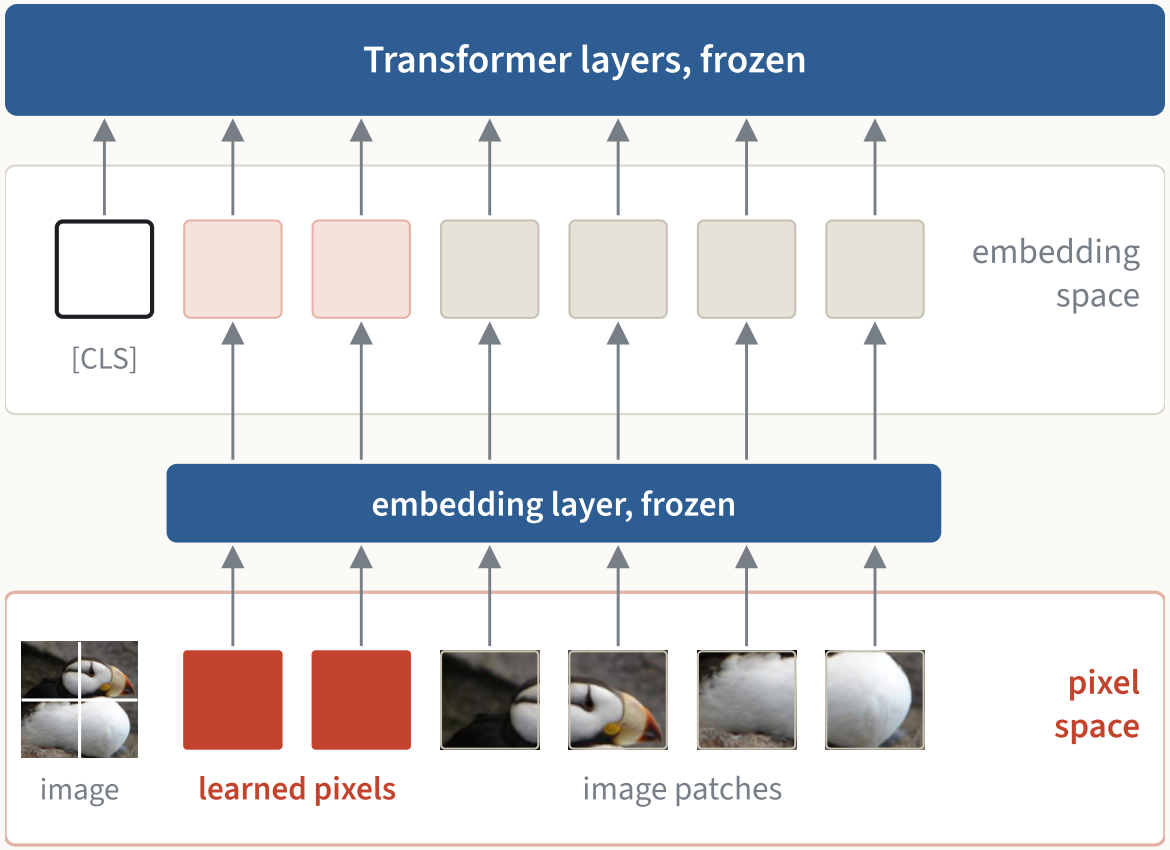
Here each prompt vector is added onto a patch vector, so the sequence keeps its length. This scores lower in all three groups, by 8.4 points on Structured.



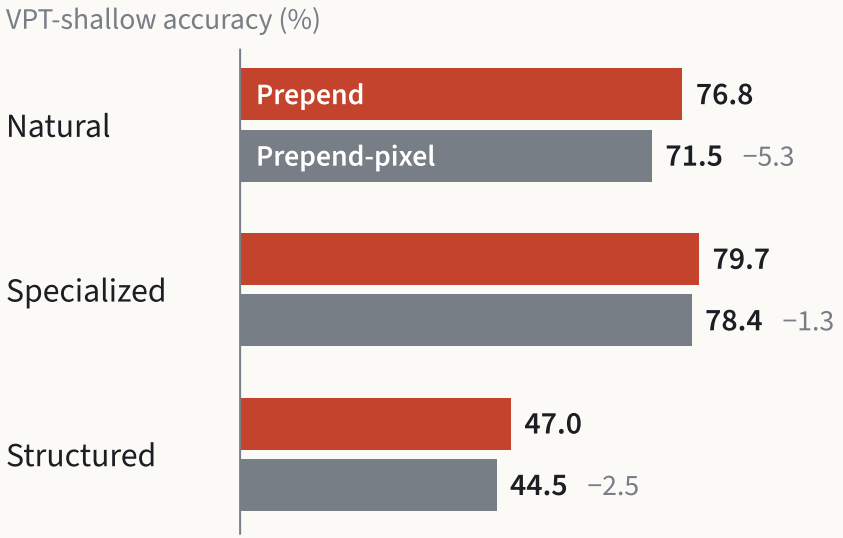
Numbers from Jia et al. 2022, Fig. 5 (ViT-B/16, prompts in the first layer only); "All four" is the mean of the three groups. Diagrams redrawn.

Four places to put the prompt

Prepend | Add | **Prepend-pixel** | Concat-channel | All four



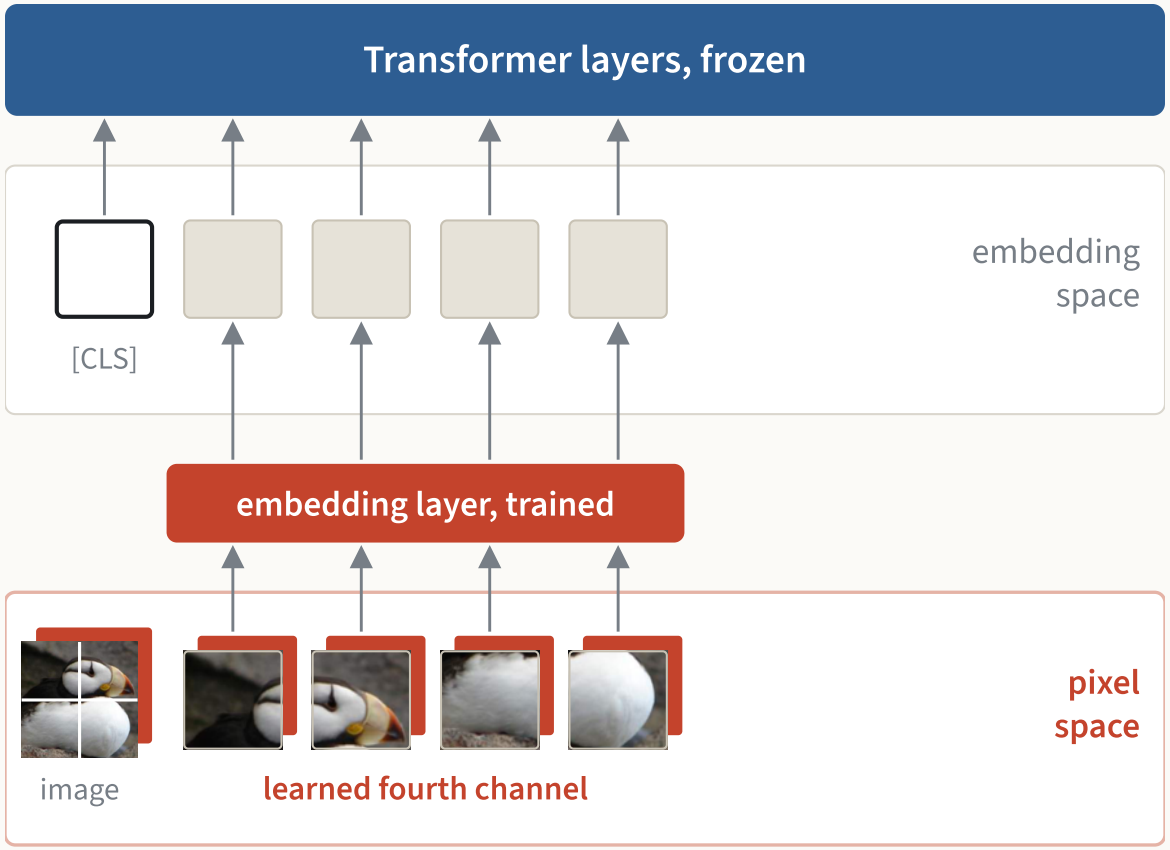
Here the prompts are learned pixel patches, placed next to the image patches before the frozen embedding layer. This also scores lower in every group, by 5.3 points on Natural.



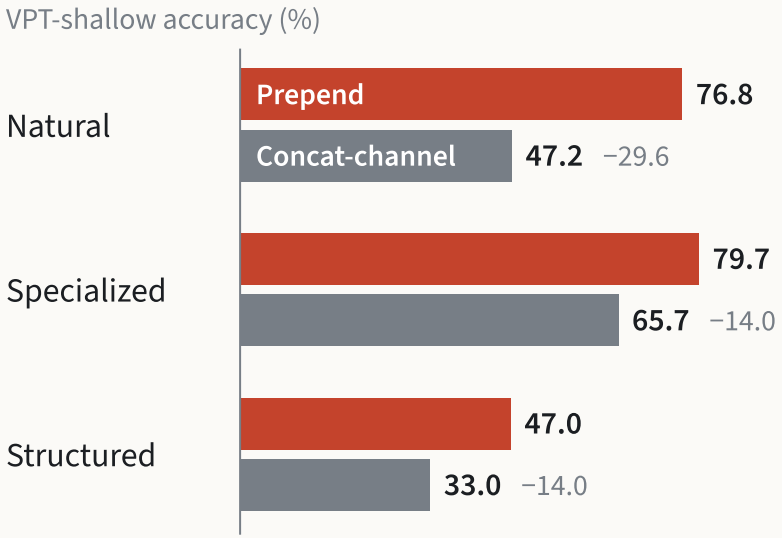
Numbers from Jia et al. 2022, Fig. 5 (ViT-B/16, prompts in the first layer only); "All four" is the mean of the three groups. Diagrams redrawn.

Four places to put the prompt

Prepend | Add | Prepend-pixel | **Concat-channel** | All four



Here the prompt is a learned fourth colour channel on the image, and the embedding layer is trained to read it. This is by far the worst: 29.6 points lower on Natural, below even a linear probe there.



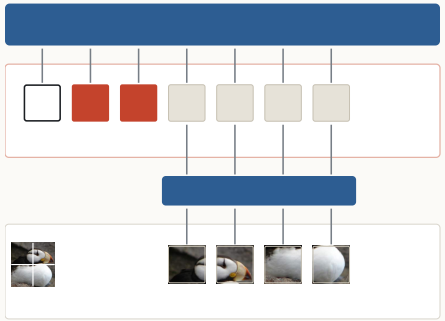
Numbers from Jia et al. 2022, Fig. 5 (ViT-B/16, prompts in the first layer only); "All four" is the mean of the three groups. Diagrams redrawn.

Four places to put the prompt

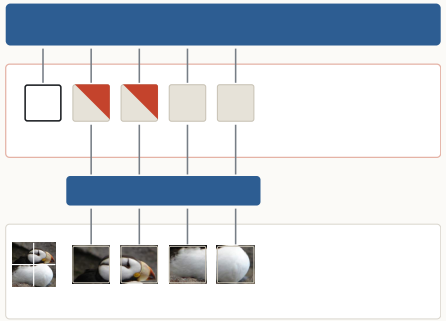
Prepend | Add | Prepend-pixel | Concat-channel | **All four**

Prompt in the embedding space

Prepend

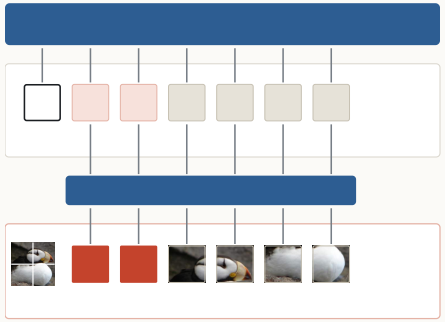


Add

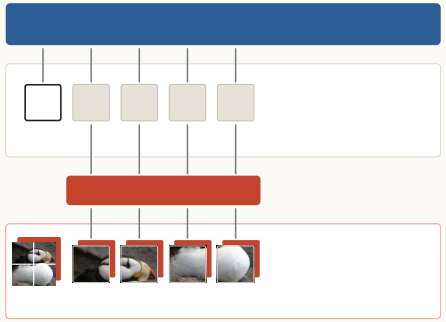


Prompt in pixel space

Prepend-pixel



Concat-channel



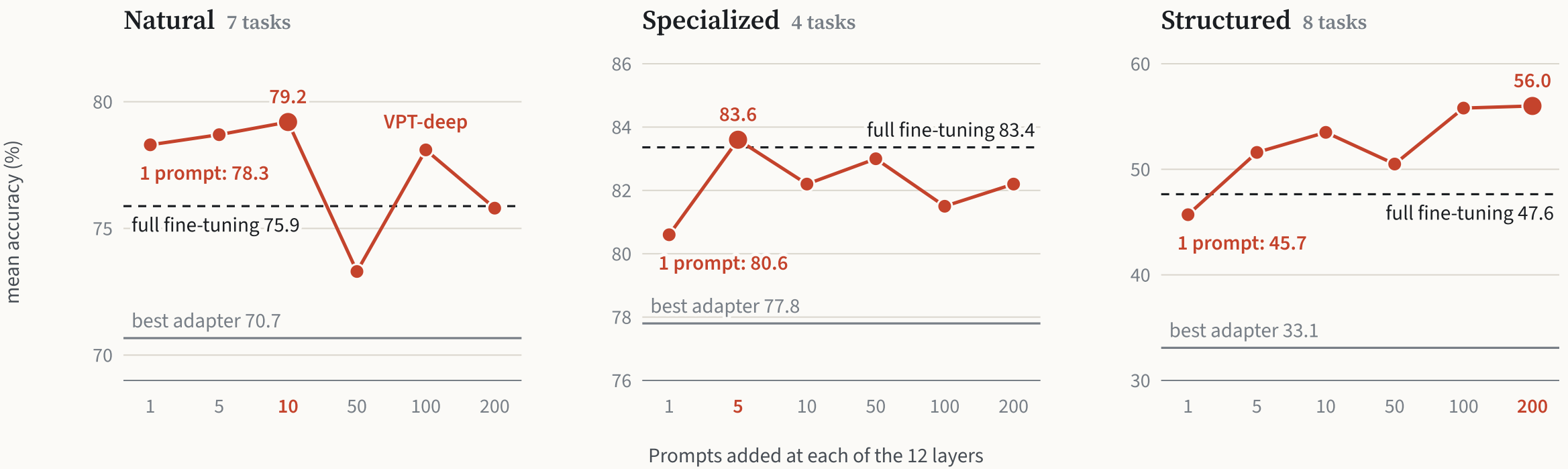
Prepend scores highest in every group. A prompt works best as separate vectors in the embedding space; adding it onto the patches or moving it into the pixels costs accuracy.

VPT-shallow accuracy (%), mean of the three groups

Numbers from Jia et al. 2022, Fig. 5 (ViT-B/16, prompts in the first layer only); "All four" is the mean of the three groups. Diagrams redrawn.

How many prompts, and at which layers

Number of prompts | Which layers

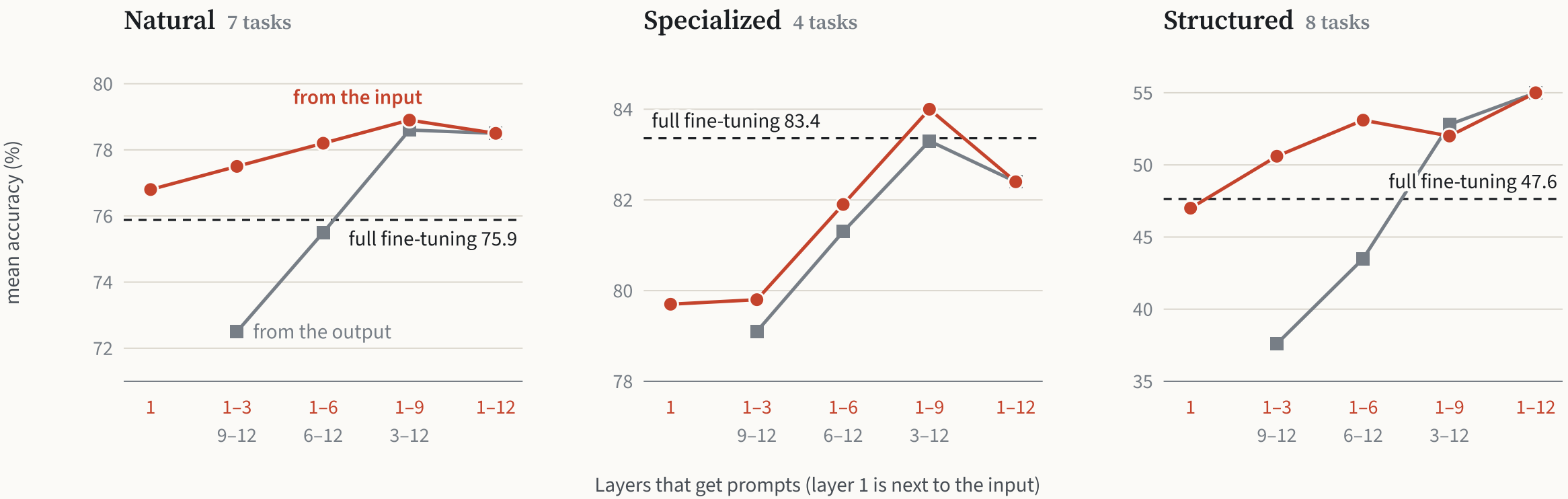


Each group prefers a different number: Natural does best with 10 prompts per layer, Specialized with 5 and Structured with 200. Even a single prompt per layer beats the best adapter in every group, and on Natural it already beats full fine-tuning.

Jia et al. 2022, Fig. 6: VPT-deep on ViT-B/16, mean test accuracy per VTAB group. The paper gives these only as a figure; values are read off its plotted points. Full fine-tuning and the adapter (best of three sizes): Table 13.

How many prompts, and at which layers

Number of prompts | **Which layers**



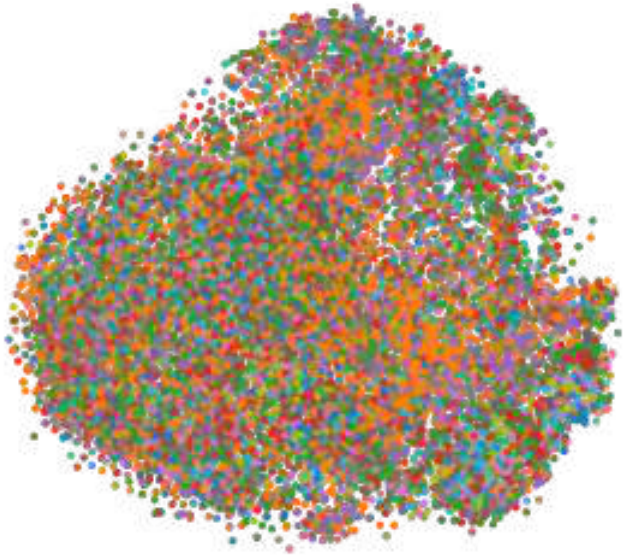
Prompting more layers generally helps: layer 1 alone is VPT-shallow, all twelve is VPT-deep. Layers next to the input matter most: prompts placed only in the layers near the output score lower, the opposite of what was found for text prompts.

Jia et al. 2022, Fig. 7, each variant with its best prompt length. Values are read off the figure's plotted points; layer 1 alone and layers 1–12 match VPT-shallow and VPT-deep in Table 13.

With prompts, the same frozen network separates the classes

House numbers (SVHN) | Satellite land use (EuroSAT) | Counting objects (CLEVR)

Linear probe 36.6%
frozen network, no prompts



VPT-deep 78.1%
same frozen network, with prompts

Full fine-tuning 87.4%
every weight retrained

Each dot is one test image, coloured by its true class; the number is test accuracy. With the frozen features alone, the classes overlap. With prompts, the same frozen network produces clearly separated groups, so a simple linear head can tell them apart.

t-SNE maps of the final [CLS] vector, test set, from Jia et al. 2022, Fig. 9. Accuracies from Fig. 9 and Table 13.

With prompts, the same frozen network separates the classes

House numbers (SVHN) | **Satellite land use (EuroSAT)** | Counting objects (CLEVR)

Linear probe 87.5%
frozen network, no prompts

VPT-deep 96.1%
same frozen network, with prompts

Full fine-tuning 95.7%
every weight retrained

Each dot is one test image, coloured by its true class; the number is test accuracy. With the frozen features alone, the classes overlap. With prompts, the same frozen network produces clearly separated groups, so a simple linear head can tell them apart.

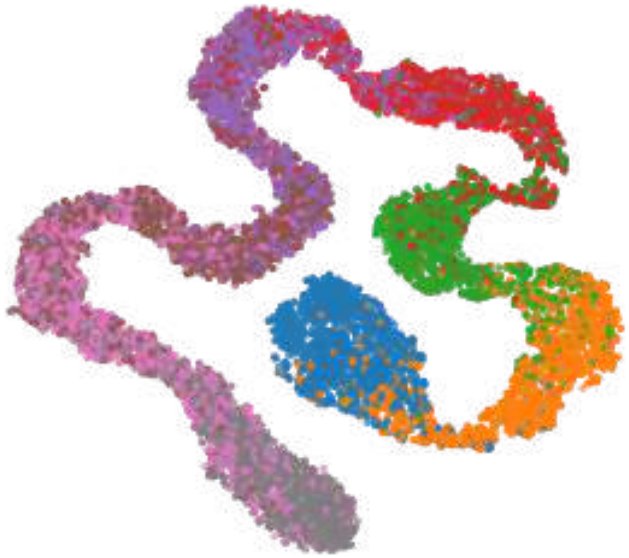
t-SNE maps of the final [CLS] vector, test set, from Jia et al. 2022, Fig. 9. Accuracies from Fig. 9 and Table 13.

With prompts, the same frozen network separates the classes

House numbers (SVHN) | Satellite land use (EuroSAT) | **Counting objects (CLEVR)**

Linear probe 34.3%
frozen network, no prompts

VPT-deep 68.5%
same frozen network, with prompts



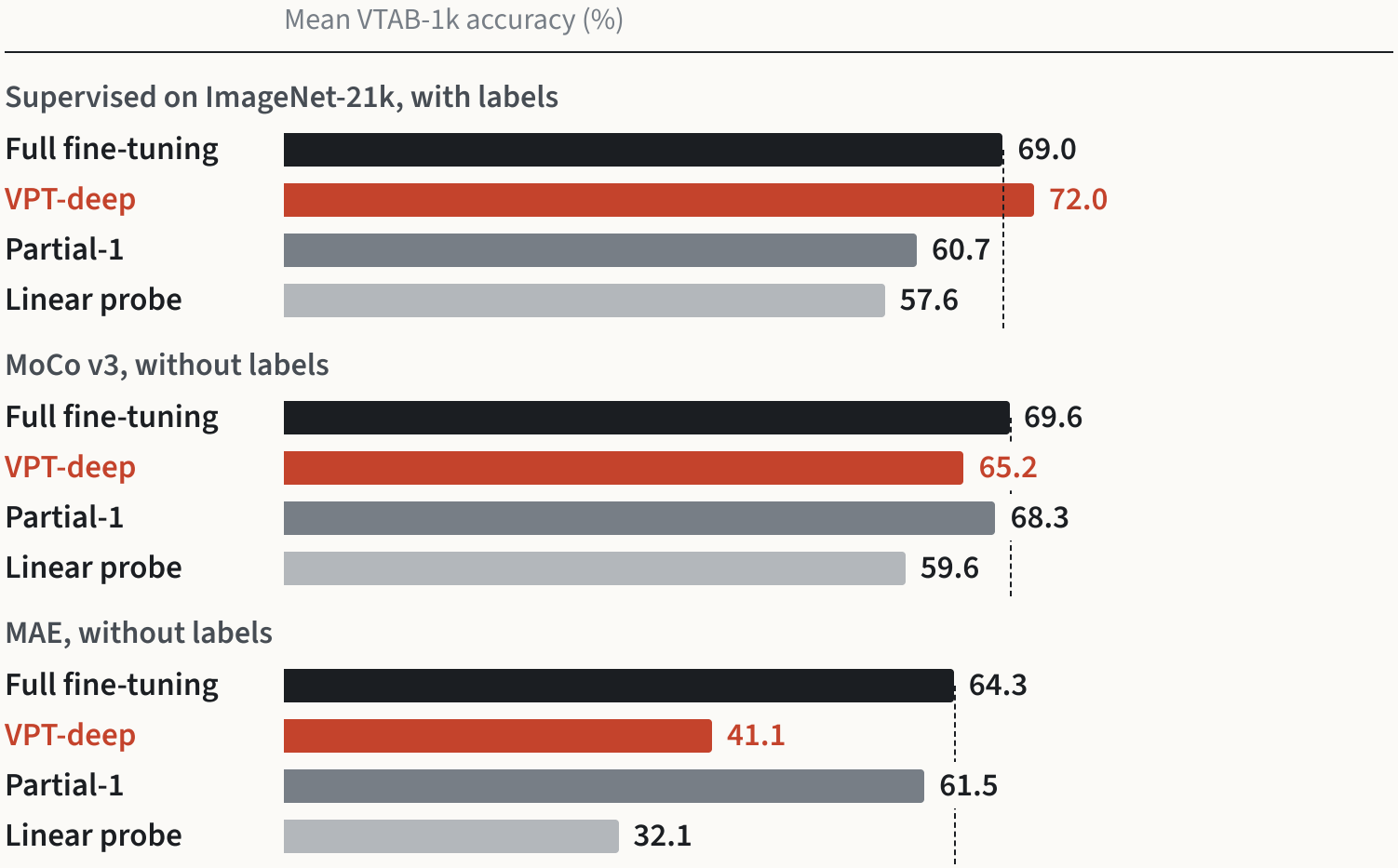
Full fine-tuning 56.3%
every weight retrained

Each dot is one test image, coloured by its true class; the number is test accuracy. With the frozen features alone, the classes overlap. With prompts, the same frozen network produces clearly separated groups, so a simple linear head can tell them apart.

t-SNE maps of the final [CLS] vector, test set, from Jia et al. 2022, Fig. 9. Accuracies from Fig. 9 and Table 13.

Where VPT falls short

Pretrained without labels | Swin Transformer | Segmentation



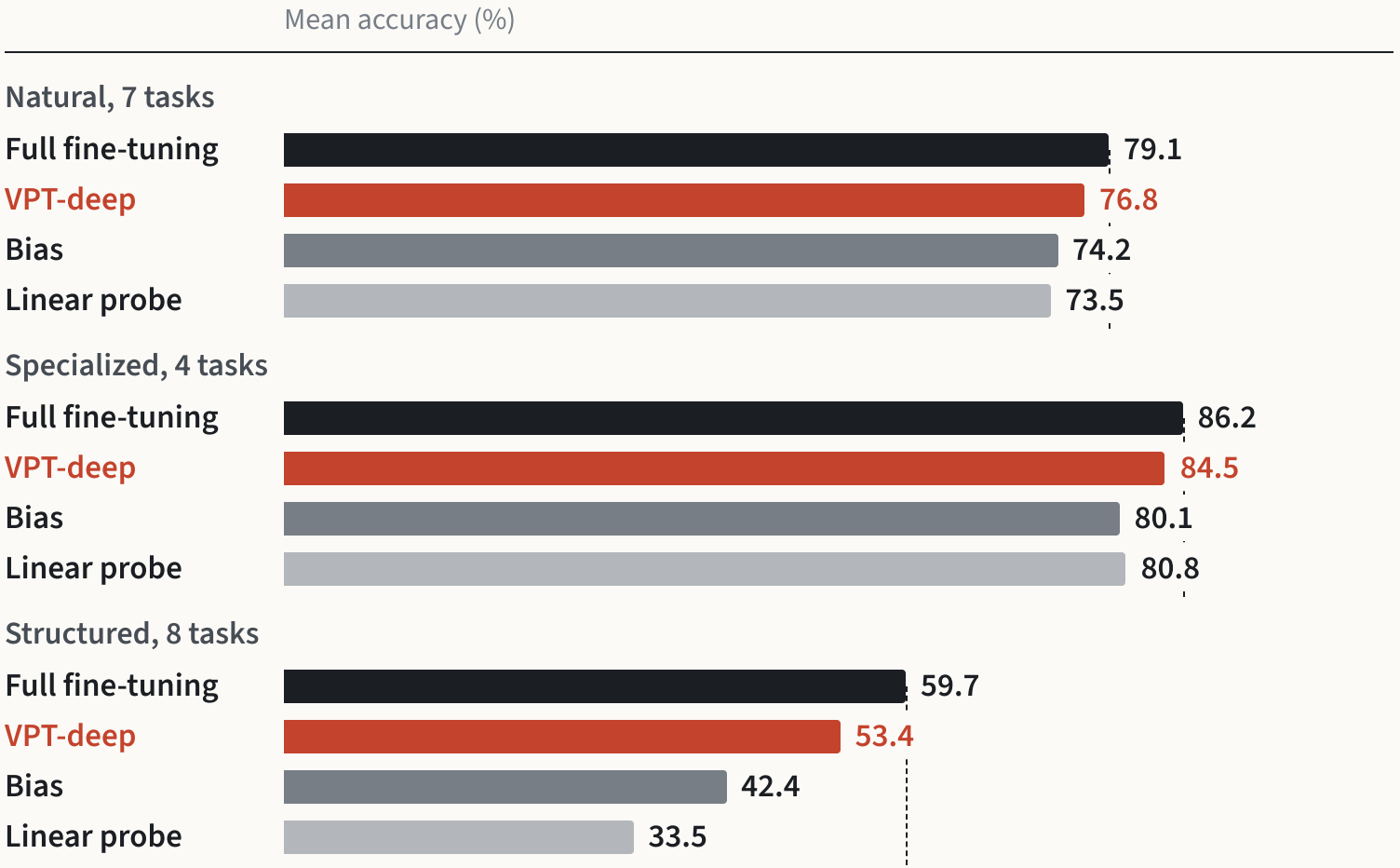
Everything so far used a ViT pretrained with labels. The paper also tries the same ViT-B pretrained without labels, and there VPT loses its lead.

With MoCo v3 it ends four points behind full fine-tuning. With MAE it drops towards the linear probe, while Partial-1, which retrains only the last block, stays close to full fine-tuning. The authors leave the reason open.

Jia et al. 2022, Table 1 (with labels) and Table 4 (MoCo v3, MAE): ViT-B/16, mean of the three VTAB-1k group averages.

Where VPT falls short

Pretrained without labels | **Swin Transformer** | Segmentation



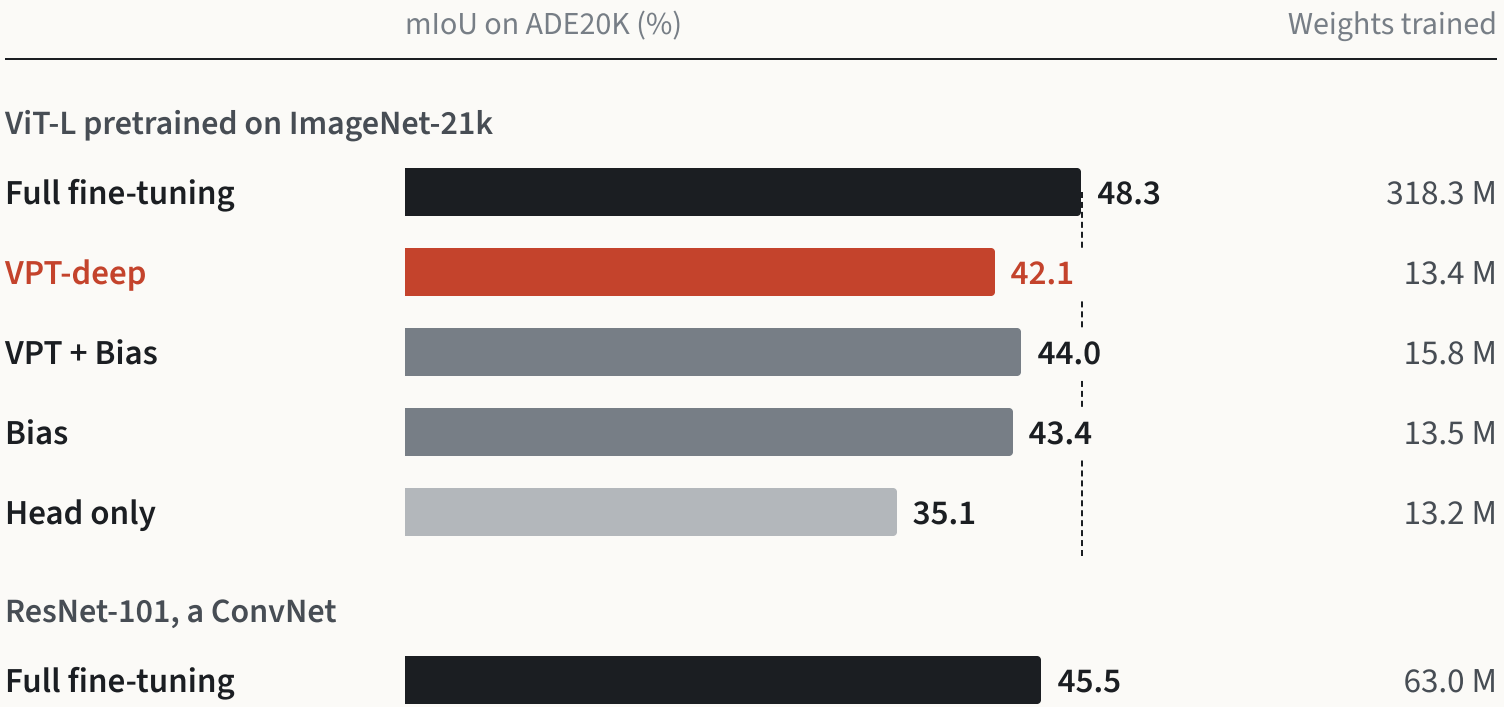
Swin is a vision Transformer that looks at small windows of the image and merges patches as it goes deeper.

On Swin, VPT still beats the other cheap methods. But full fine-tuning is ahead of VPT-deep in all three groups, by 6.3 points on Structured.

Jia et al. 2022, Table 2: Swin-B pretrained on ImageNet-21k with labels, mean test accuracy per VTAB-1k group.

Where VPT falls short

Pretrained without labels | Swin Transformer | **Segmentation**



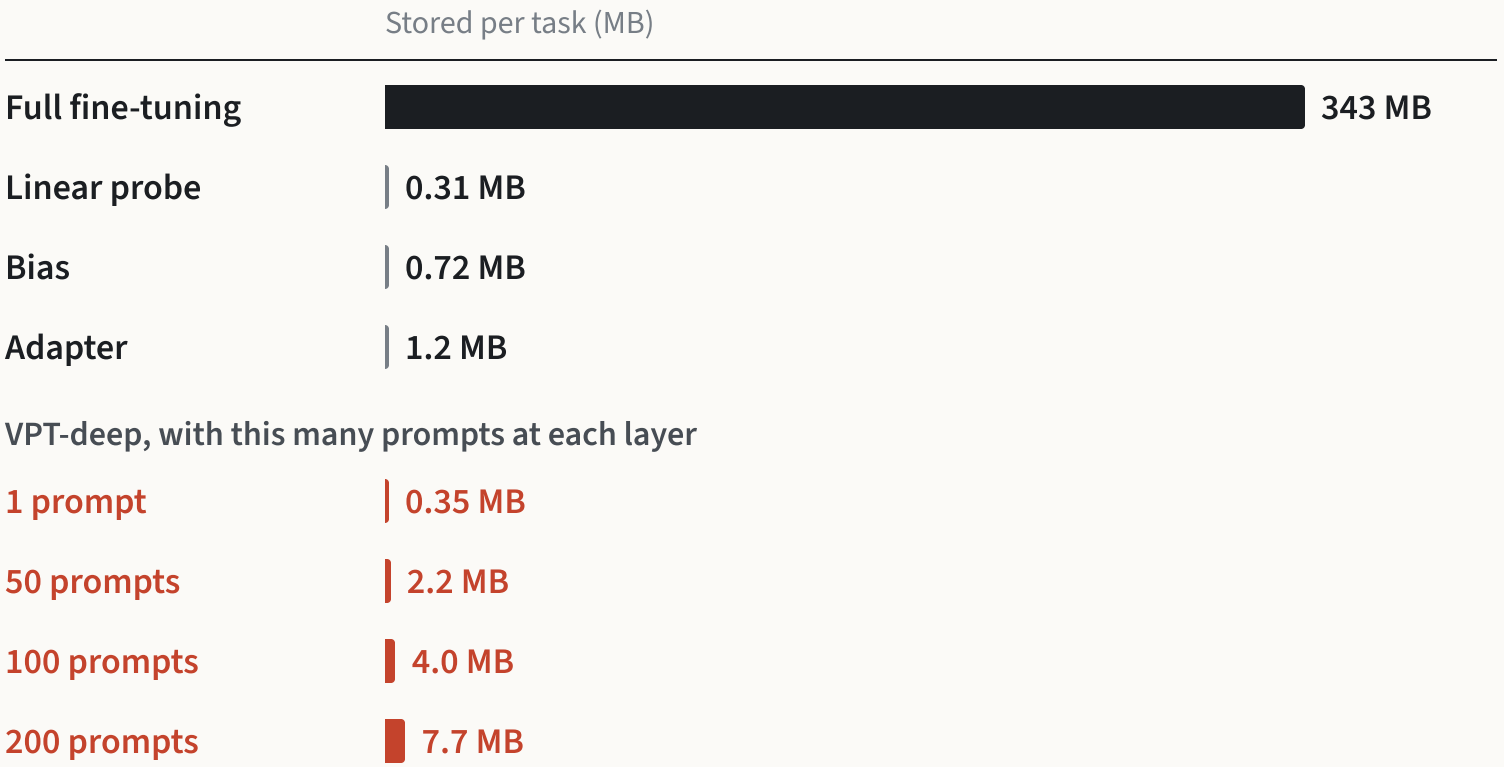
Segmentation labels every pixel of a scene. The score, mIoU, measures how well the predicted regions overlap the true ones.

Here VPT-deep reaches 42.1 against 48.3 for full fine-tuning, about the same as tuning only the biases. Prompts and biases together come within 1.5 points of a fully fine-tuned ResNet-101.

Jia et al. 2022, Table 3: ADE20K validation, SETR decoder, single-scale mIoU. ResNet-101: DeepLab v3+ (Chen et al. 2018).

VPT saves storage, not training memory or inference time

Storage per task | Training memory | Inference time



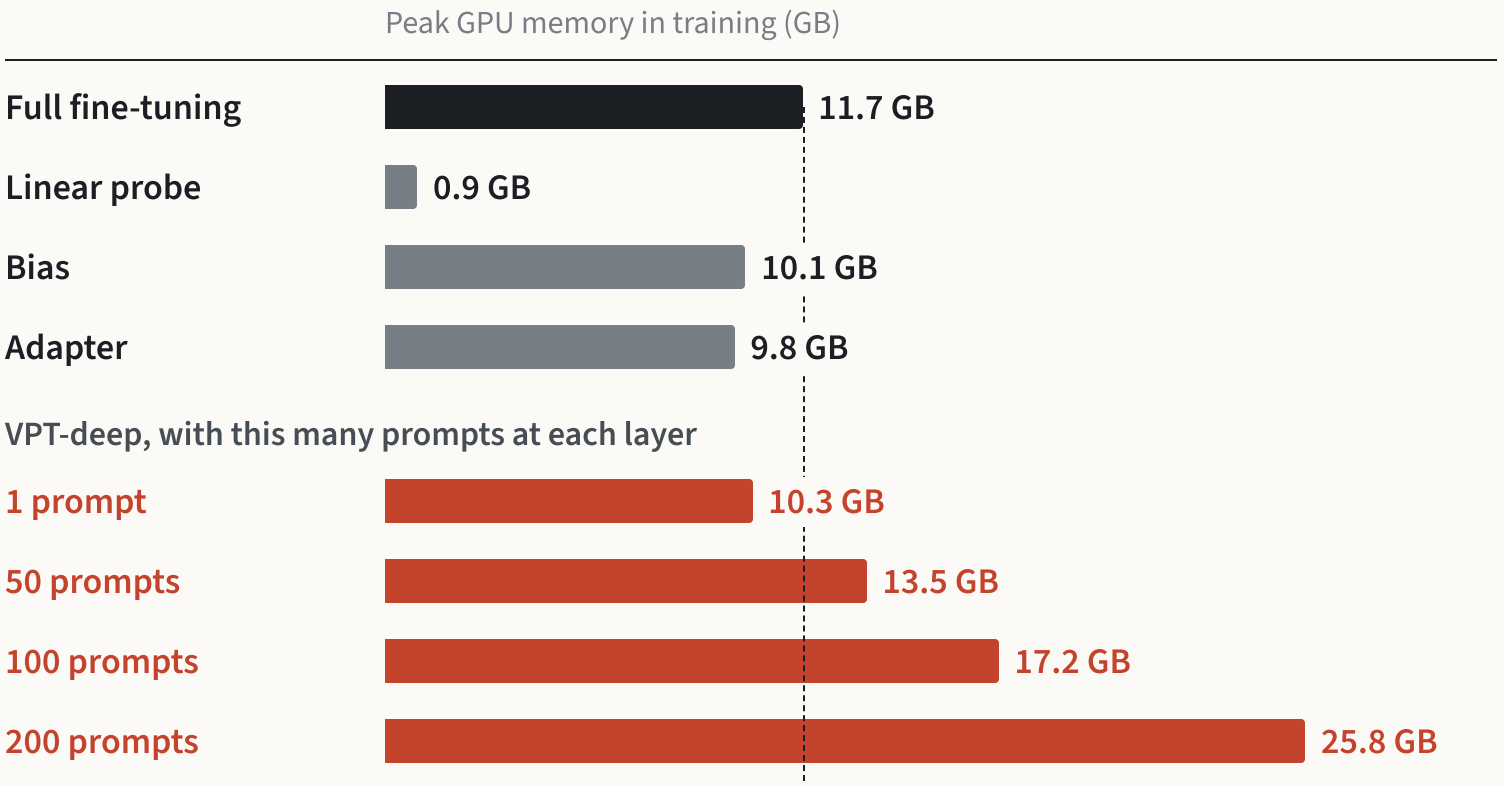
A task trained with VPT-deep keeps only its prompts and its head: 0.35 MB with one prompt per layer, 7.7 MB with 200. A fully fine-tuned copy is 343 MB for every task.

All tasks share one frozen backbone, so this is the real saving, and it grows with every task.

Trained numbers at 4 bytes each. Bias and adapter ($r = 64$): their share of tuned weights in Jia et al. 2022, Table 12. VPT-deep: $12 \times p \times 768$ prompt numbers plus a 102-class head.

VPT saves storage, not training memory or inference time

Storage per task | **Training memory** | Inference time



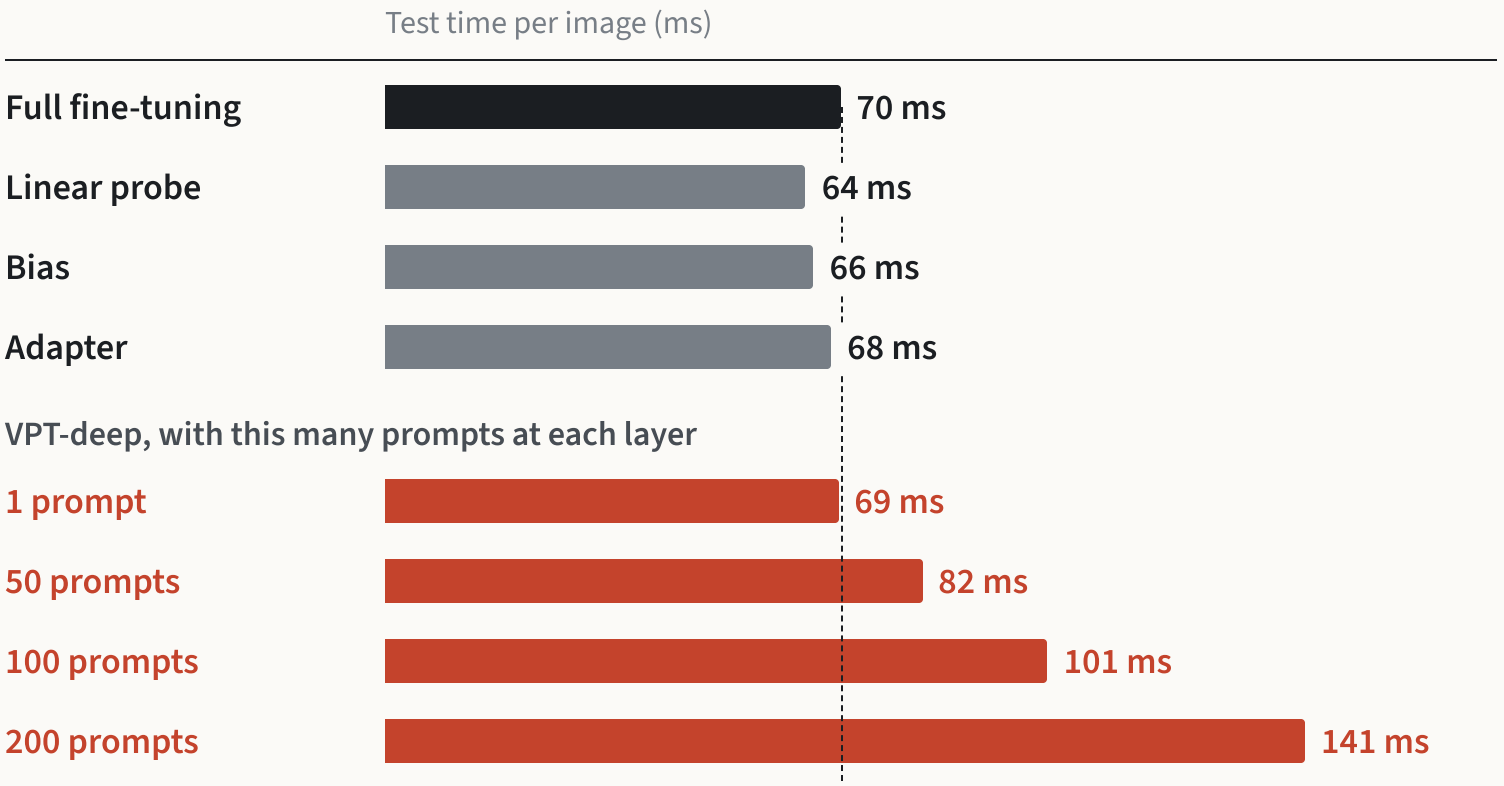
In training, VPT-deep needs about as much GPU memory as full fine-tuning, and with 200 prompts more than twice as much.

To train the prompts in the first layer, the gradient has to travel back through all twelve frozen layers, so their activations must be kept, as in full fine-tuning. Only the linear probe avoids this.

Jia et al. 2022, Appendix Table 12: ViT-B/16 on one A100 GPU, batch 64. The table has no rows for 50 and 100 prompts; those values are read from its Fig. 18.

VPT saves storage, not training memory or inference time

Storage per task | Training memory | **Inference time**



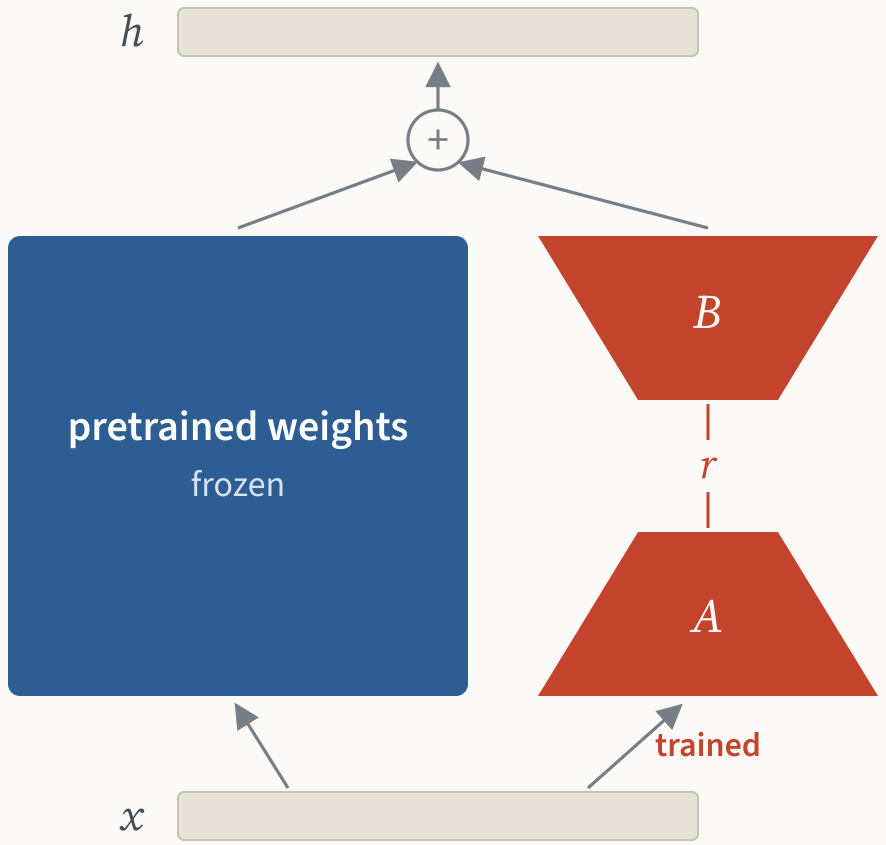
At test time every image carries the prompts through all twelve layers, so the sequence is longer. Fifty prompts make it 82 ms instead of 70, and 200 prompts double it.

The prompts cannot be folded into the weights, so this cost stays. LoRA, the next paper, avoids it.

Jia et al. 2022, Appendix Table 12: ViT-B/16 on one A100 GPU, batch 64. The table has no rows for 50 and 100 prompts; those values are read from its Fig. 18.

Low-Rank Adaptation

Teach a frozen model a new task by learning a small, low-rank change to its weights.



Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen
Microsoft. ICLR 2022. Results on images from Zhang, Zhou & Liu 2022 and Zhu et al. 2024.

Presented by **Ertugrul Senturk**
October 2026

From VPT to LoRA

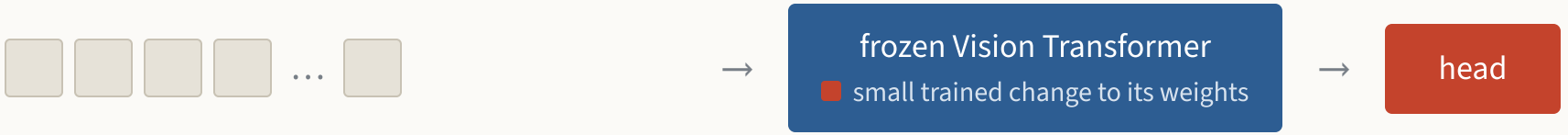
VPT

Jia et al. 2022

Learned vectors join the image patches, so every image carries a longer sequence through every layer.

LoRA

Hu et al. 2022



The input stays as it is. **A small change to the weights** is learned, and after training it is added into them.

VPT trained 0.6% of the weights and beat full fine-tuning on 20 of 24 tasks. But its prompts make every image slower, and they can never be folded into the model.

LoRA leaves the input alone and changes the weights instead. It ends up storing about as little per task as VPT, while the model runs exactly as fast as before.

VPT: Jia et al., ECCV 2022 (0.6% of the weights with 50 prompts per layer and a 102-class head; 20 of 24 tasks, Table 1). LoRA: Hu et al., ICLR 2022, first posted in June 2021.

Every fine-tuned copy is as big as the model

The paper starts from the largest model of its time, GPT-3, with 175 billion weights. Each fine-tuned copy takes 350 GB, so a service with a hundred tasks would store 35 TB. LoRA keeps one shared model and only a small file per task, and the frozen weights also need no optimizer state, which cuts training memory from 1.2 TB to 350 GB.

Shown for **1 task** on **ViT-B/16, 86 million weights**. The next pages show 100 tasks, then GPT-3.

Full fine-tuning: 1 retrained ViT-B copy

343 MB

1 copy at 343 MB each

LoRA: one shared ViT-B plus 1 small file

each red speck is one task file, to scale

345 MB

343 MB once, plus 1.5 MB per task

GPT-3 in 16-bit numbers: 350 GB per copy, 35 MB per LoRA file at rank 4 on the query and value weights (Hu et al. 2022, Sec. 4.2). ViT-B/16 in 32-bit: 343 MB per copy; its LoRA file is rank 8 on the same weights plus a 102-class head. Squares are sized by area.

Every fine-tuned copy is as big as the model

The paper starts from the largest model of its time, GPT-3, with 175 billion weights. Each fine-tuned copy takes 350 GB, so a service with a hundred tasks would store 35 TB. LoRA keeps one shared model and only a small file per task, and the frozen weights also need no optimizer state, which cuts training memory from 1.2 TB to 350 GB.

Shown for **100 tasks** on **ViT-B/16, 86 million weights**.

Full fine-tuning: 100 retrained ViT-B copies

34 GB

100 copies at 343 MB each

LoRA: one shared ViT-B plus 100 small files

493 MB

343 MB once, plus 1.5 MB per task

GPT-3 in 16-bit numbers: 350 GB per copy, 35 MB per LoRA file at rank 4 on the query and value weights (Hu et al. 2022, Sec. 4.2). ViT-B/16 in 32-bit: 343 MB per copy; its LoRA file is rank 8 on the same weights plus a 102-class head. Squares are sized by area.

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Shown for **100 tasks** on **GPT-3, 175 billion weights**.

Full fine-tuning: 100 retrained GPT-3 copies

35 TB

100 copies at 350 GB each

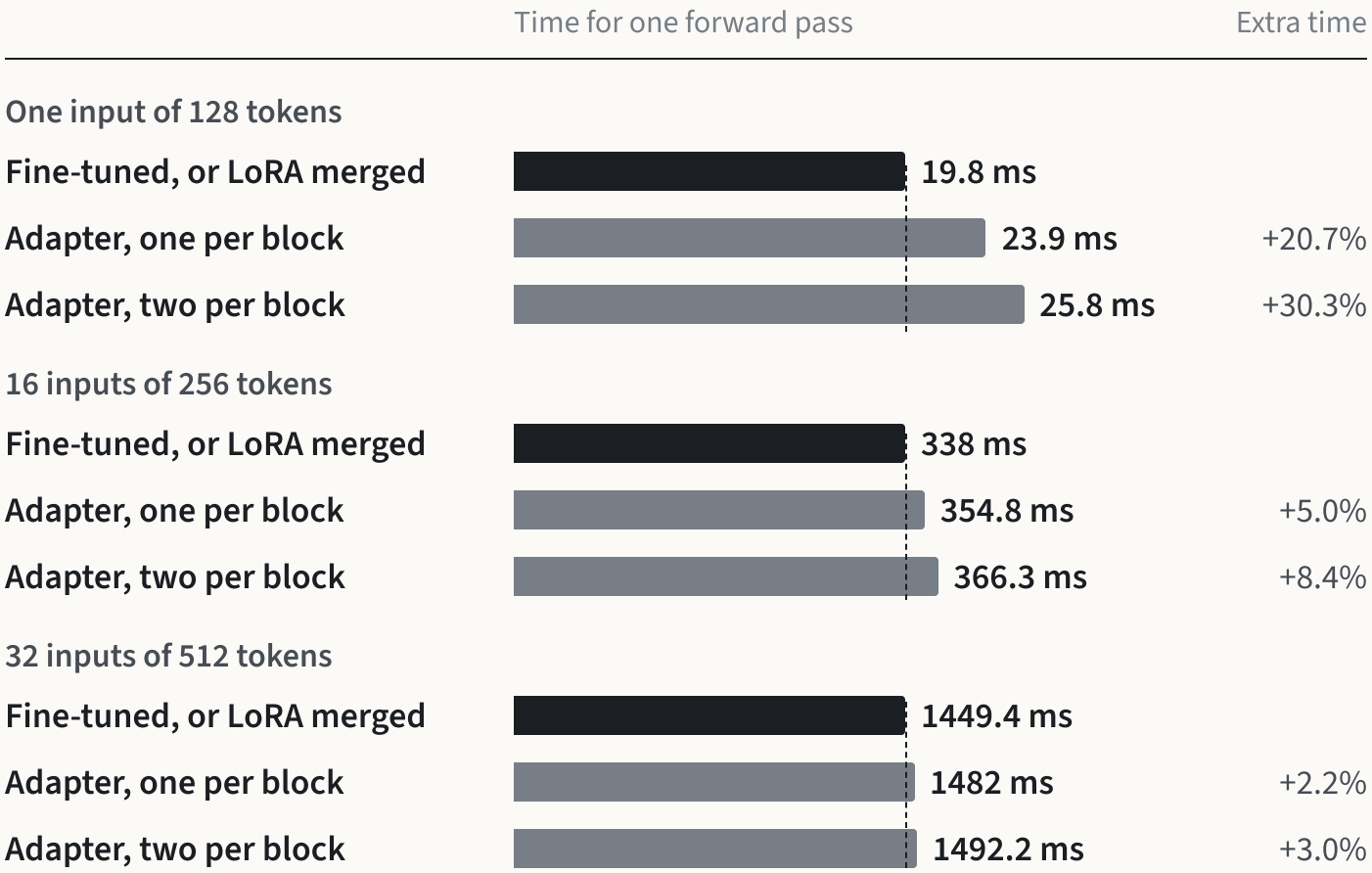
LoRA: one shared GPT-3 plus 100 small files

354 GB

350 GB once, plus 35 MB per task

GPT-3 in 16-bit numbers: 350 GB per copy, 35 MB per LoRA file at rank 4 on the query and value weights (Hu et al. 2022, Sec. 4.2). ViT-B/16 in 32-bit: 343 MB per copy; its LoRA file is rank 8 on the same weights plus a 102-class head. Squares are sized by area.

Earlier cheap methods cost time or sequence length



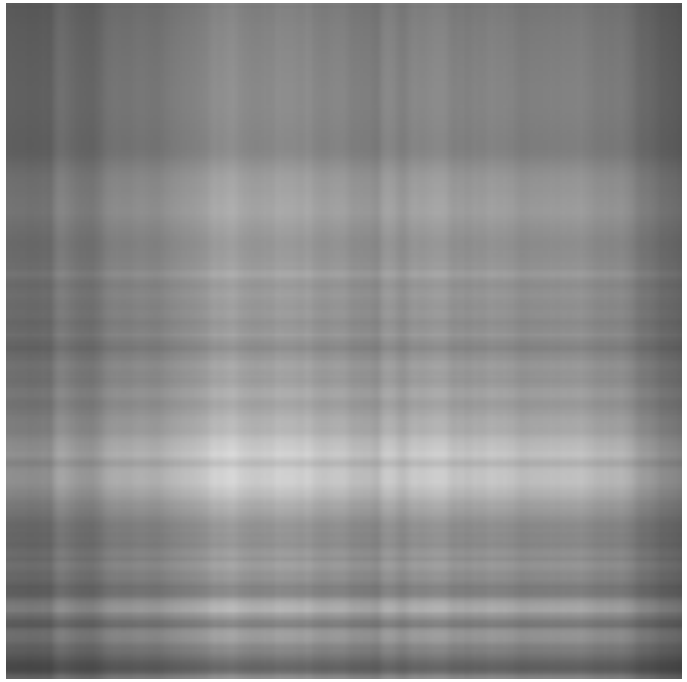
Adapters insert small extra layers into every block. They have few weights, but an image or a sentence must pass through them one after another, so serving a single input takes 20 to 30% longer. With large batches the gap shrinks to 2 or 3%.

Prompt methods such as prefix tuning and VPT lengthen the input instead. That leaves less room for the real input and costs compute in every layer, and on GPT-3 prefix tuning got worse beyond 256 prompt tokens.

LoRA adds no layer and no token at test time.

Hu et al. 2022, Table 1: one forward pass of GPT-2 medium, mean of 100 trials on one NVIDIA Quadro RTX 8000; bars are drawn relative to the fine-tuned model in each setting. Prefix tuning: Sec. 5.5 and Table 15.

What a low-rank matrix is



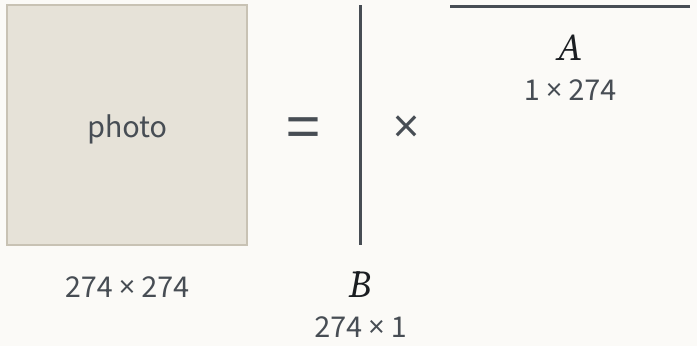
Rank 1: 1 × (274 + 274) = **548** numbers, 0.7% of the photo's 75,076

1	2	4	8	16	32	64	all 274

A grayscale photo is a matrix of 274 × 274 brightness values. Any matrix can be written as a sum of simple pieces, where each piece is one column of numbers times one row. The rank is the number of pieces.

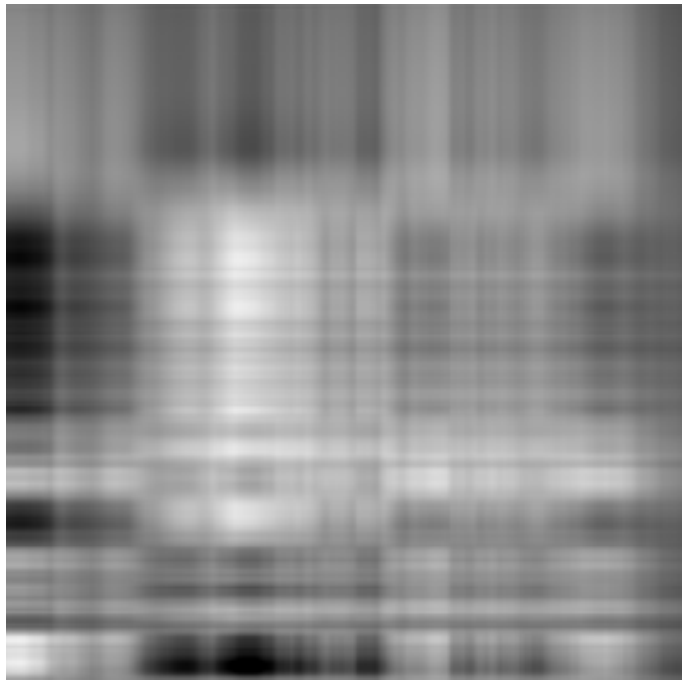
One piece gives only bands of light and dark, because every row of it is the same pattern, scaled.

> LoRA bets that the change fine-tuning makes to a weight matrix is like this photo: a few pieces carry almost all of it.



Not from the paper: the best rank-r approximation (SVD) of an Oxford Pets photo, Jia et al. 2022, Fig. 10. The low-rank bet follows Li et al. 2018 and Aghajanyan et al. 2021.

What a low-rank matrix is



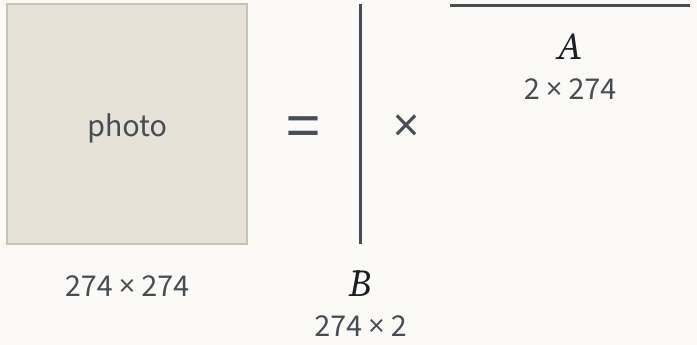
Rank 2: 2 × (274 + 274) = **1,096** numbers, 1.5% of the photo's 75,076

1	2	4	8	16	32	64	all 274

A grayscale photo is a matrix of 274 × 274 brightness values. Any matrix can be written as a sum of simple pieces, where each piece is one column of numbers times one row. The rank is the number of pieces.

Two pieces are still mostly bands, now in two patterns.

> LoRA bets that the change fine-tuning makes to a weight matrix is like this photo: a few pieces carry almost all of it.



Not from the paper: the best rank-r approximation (SVD) of an Oxford Pets photo, Jia et al. 2022, Fig. 10. The low-rank bet follows Li et al. 2018 and Aghajanyan et al. 2021.

What a low-rank matrix is

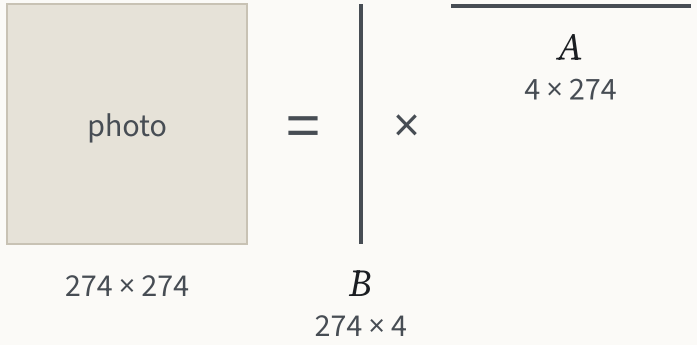
Rank 4: 4 × (274 + 274) = **2,192** numbers, 2.9% of the photo's 75,076

1	2	**4**	8	16	32	64	all 274

A grayscale photo is a matrix of 274 × 274 brightness values. Any matrix can be written as a sum of simple pieces, where each piece is one column of numbers times one row. The rank is the number of pieces.

With four pieces the main light and dark areas appear.

> LoRA bets that the change fine-tuning makes to a weight matrix is like this photo: a few pieces carry almost all of it.



Not from the paper: the best rank-r approximation (SVD) of an Oxford Pets photo, Jia et al. 2022, Fig. 10. The low-rank bet follows Li et al. 2018 and Aghajanyan et al. 2021.

What a low-rank matrix is

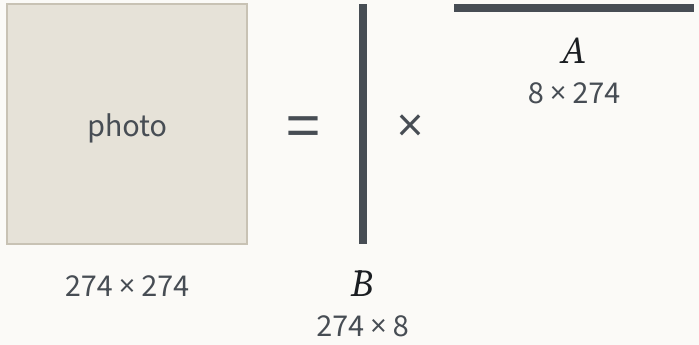
Rank 8: 8 × (274 + 274) = **4,384** numbers, 5.8% of the photo's 75,076

1	2	4	8	16	32	64	all 274

A grayscale photo is a matrix of 274 × 274 brightness values. Any matrix can be written as a sum of simple pieces, where each piece is one column of numbers times one row. The rank is the number of pieces.

With eight pieces, 6% of the numbers, the rough shape of the cat appears, because the first pieces carry most of the picture.

> LoRA bets that the change fine-tuning makes to a weight matrix is like this photo: a few pieces carry almost all of it.



Not from the paper: the best rank-r approximation (SVD) of an Oxford Pets photo, Jia et al. 2022, Fig. 10. The low-rank bet follows Li et al. 2018 and Aghajanyan et al. 2021.

What a low-rank matrix is

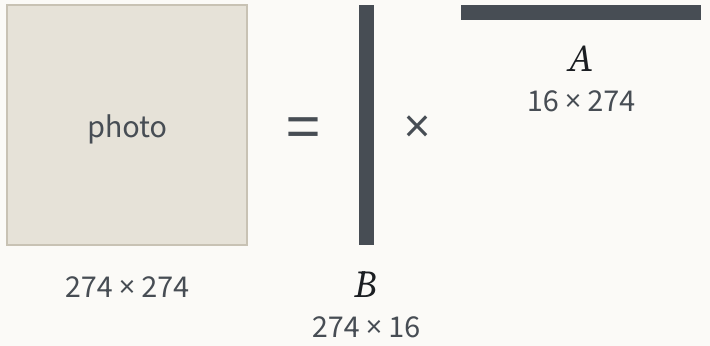
Rank 16: 16 × (274 + 274) = **8,768** numbers,
11.7% of the photo's 75,076

1	2	4	8	**16**	32	64	all 274

A grayscale photo is a matrix of 274 × 274 brightness values. Any matrix can be written as a sum of simple pieces, where each piece is one column of numbers times one row. The rank is the number of pieces.

With 16 pieces the cat is recognisable, from 12% of the numbers.

> LoRA bets that the change fine-tuning makes to a weight matrix is like this photo: a few pieces carry almost all of it.



Not from the paper: the best rank-r approximation (SVD) of an Oxford Pets photo, Jia et al. 2022, Fig. 10. The low-rank bet follows Li et al. 2018 and Aghajanyan et al. 2021.

What a low-rank matrix is

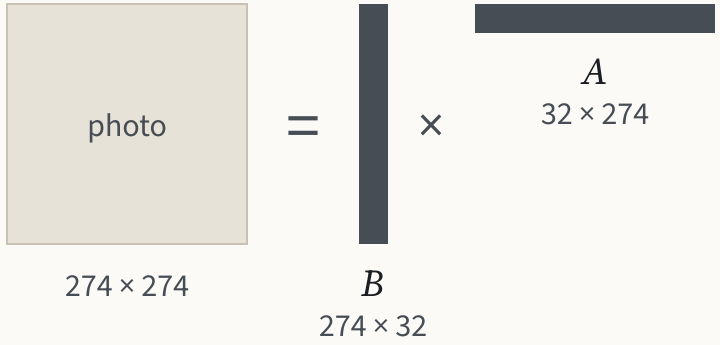
Rank 32: 32 × (274 + 274) = **17,536** numbers, 23.4% of the photo's 75,076

1	2	4	8	16	**32**	64	all 274

A grayscale photo is a matrix of 274 × 274 brightness values. Any matrix can be written as a sum of simple pieces, where each piece is one column of numbers times one row. The rank is the number of pieces.

With 32 pieces the face and the fur appear.

> LoRA bets that the change fine-tuning makes to a weight matrix is like this photo: a few pieces carry almost all of it.



Not from the paper: the best rank-r approximation (SVD) of an Oxford Pets photo, Jia et al. 2022, Fig. 10. The low-rank bet follows Li et al. 2018 and Aghajanyan et al. 2021.

What a low-rank matrix is

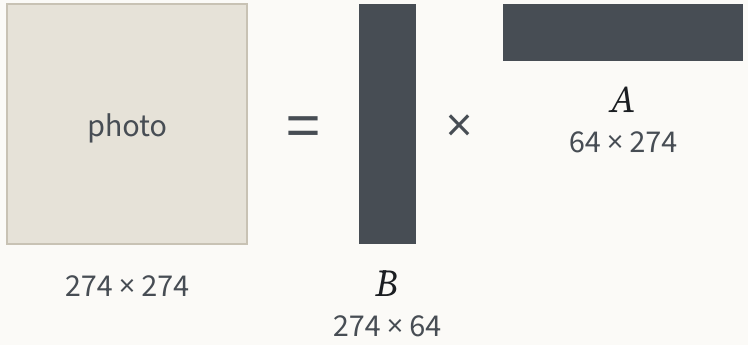
Rank 64: 64 × (274 + 274) = **35,072** numbers, 46.7% of the photo's 75,076

1	2	4	8	16	32	64	all 274

A grayscale photo is a matrix of 274 × 274 brightness values. Any matrix can be written as a sum of simple pieces, where each piece is one column of numbers times one row. The rank is the number of pieces.

With 64 pieces it is hard to tell from the original.

> LoRA bets that the change fine-tuning makes to a weight matrix is like this photo: a few pieces carry almost all of it.



Not from the paper: the best rank-r approximation (SVD) of an Oxford Pets photo, Jia et al. 2022, Fig. 10. The low-rank bet follows Li et al. 2018 and Aghajanyan et al. 2021.

What a low-rank matrix is

75,076 numbers: the full photo

1	2	4	8	16	32	64	all 274

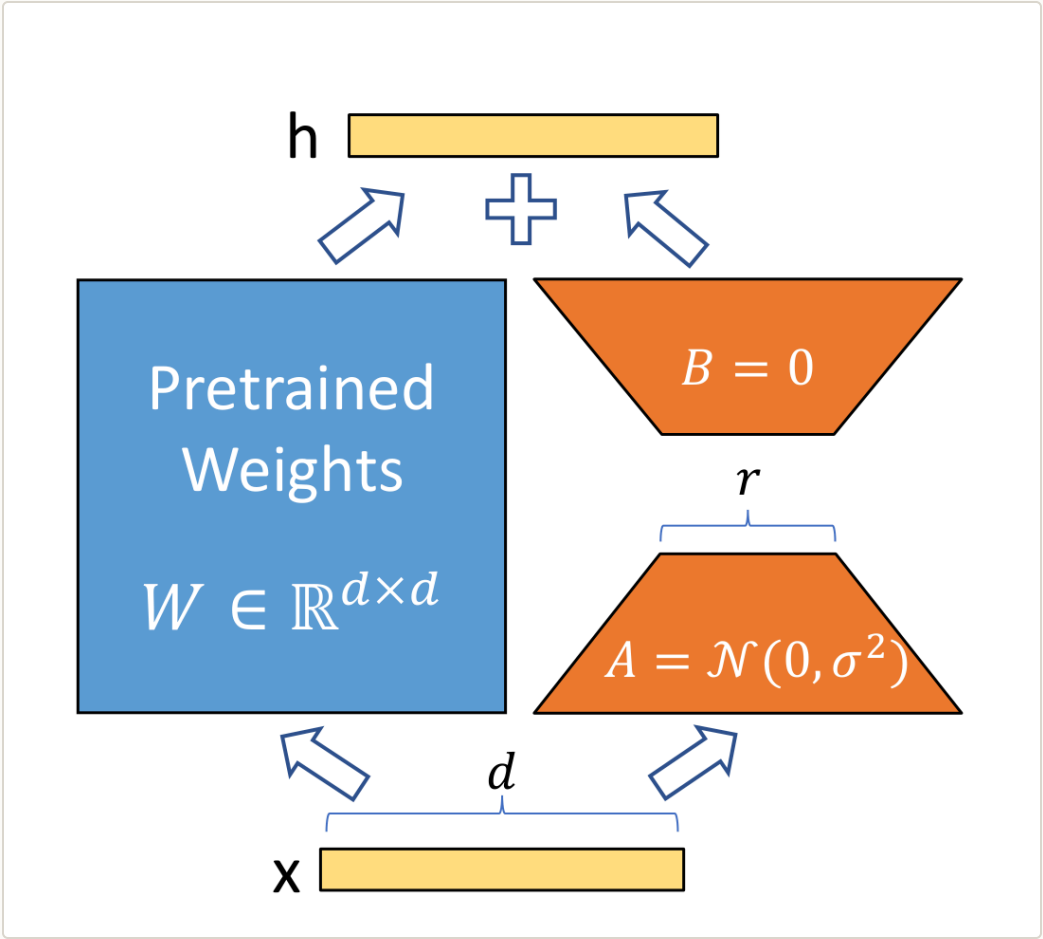
A grayscale photo is a matrix of 274 × 274 brightness values. Any matrix can be written as a sum of simple pieces, where each piece is one column of numbers times one row. The rank is the number of pieces.

All 274 pieces give back the photo exactly.

> LoRA bets that the change fine-tuning makes to a weight matrix is like this photo: a few pieces carry almost all of it.

Not from the paper: the best rank-r approximation (SVD) of an Oxford Pets photo, Jia et al. 2022, Fig. 10. The low-rank bet follows Li et al. 2018 and Aghajanyan et al. 2021.

LoRA adds a trainable bypass to a frozen weight



During training | After training

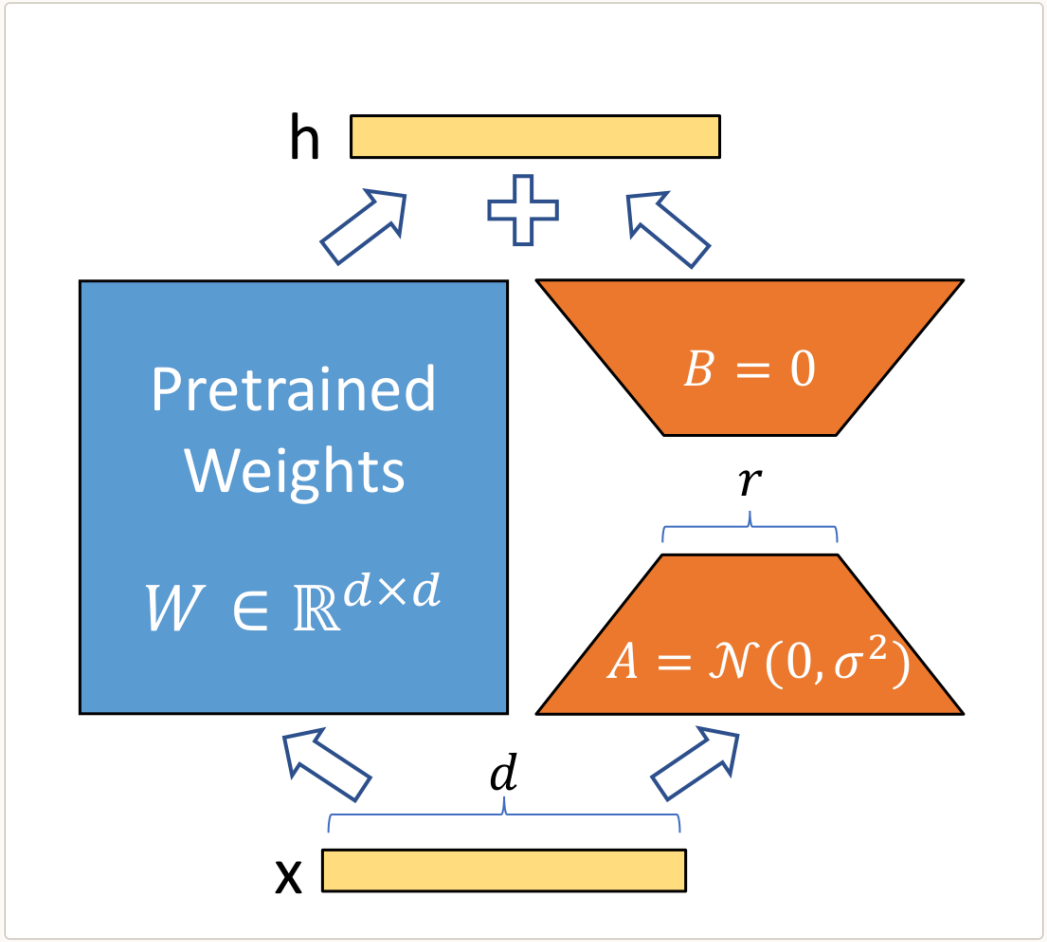
The input goes through the frozen weight W_0 and, in parallel, through A and B, which are trained. The two results are added. The full change ΔW is never built; LoRA trains the thin matrices directly, with a rank r chosen by hand.

$$h = W_0 x + (\alpha / r)\, BAx$$

B starts at zero, so training begins exactly at the pretrained model. Only A and B get gradients and optimizer state.

Figure: Hu et al. 2022, Fig. 1, where orange is trained. Equation (3) and Sec. 4.1. The paper sets α to the first r it tries and does not tune it.

LoRA adds a trainable bypass to a frozen weight



During training | **After training**

After training, the product BA is added into the weight once. The model is again one matrix per layer, of the same size, so it runs exactly as fast as before.

$$W = W_0 + (\alpha/r)\, BA$$

$$h = Wx$$

To switch task, subtract this BA and add that task's $B'A'$. Per task, only A and B are stored.

Figure: Hu et al. 2022, Fig. 1, where orange is trained. Equation (3) and Sec. 4.1. The paper sets α to the first r it tries and does not tune it.

LoRA on a 3 × 3 weight matrix

Frozen path | LoRA path | Merging | Merged layer

During training

$$\underbrace{\begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{bmatrix}}_{W} \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 5 \\ 7 \\ 8 \end{bmatrix}}_{Wx}$$

After training, once

At test time

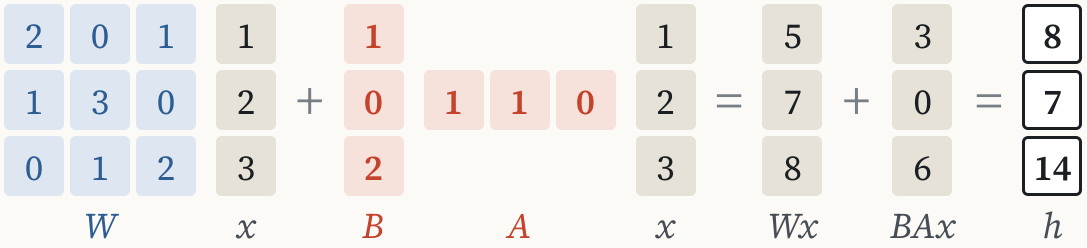
Take a tiny weight W and an input x. The frozen path is just the pretrained layer, and Wx gives 5, 7 and 8.

Made-up numbers with rank $r = 1$ and $\alpha/r = 1$, so every product can be checked by eye. In ViT-B the same happens with 768 × 768 weights and $r = 8$.

LoRA on a 3 × 3 weight matrix

Frozen path | LoRA path | Merging | Merged layer

During training



After training, once

At test time

LoRA adds a second path. A squeezes the input to r numbers, here a single 3, and B expands it back to three; the two paths are added. In training, only the six red numbers change.

Made-up numbers with rank $r = 1$ and $\alpha/r = 1$, so every product can be checked by eye. In ViT-B the same happens with 768 × 768 weights and $r = 8$.

LoRA on a 3 × 3 weight matrix

Frozen path | LoRA path | **Merging** | Merged layer

During training

$$\underset{W}{\begin{bmatrix}2&0&1\\1&3&0\\0&1&2\end{bmatrix}}\underset{x}{\begin{bmatrix}1\\2\\3\end{bmatrix}}+\underset{B}{\begin{bmatrix}1\\0\\2\end{bmatrix}}\underset{A}{\begin{bmatrix}1&1&0\end{bmatrix}}\underset{x}{\begin{bmatrix}1\\2\\3\end{bmatrix}}=\underset{Wx}{\begin{bmatrix}5\\7\\8\end{bmatrix}}+\underset{BAx}{\begin{bmatrix}3\\0\\6\end{bmatrix}}=\underset{h}{\boxed{\begin{bmatrix}8\\7\\14\end{bmatrix}}}$$

After training, once

$$\underset{B}{\begin{bmatrix}1\\0\\2\end{bmatrix}}\underset{A}{\begin{bmatrix}1&1&0\end{bmatrix}}=\underset{BA}{\begin{bmatrix}1&1&0\\0&0&0\\2&2&0\end{bmatrix}}\qquad\underset{W}{\begin{bmatrix}2&0&1\\1&3&0\\0&1&2\end{bmatrix}}+\underset{BA}{\begin{bmatrix}1&1&0\\0&0&0\\2&2&0\end{bmatrix}}=\underset{W'}{\begin{bmatrix}3&1&1\\1&3&0\\2&3&2\end{bmatrix}}$$

At test time

After training, B times A is worked out once. A column times a row is a full 3 × 3 matrix, the same shape as W, so it is simply added to it; the red cells are the ones that changed.

Made-up numbers with rank $r = 1$ and $\alpha/r = 1$, so every product can be checked by eye. In ViT-B the same happens with 768 × 768 weights and $r = 8$.

LoRA on a 3 × 3 weight matrix

Frozen path | LoRA path | Merging | **Merged layer**

During training

$$\underset{W}{\begin{bmatrix}2&0&1\\1&3&0\\0&1&2\end{bmatrix}}\underset{x}{\begin{bmatrix}1\\2\\3\end{bmatrix}}+\underset{B}{\begin{bmatrix}1\\0\\2\end{bmatrix}}\underset{A}{\begin{bmatrix}1&1&0\end{bmatrix}}\underset{x}{\begin{bmatrix}1\\2\\3\end{bmatrix}}=\underset{Wx}{\begin{bmatrix}5\\7\\8\end{bmatrix}}+\underset{BAx}{\begin{bmatrix}3\\0\\6\end{bmatrix}}=\underset{h}{\boxed{\begin{bmatrix}8\\7\\14\end{bmatrix}}}$$

After training, once

$$\underset{B}{\begin{bmatrix}1\\0\\2\end{bmatrix}}\underset{A}{\begin{bmatrix}1&1&0\end{bmatrix}}=\underset{BA}{\begin{bmatrix}1&1&0\\0&0&0\\2&2&0\end{bmatrix}}\qquad\underset{W}{\begin{bmatrix}2&0&1\\1&3&0\\0&1&2\end{bmatrix}}+\underset{BA}{\begin{bmatrix}1&1&0\\0&0&0\\2&2&0\end{bmatrix}}=\underset{W'}{\begin{bmatrix}3&1&1\\1&3&0\\2&3&2\end{bmatrix}}$$

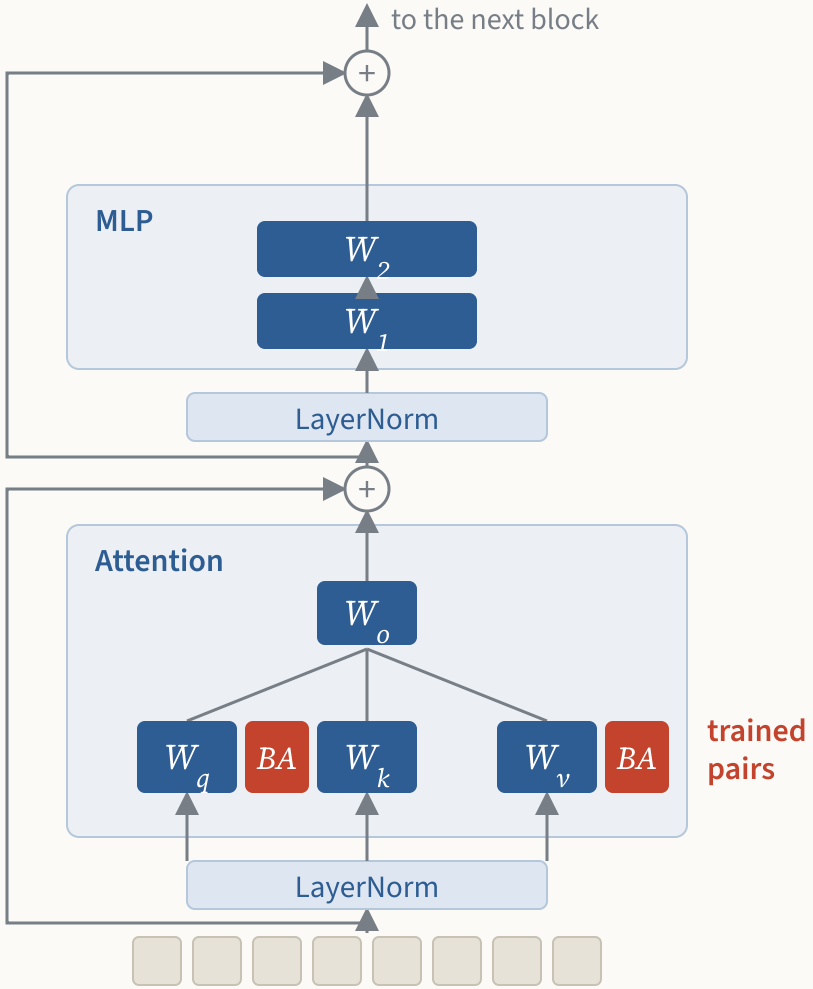
At test time

$$\underset{W'}{\begin{bmatrix}3&1&1\\1&3&0\\2&3&2\end{bmatrix}}\underset{x}{\begin{bmatrix}1\\2\\3\end{bmatrix}}=\underset{h}{\boxed{\begin{bmatrix}8\\7\\14\end{bmatrix}}}$$

The merged weight gives exactly the same output, because $Wx + B(Ax) = (W + BA)x$. B and A are deleted, and the layer is one ordinary matrix again.

Made-up numbers with rank $r = 1$ and $\alpha/r = 1$, so every product can be checked by eye. In ViT-B the same happens with 768 × 768 weights and $r = 8$.

Where LoRA goes in a Vision Transformer

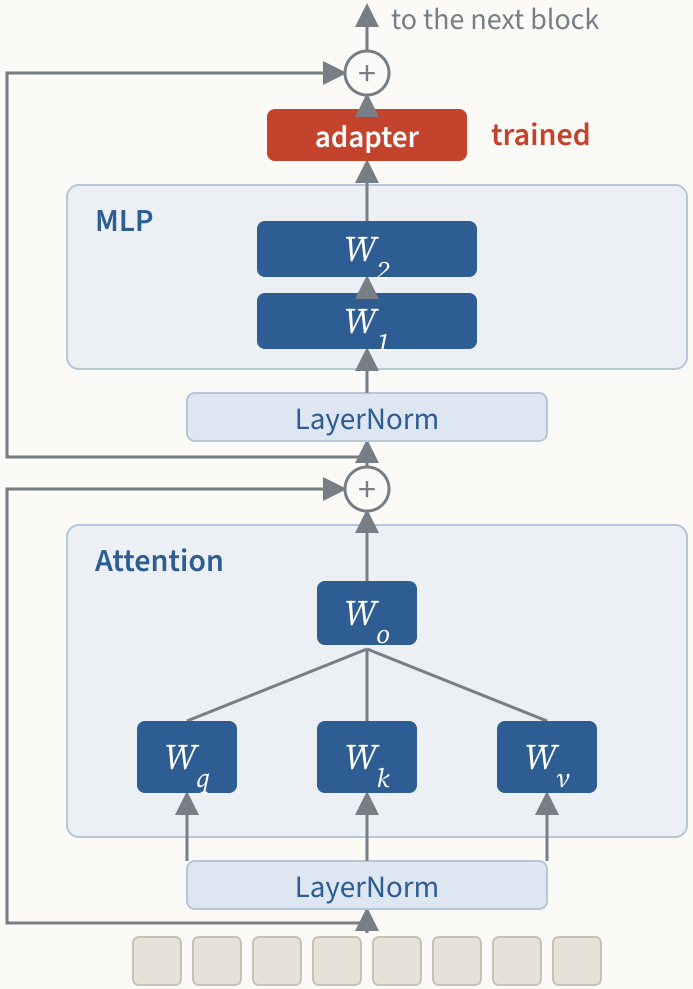


LoRA | Adapter

LoRA puts a trained pair $B\ A$ beside the query and value weights of the attention, in all 12 blocks. The key and output weights, the MLP and the layer norms stay frozen. After merging, the block is exactly the original.

One of the 12 blocks of ViT-B/16. Placement as in Hu et al. 2022, Sec. 4.2, and Houlsby et al. 2019. Zhang et al. 2022, Fig. 1a, draws LoRA, the adapter and VPT in one block.

Where LoRA goes in a Vision Transformer



LoRA | Adapter

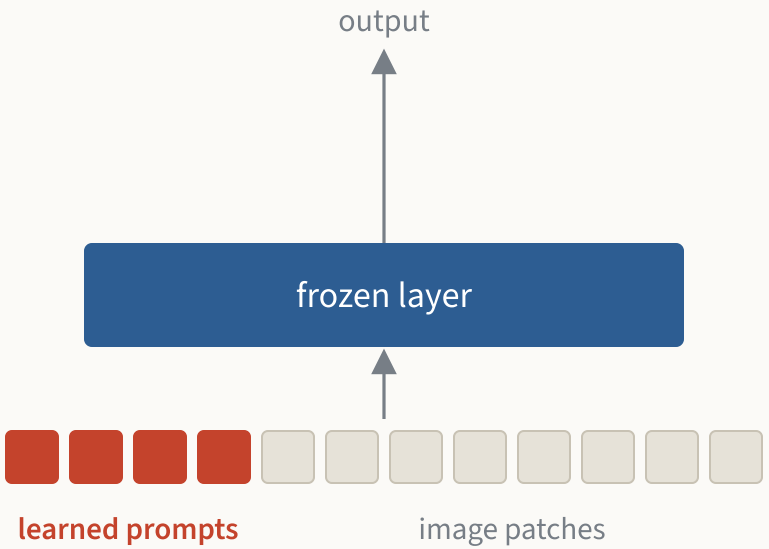
An adapter, for contrast, is a small trained two-layer network inserted after the MLP. It is an extra layer that every input must pass through, which is where its extra time comes from; LoRA sits beside weights that already exist.

One of the 12 blocks of ViT-B/16. Placement as in Hu et al. 2022, Sec. 4.2, and Houlsby et al. 2019. Zhang et al. 2022, Fig. 1a, draws LoRA, the adapter and VPT in one block.

VPT and LoRA side by side

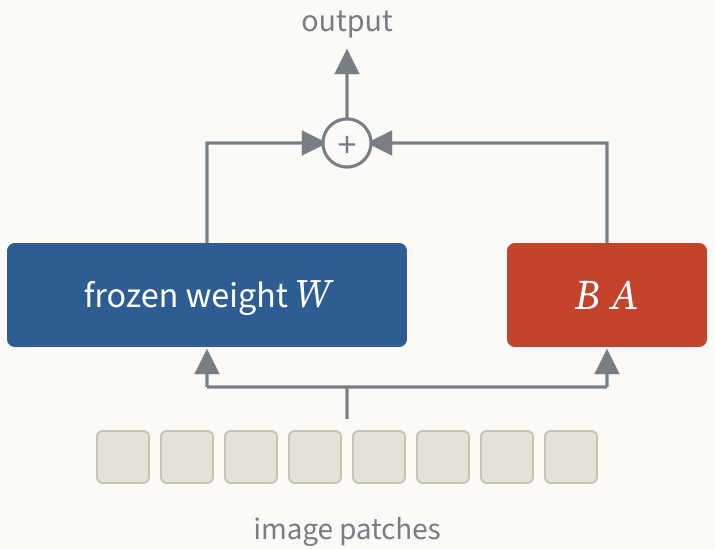
What is learned	During training	At test time

VPT



VPT learns extra input vectors. The same 50 per layer are placed in front of every image of the task; they start random and are not taken from any image.

LoRA



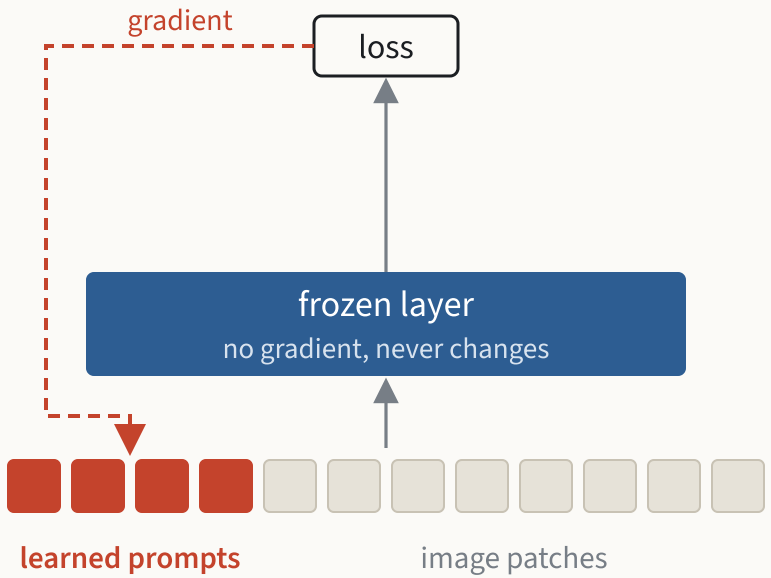
LoRA learns a correction to the weights, the pair B and A beside the query and value weights. It is also one pair for the whole task, and B starts at zero.

One layer of ViT-B/16 drawn for each. VPT-deep with 50 prompts per layer (Jia et al. 2022); LoRA with rank 8 on the query and value weights (Hu et al. 2022). Both train the same 102-class head.

VPT and LoRA side by side

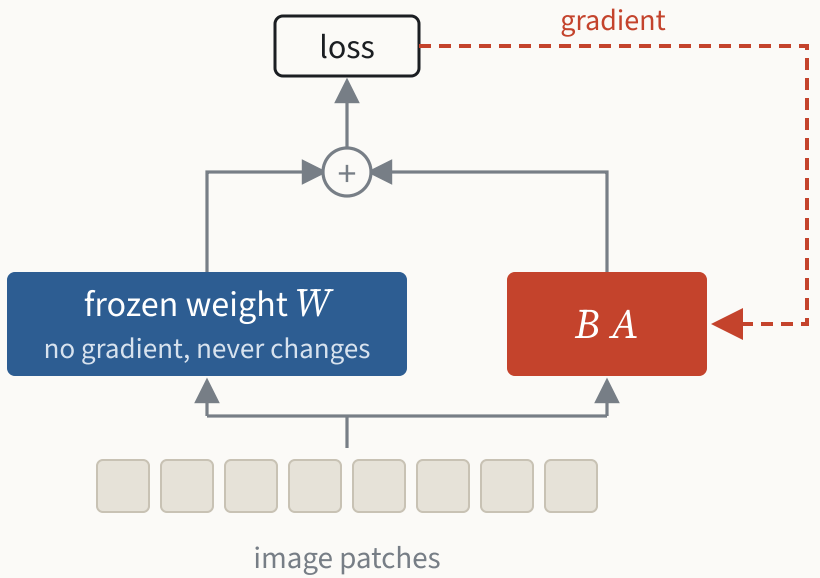
What is learned | During training | At test time

VPT



Each training image goes through the frozen model with the prompts attached. The loss sends gradients back, and only the prompts and the head change.

LoRA



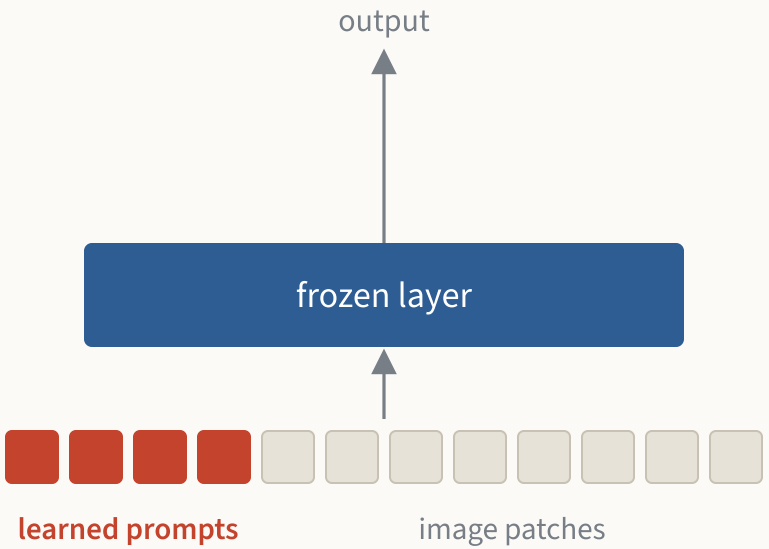
Each training image takes both paths, Wx and BAx. Only B, A and the head change; W never receives a gradient.

One layer of ViT-B/16 drawn for each. VPT-deep with 50 prompts per layer (Jia et al. 2022); LoRA with rank 8 on the query and value weights (Hu et al. 2022). Both train the same 102-class head.

VPT and LoRA side by side

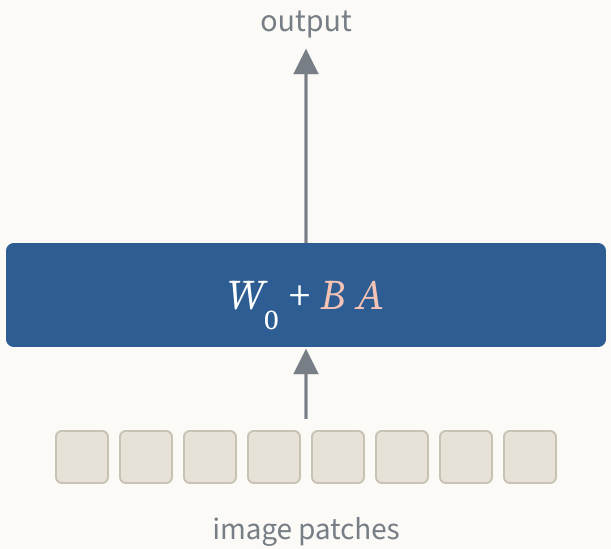
What is learned | During training | **At test time**

VPT



The prompts must be attached to every test image, so each layer works on 247 tokens instead of 197. They are inputs, so they can never be folded into the weights.

LoRA



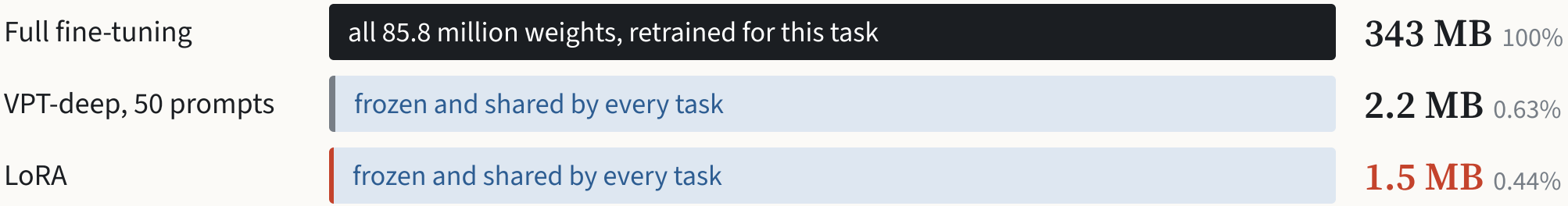
BA is added into W once, and the layer is one plain matrix again. The test image goes through exactly the original model.

One layer of ViT-B/16 drawn for each. VPT-deep with 50 prompts per layer (Jia et al. 2022); LoRA with rank 8 on the query and value weights (Hu et al. 2022). Both train the same 102-class head.

How much LoRA trains and stores per task

$2 \cdot N \cdot m \cdot d \cdot r = 2 \times 12 \times 2 \times 768 \times 8 =$ **294,912** numbers in the pairs, plus 78,438 for a 102-class head

Numbers trained and stored per task



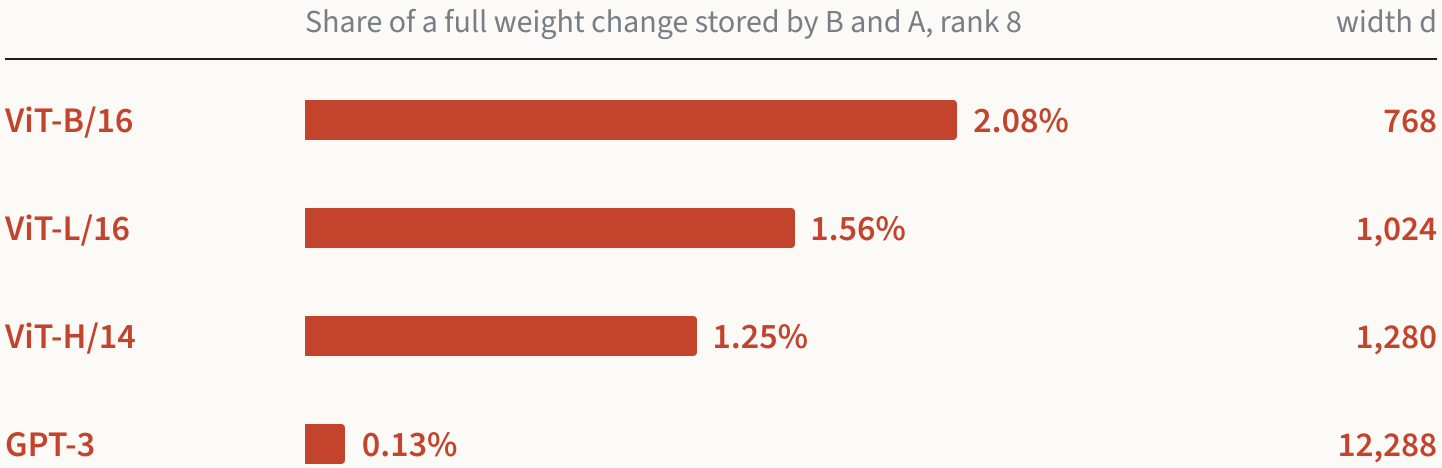
Compute per image at test time

Shown for rank 8 on the query and value weights. At rank 64 on all four attention weights the file grows to 19 MB, 5.6% of the model, and the compute per image does not change.

ViT-B/16: $N = 12$ blocks, $d = 768$, m adapted weights per block, 32-bit numbers. VPT-deep as before. Compute counts multiply-adds per layer for n tokens ($12nd^2 + 2n^2d$).

LoRA's saving grows with the width of the model

Full change ΔW	$d \times d = d^2$	589,824 numbers on ViT-B
B and A together	$d \times r + r \times d = 2rd$	12,288 numbers on ViT-B



A full change to one $d \times d$ weight has d^2 numbers, while B and A together have $2rd$. Since d is the square root of d^2, LoRA's cost grows like the square root of the full size, times the small constant $2r$.

So the wider the model, the smaller LoRA's share. Together with adapting only the query and value weights, this is why GPT-3's task file is 10,000 times smaller than a copy.

One $d \times d$ attention weight, rank $r = 8$ for every model. Widths: ViT-B/16 768, ViT-L/16 1,024, ViT-H/14 1,280 (Dosovitskiy et al. 2021); GPT-3 12,288 (Hu et al. 2022, Sec. 1). 10,000 times: Hu et al. 2022, Sec. 4.2.

LoRA on a ViT in a few lines

```
vit = timm.create_model(BACKBONE, pretrained=True, num_classes=C)
vit.requires_grad_(False); vit.head.requires_grad_(True)

class LoRAqv(nn.Module):                # wraps qkv = Linear(768, 2304)
    def __init__(self, qkv, r=8, alpha=8):
        super().__init__()
        d = qkv.in_features
        self.qkv, self.s = qkv, alpha / r    # frozen W0, scale α/r
        self.Aq = nn.Parameter(torch.randn(r, d) / d ** 0.5)
        self.Av = nn.Parameter(torch.randn(r, d) / d ** 0.5)
        self.Bq = nn.Parameter(torch.zeros(d, r))   # B = 0
        self.Bv = nn.Parameter(torch.zeros(d, r))
    def forward(self, x):
        q, k, v = self.qkv(x).chunk(3, dim=-1)      # W0 x
        q = q + self.s * x @ self.Aq.T @ self.Bq.T  # + B A x
        v = v + self.s * x @ self.Av.T @ self.Bv.T
        return torch.cat([q, k, v], dim=-1)

for blk in vit.blocks: blk.attn.qkv = LoRAqv(blk.attn.qkv)
```

B starts at zero and A at small random values, so the first forward pass is exactly the pretrained model.

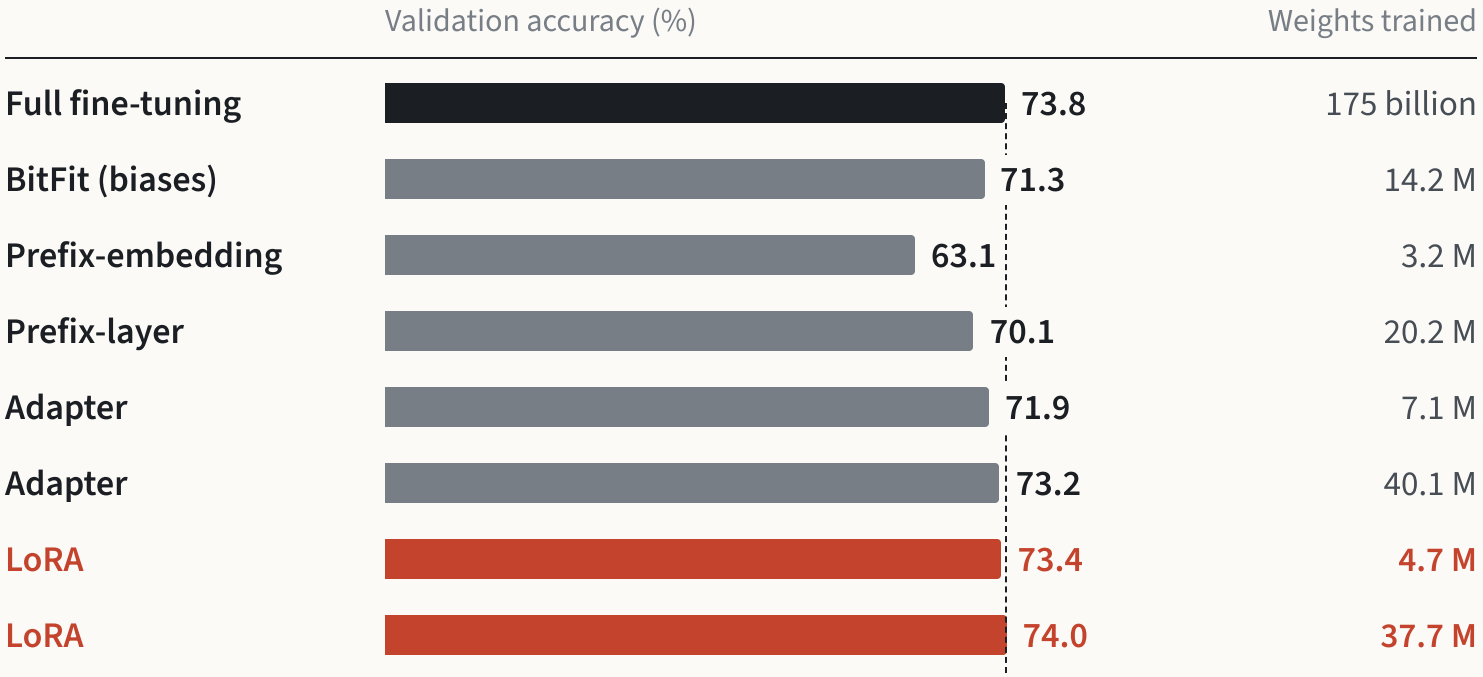
The two new choices are the rank r and which weights get a pair. On ViT-B, Zhang et al. use rank 8 with AdamW at a learning rate of 0.001 for 100 epochs.

After training, each pair is added into its qkv weight once, and the model is a plain timm ViT again.

My sketch for timm's ViT-B/16: 294,912 trained numbers in the pairs, and identical outputs before and after merging. Official code: loralib, github.com/microsoft/LoRA. Settings: Hu et al. 2022, Sec. 4.1; Zhang et al. 2022, App. A.2.

Results in the paper: language models

GPT-3, WikiSQL | GPT-3, MNLI | RoBERTa-base, GLUE



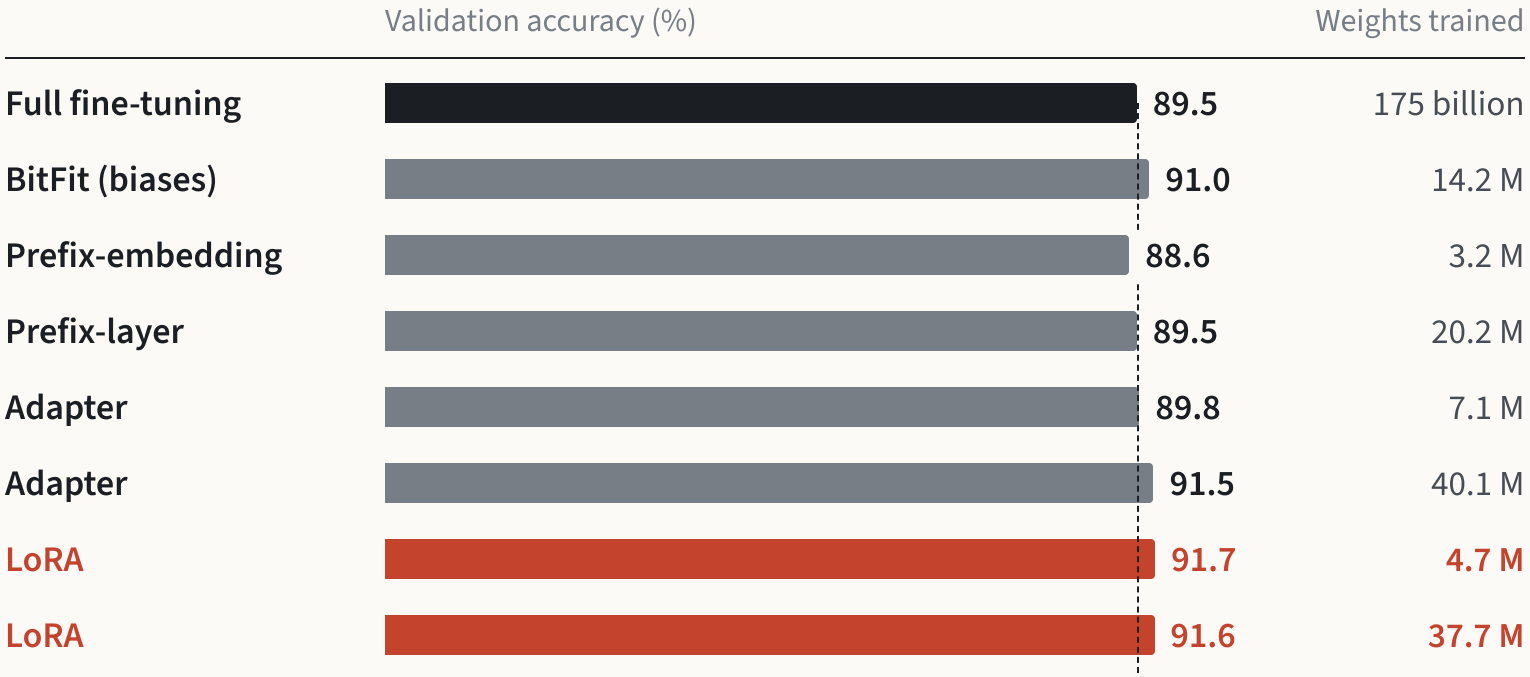
WikiSQL turns a question into an SQL query. On GPT-3, LoRA with 4.7 million trained weights, 0.003% of the model, is within the run-to-run noise of full fine-tuning, and slightly above it with 37.7 million.

The prefix methods are far behind, and the adapter needs more weights to come close.

Hu et al. 2022, Table 4: GPT-3 175B, validation accuracy, noise about ±0.5. Adapter: two per block (Houlsby et al.). BitFit trains only the biases.

Results in the paper: language models

GPT-3, WikiSQL | **GPT-3, MNLI** | RoBERTa-base, GLUE

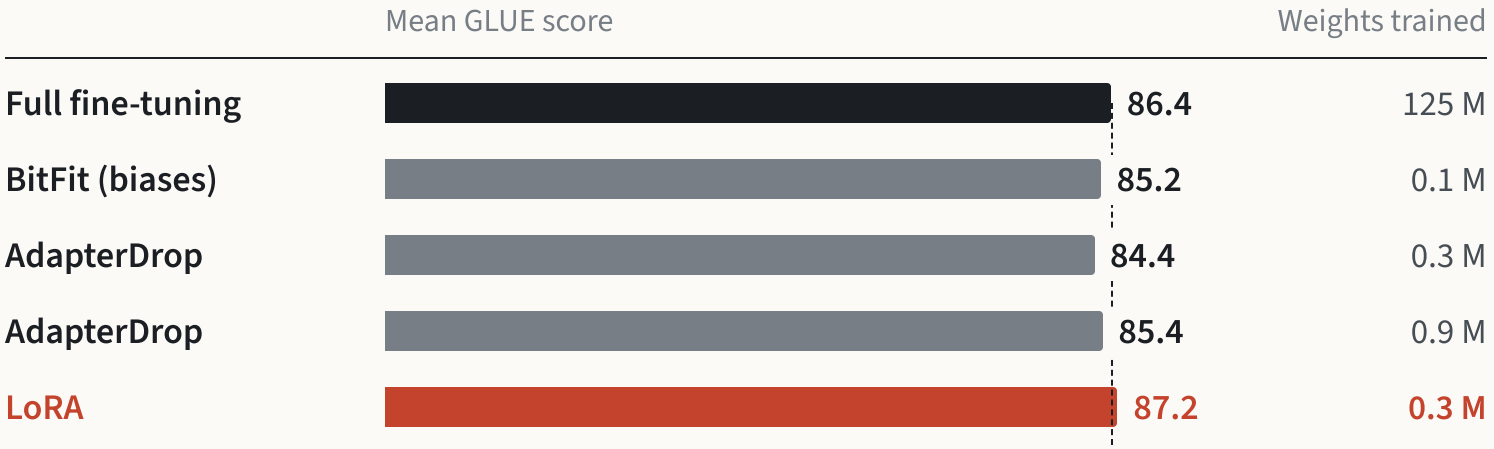


MNLI asks whether one sentence follows from another. Here LoRA is two points above full fine-tuning, 91.7 against 89.5, and above every other cheap method.

Hu et al. 2022, Table 4: GPT-3 175B, validation accuracy on MNLI-matched, noise about ±0.1. Adapter: two per block (Houlsby et al.). BitFit trains only the biases.

Results in the paper: language models

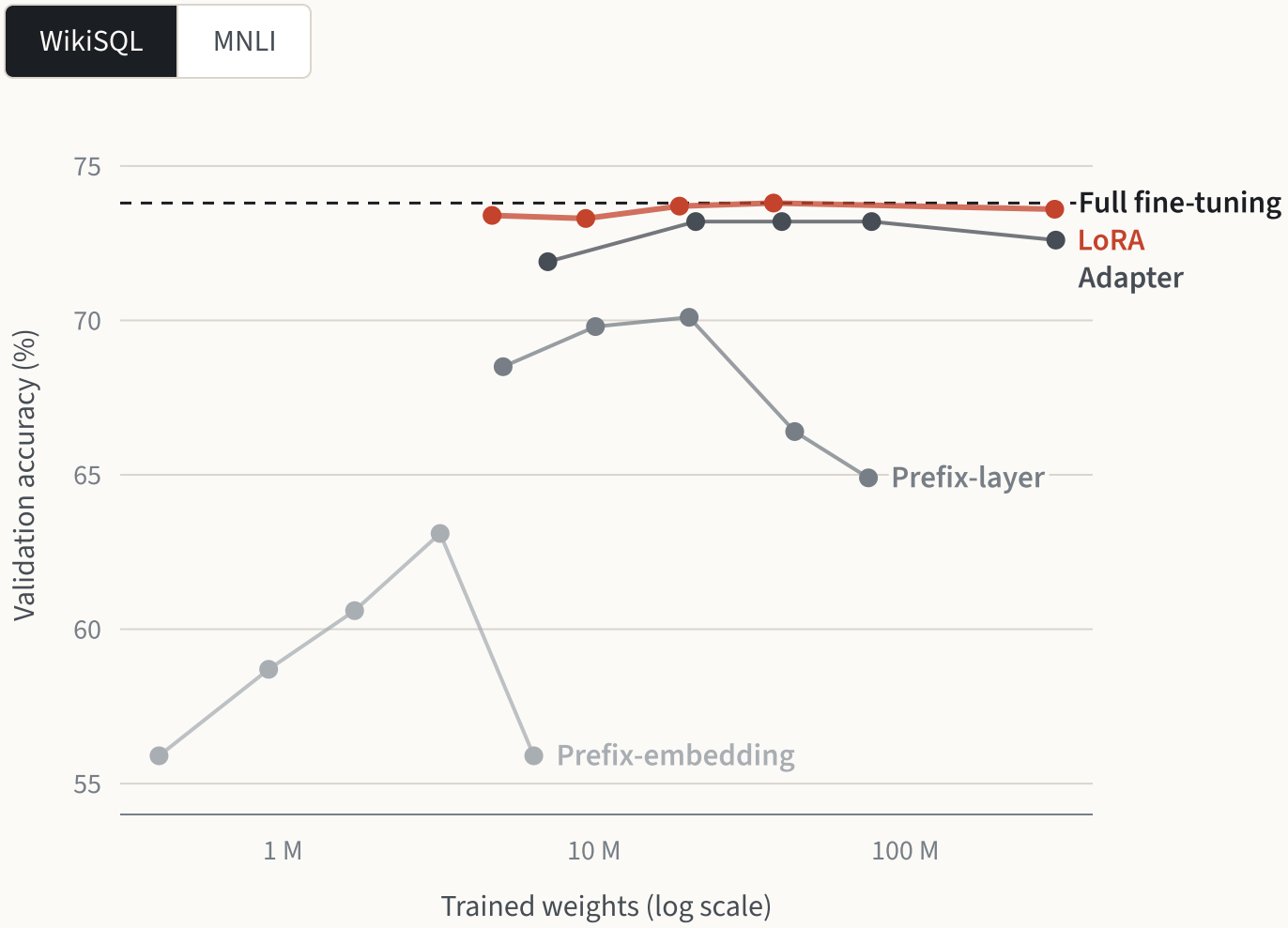
GPT-3, WikiSQL | GPT-3, MNLI | **RoBERTa-base, GLUE**



RoBERTa-base has 125 million weights, close to ViT-B. Averaged over the eight GLUE language tasks, LoRA with 0.3 million trained weights scores 87.2, against 86.4 for full fine-tuning.

Hu et al. 2022, Table 2: RoBERTa-base, mean over the eight GLUE tasks. Full fine-tuning, BitFit and AdapterDrop (Rücklé et al.) as published in earlier work.

What happens when more weights are trained

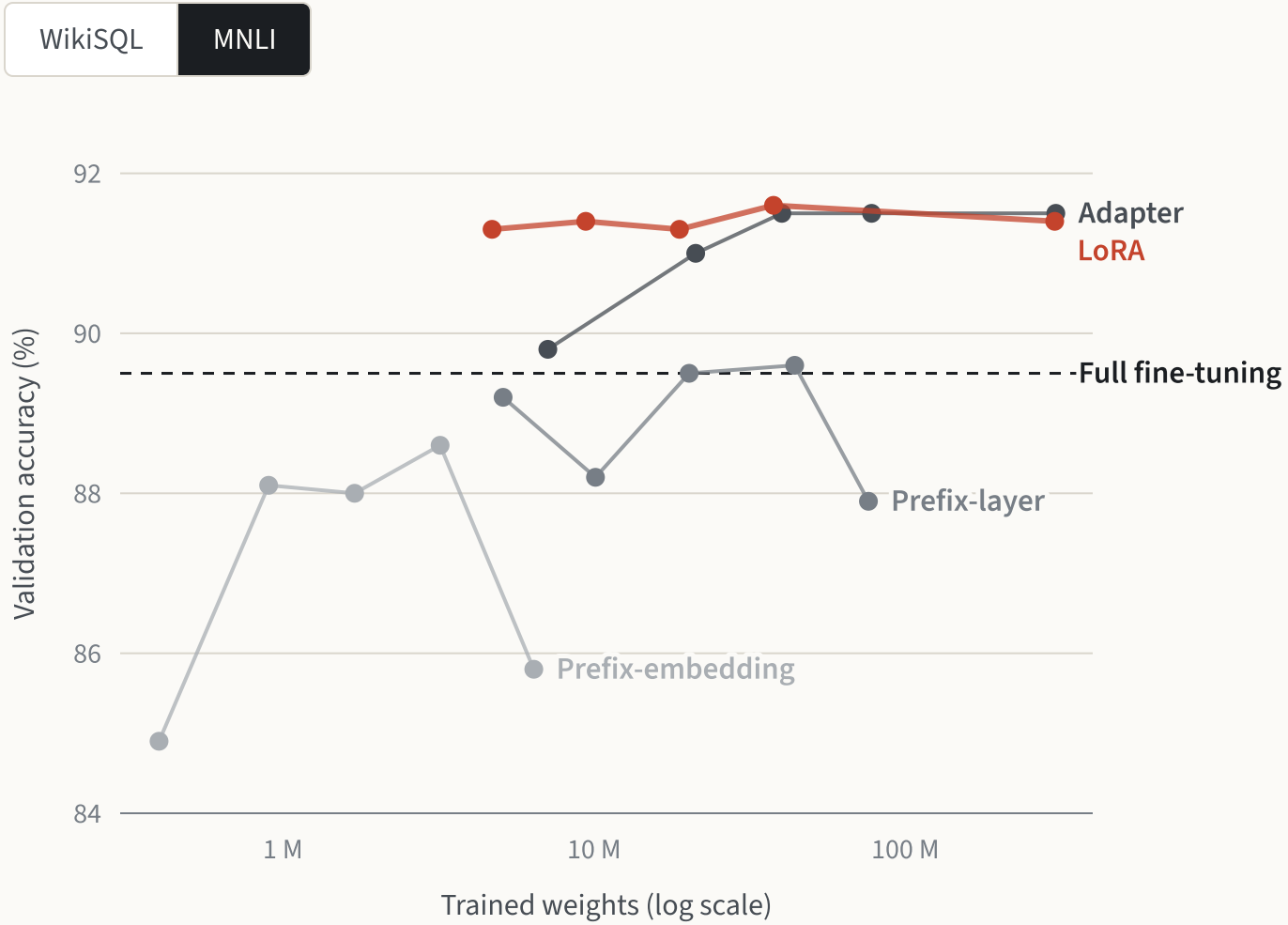


Each point is one size of a method. Prefix-embedding tuning improves up to 256 prompt tokens, then drops from 63.1 to 55.9. Prefix-layer tuning also falls after 20 million weights.

LoRA is flat from rank 1 to rank 64, at the level of full fine-tuning.

Hu et al. 2022, Table 15 (the points of Fig. 2): GPT-3 175B, validation accuracy; LoRA with equal rank on query and value, from rank 1 to 64. Full fine-tuning trains all 175 billion weights.

What happens when more weights are trained

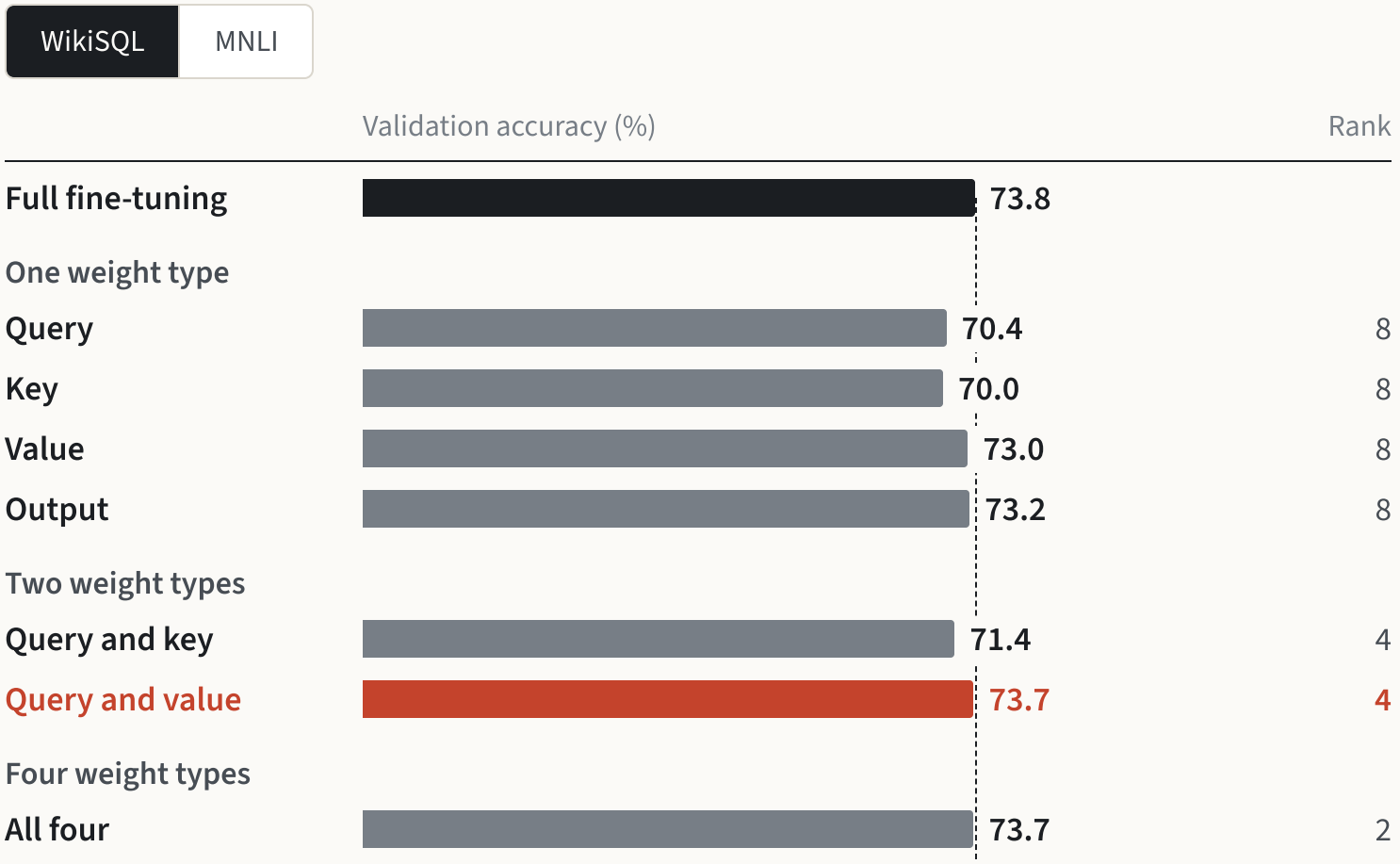


On MNLI the prompt methods again rise and fall, and the adapter needs about 40 million weights to catch up.

LoRA is above full fine-tuning at every size, so its rank is an easy choice.

Hu et al. 2022, Table 15 (the points of Fig. 2): GPT-3 175B, validation accuracy; LoRA with equal rank on query and value, from rank 1 to 64. Full fine-tuning trains all 175 billion weights.

Which weight matrices to adapt

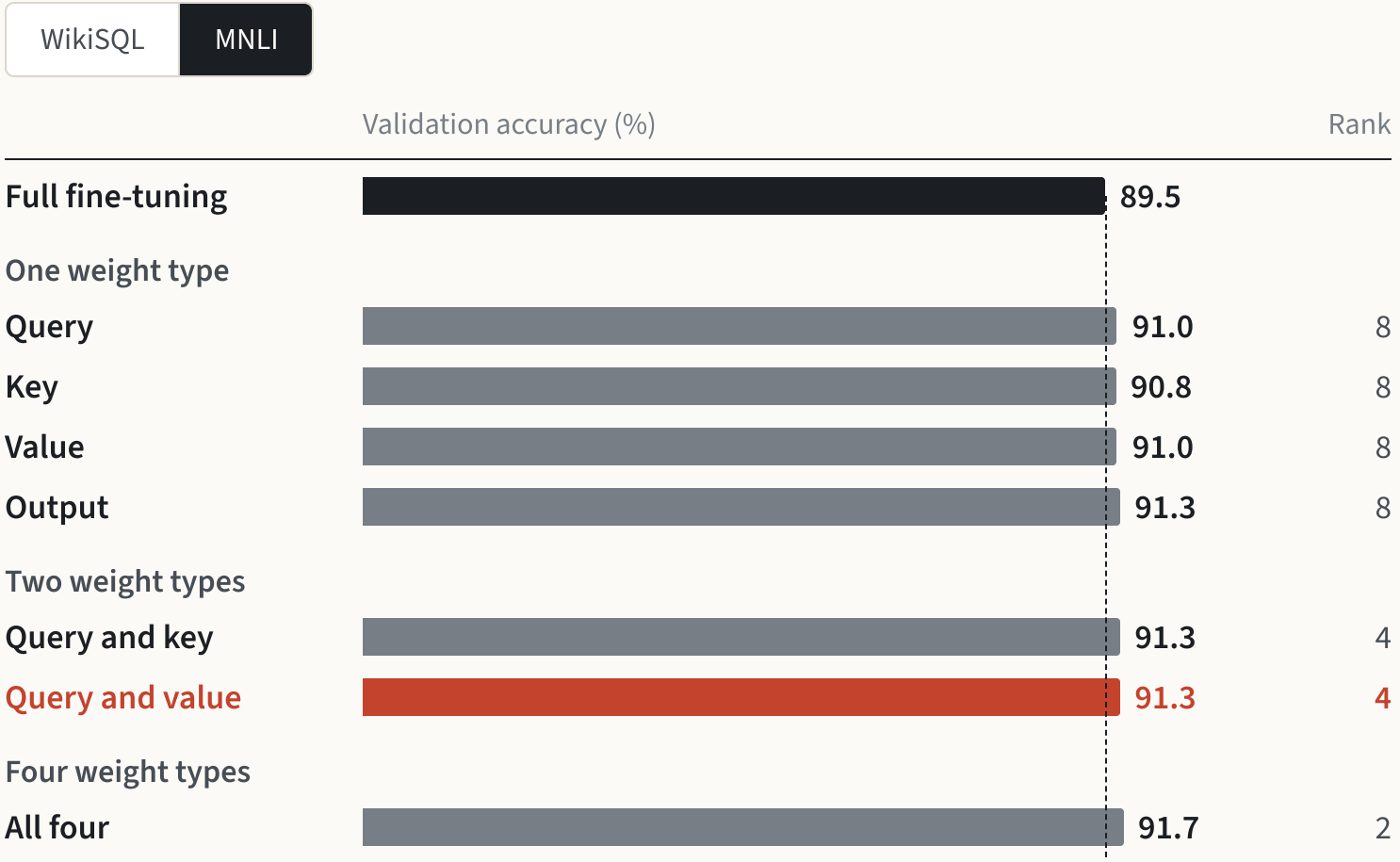


Every LoRA row trains the same 18 million weights. Putting them all into the query or the key weight is about three points worse than spreading them over query and value at rank 4.

Adapting all four at rank 2 does as well, so it pays to adapt more matrices with a smaller rank.

Hu et al. 2022, Table 5: GPT-3 175B with the same 18 million trained weights in every LoRA row, validation accuracy (noise ±0.5 on WikiSQL, ±0.1 on MNLI). Full fine-tuning: Table 4.

Which weight matrices to adapt

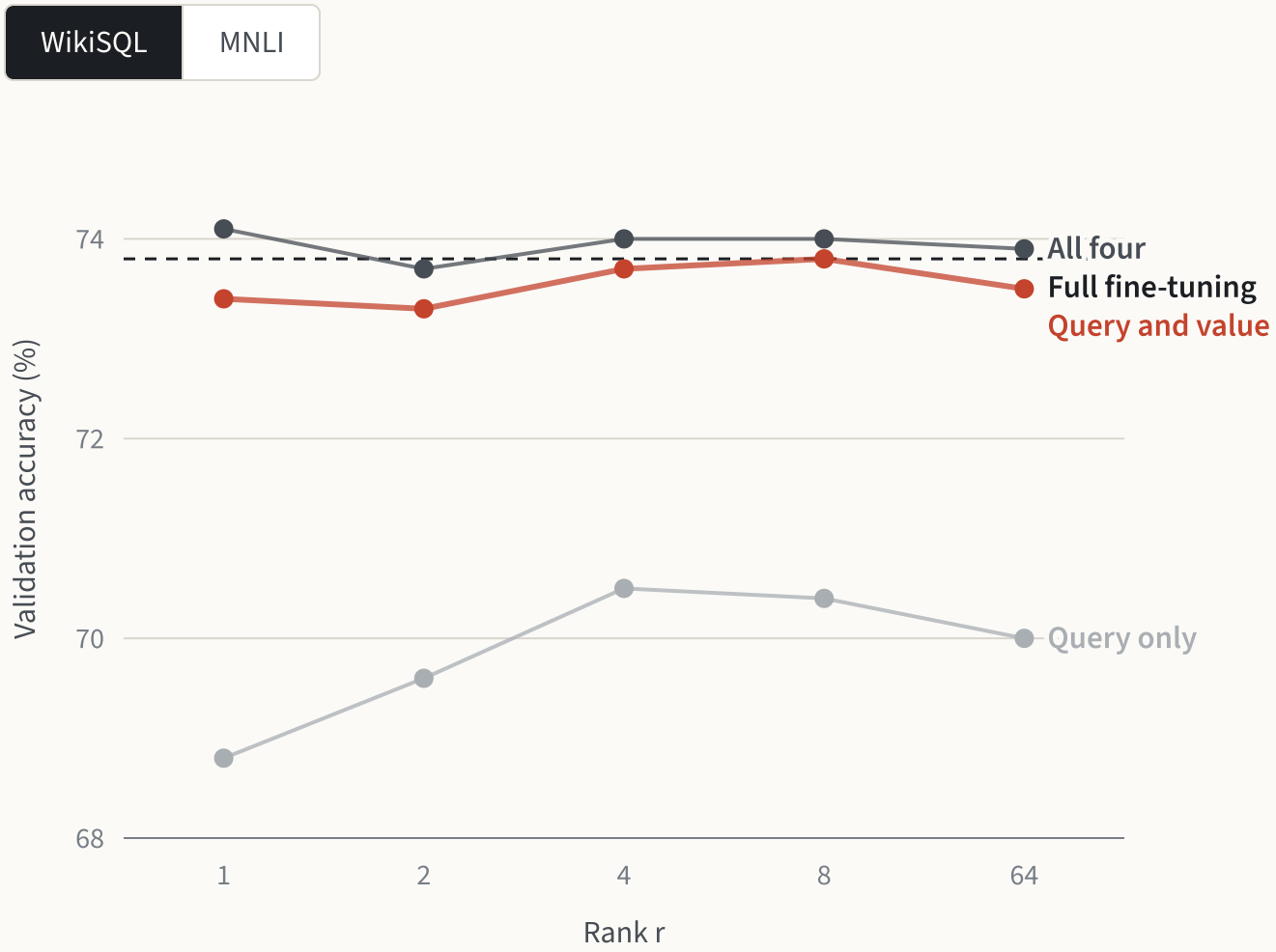


On MNLI the differences are smaller, but the order is similar: the query or key alone is lowest, and all four at rank 2 is highest, 91.7.

Every LoRA setting is above full fine-tuning here.

Hu et al. 2022, Table 5: GPT-3 175B with the same 18 million trained weights in every LoRA row, validation accuracy (noise ±0.5 on WikiSQL, ±0.1 on MNLI). Full fine-tuning: Table 4.

How small the rank can be

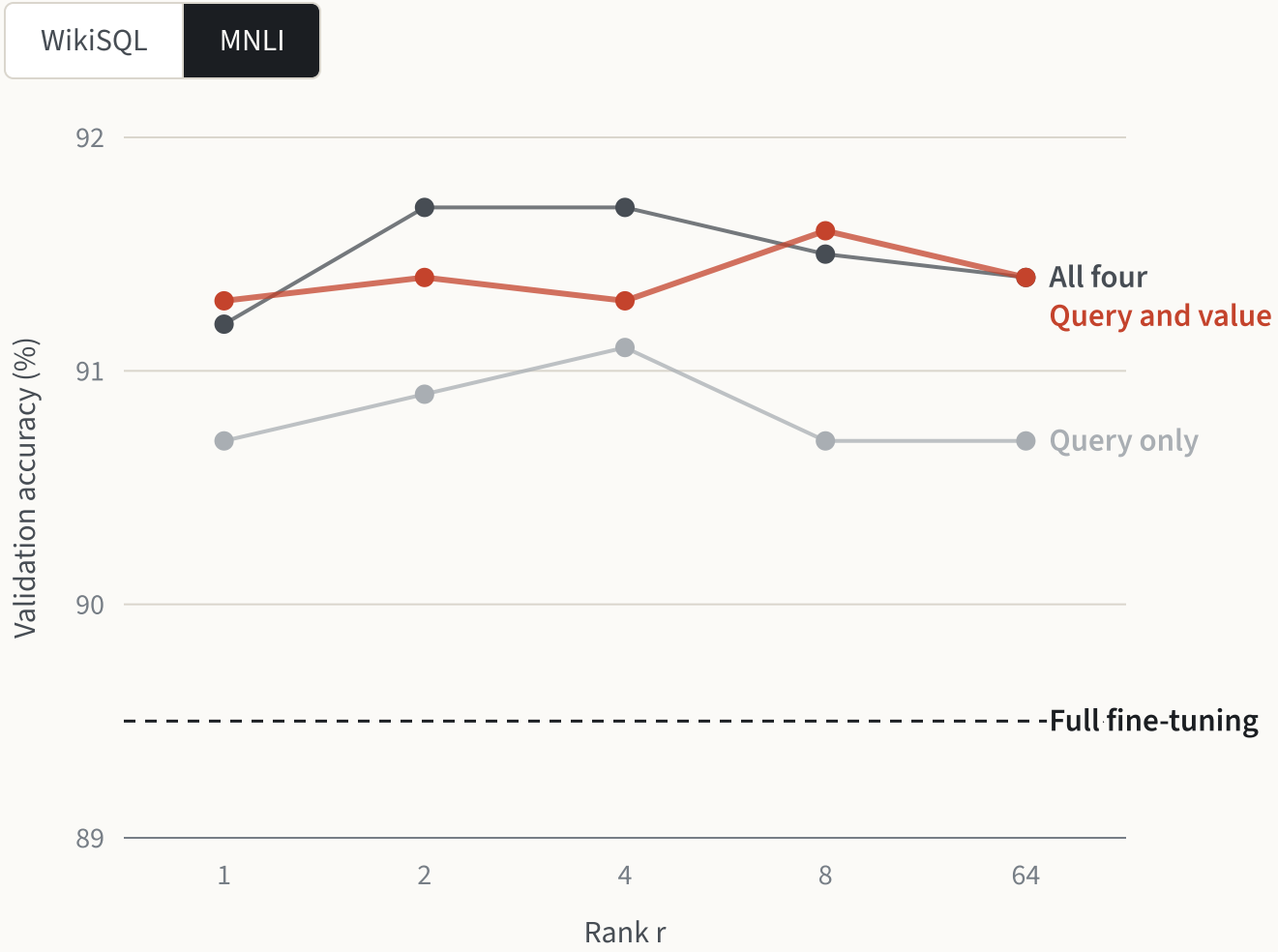


With the query and value weights together, rank 1 already gives 73.4, and no rank does better than 73.8, inside the noise of ±0.5.

The query weight alone needs rank 4 to reach its best, and stays three points lower.

Hu et al. 2022, Table 6: GPT-3 175B, validation accuracy at five ranks; noise ±0.5 on WikiSQL and ±0.1 on MNLI. On GPT-2 the best rank was between 4 and 16 (App. H.2).

How small the rank can be



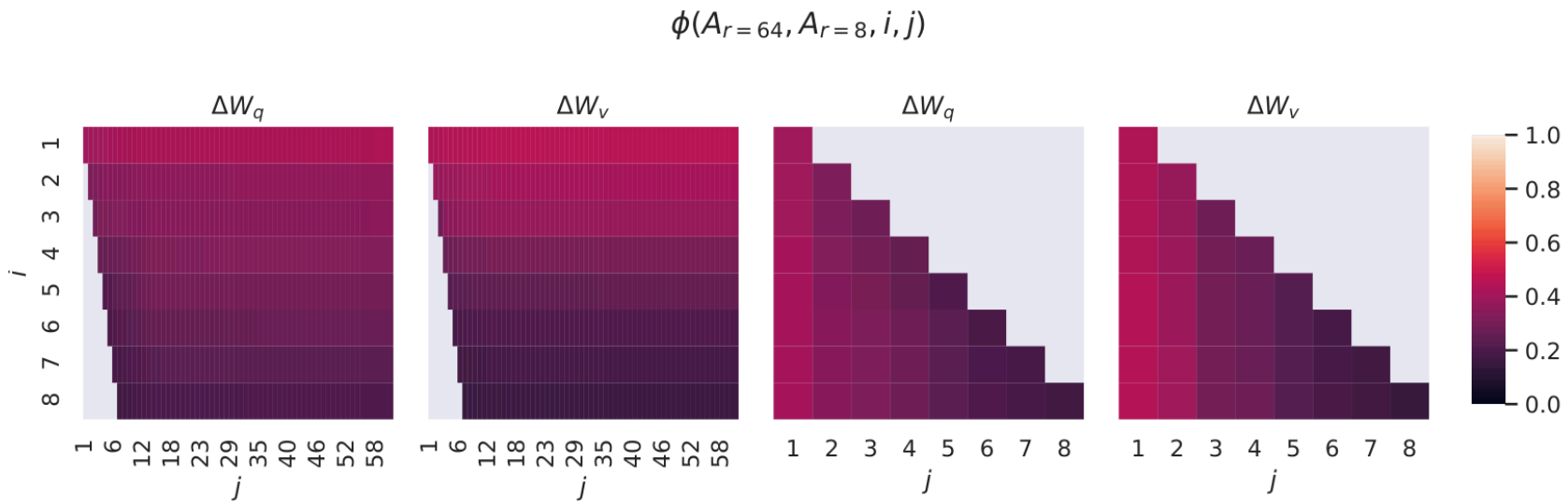
On MNLI, too, rank 1 for query and value is within 0.3 points of the best rank.

So the change fine-tuning needs has very few real directions, at least on GPT-3.

Hu et al. 2022, Table 6: GPT-3 175B, validation accuracy at five ranks; noise ±0.5 on WikiSQL and ±0.1 on MNLI. On GPT-2 the best rank was between 4 and 16 (App. H.2).

Why a small rank is enough

Rank 8 against rank 64 | The change against the weight



The authors trained rank 8 and rank 64 separately and compared the directions each learned. Bright means the two runs found the same direction. Only the first direction is clearly shared, with similarity above 0.5, and the rest overlap little, so most of the extra rank holds noise.

Hu et al. 2022, Fig. 3 (layer 48 of 96, GPT-3; values are normalized subspace similarity, 0 to 1) and Table 7 (rank 4, Frobenius norms).

Why a small rank is enough

Rank 8 against rank 64 | **The change against the weight**

21.67 is how strongly the pretrained weight W_q acts along its own top 4 directions.

0.32 is how strongly it acts along the 4 directions of the learned change ΔW_q: weak, but more than the 0.02 of random directions.

6.91 is the strength of the change along those directions, about 21 times what the pretrained weight had there.

> The change does not repeat what the model already does. It strongly amplifies directions the model learned in pretraining but used weakly.

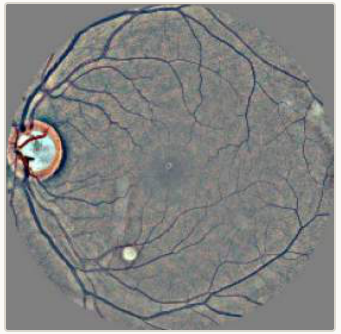
Hu et al. 2022, Fig. 3 (layer 48 of 96, GPT-3; values are normalized subspace similarity, 0 to 1) and Table 7 (rank 4, Frobenius norms).

LoRA on images

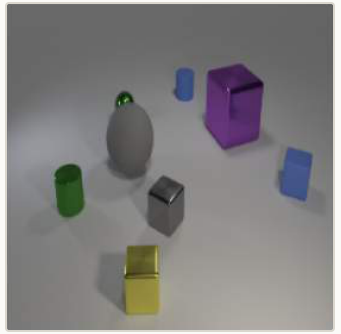
The paper itself has no image experiments. Vision groups tested it next, first on the same 19 VTAB-1k tasks as VPT, then on medical images.

Both use a ViT-B/16, so the numbers sit next to the VPT results.

Natural



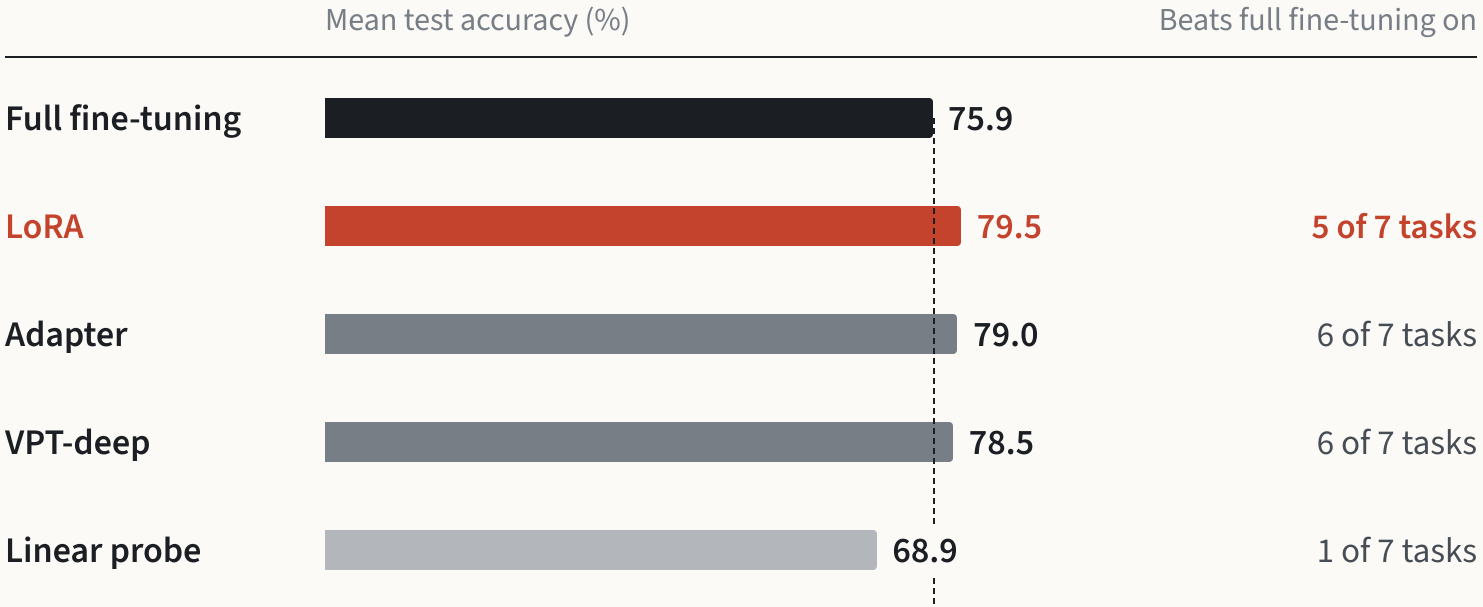
Specialized



Structured

LoRA, VPT and adapters on the same ViT

Natural | Specialized | Structured | All 19 tasks

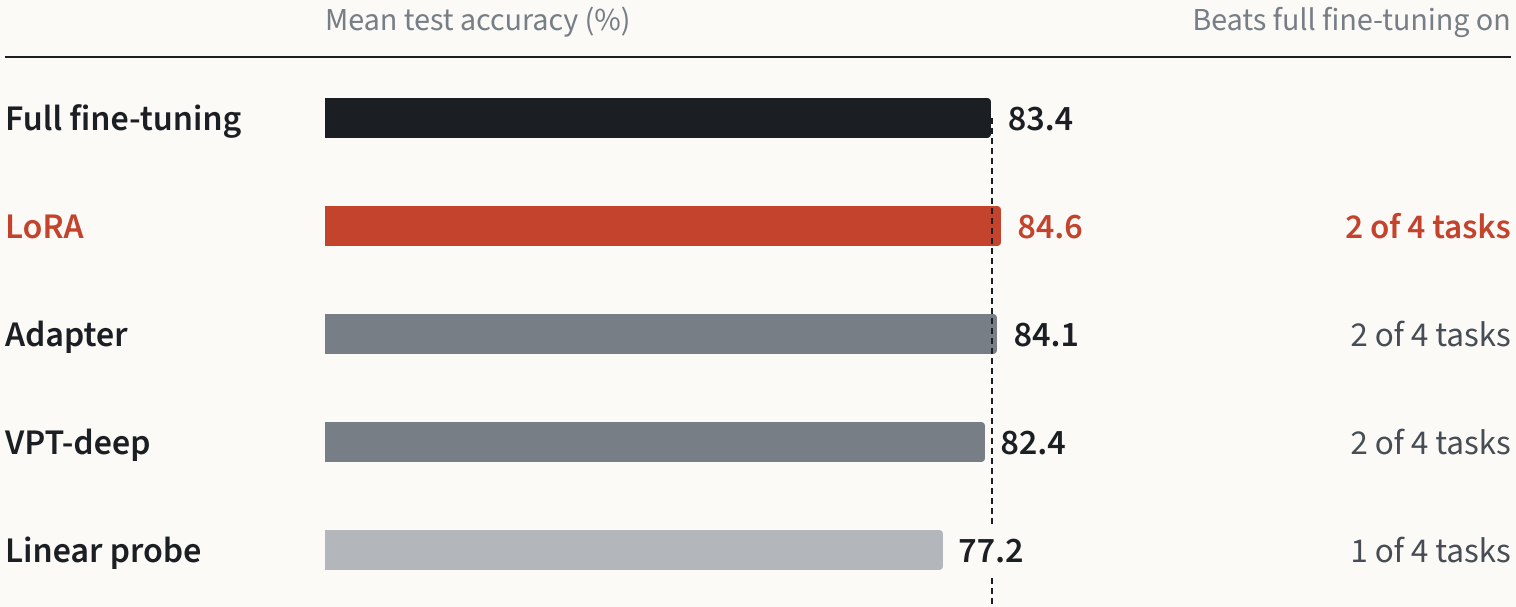


Ordinary photos, with 1,000 training images per task. LoRA, the adapter and VPT-deep all beat full fine-tuning by two to four points, and LoRA is narrowly the best.

Zhang et al. 2022, Table 1: ViT-B/16 pretrained on ImageNet-21k, VTAB-1k test accuracy. LoRA (rank 8) and the adapter were trained by Zhang et al.; the other rows are Jia et al. 2022, Table 1. Group means computed from the per-task values; "all 19" is the mean of the three groups.

LoRA, VPT and adapters on the same ViT

Natural | **Specialized** | Structured | All 19 tasks

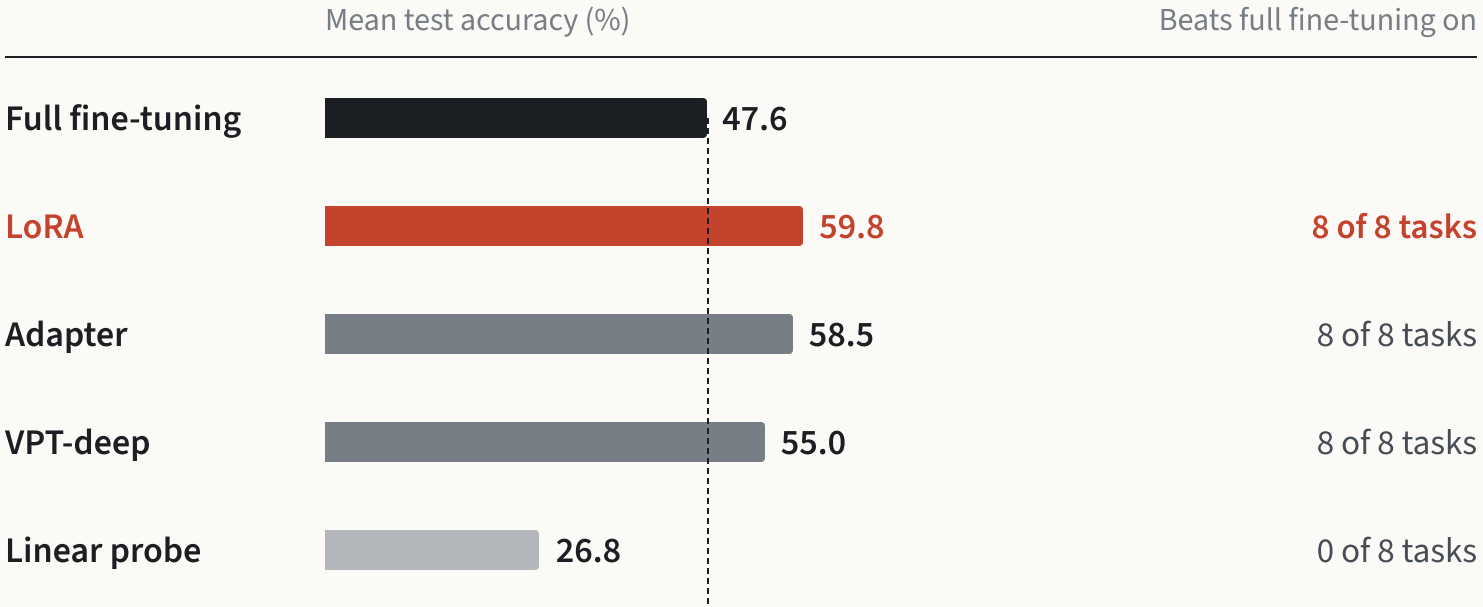


Satellite, microscope and retina images. LoRA and the adapter are slightly above full fine-tuning, 84.6 and 84.1 against 83.4, while VPT-deep is a point below.

Zhang et al. 2022, Table 1: ViT-B/16 pretrained on ImageNet-21k, VTAB-1k test accuracy. LoRA (rank 8) and the adapter were trained by Zhang et al.; the other rows are Jia et al. 2022, Table 1. Group means computed from the per-task values; "all 19" is the mean of the three groups.

LoRA, VPT and adapters on the same ViT

Natural | Specialized | **Structured** | All 19 tasks

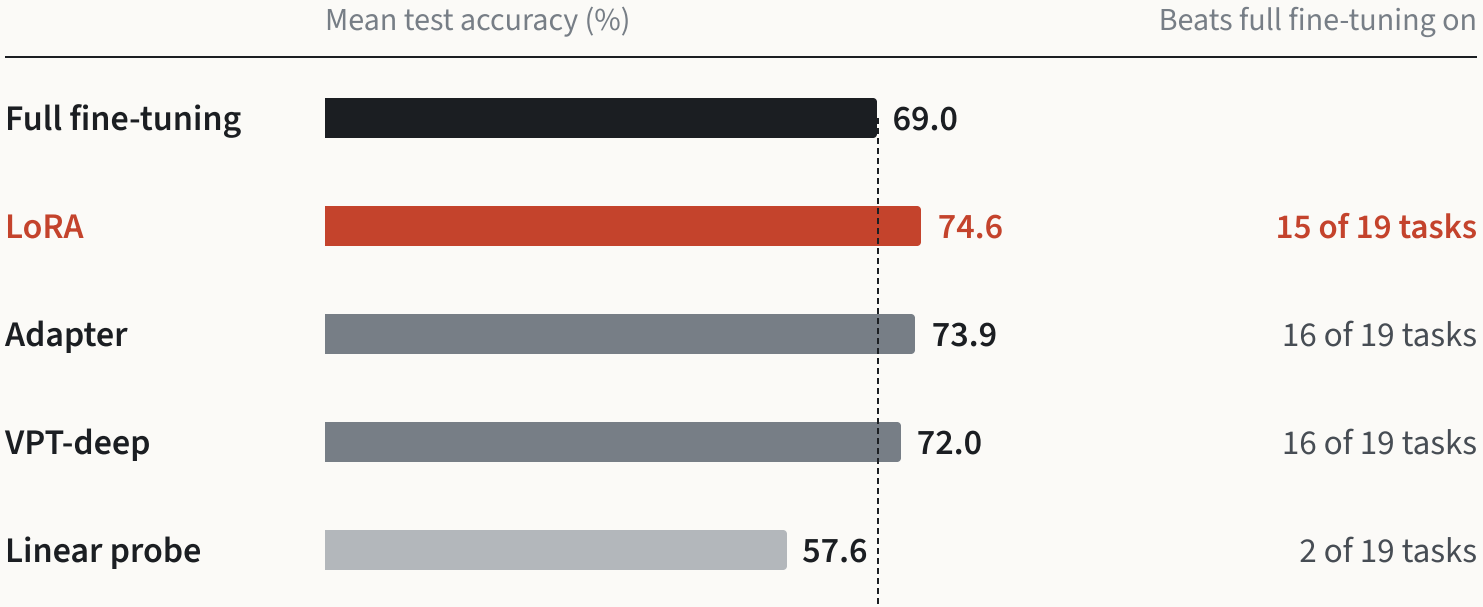


Counting, distances and angles, far from ImageNet. All three cheap methods beat full fine-tuning on all eight tasks, LoRA by the most: 59.8 against 47.6, with VPT-deep at 55.0.

Zhang et al. 2022, Table 1: ViT-B/16 pretrained on ImageNet-21k, VTAB-1k test accuracy. LoRA (rank 8) and the adapter were trained by Zhang et al.; the other rows are Jia et al. 2022, Table 1. Group means computed from the per-task values; "all 19" is the mean of the three groups.

LoRA, VPT and adapters on the same ViT

Natural | Specialized | Structured | **All 19 tasks**

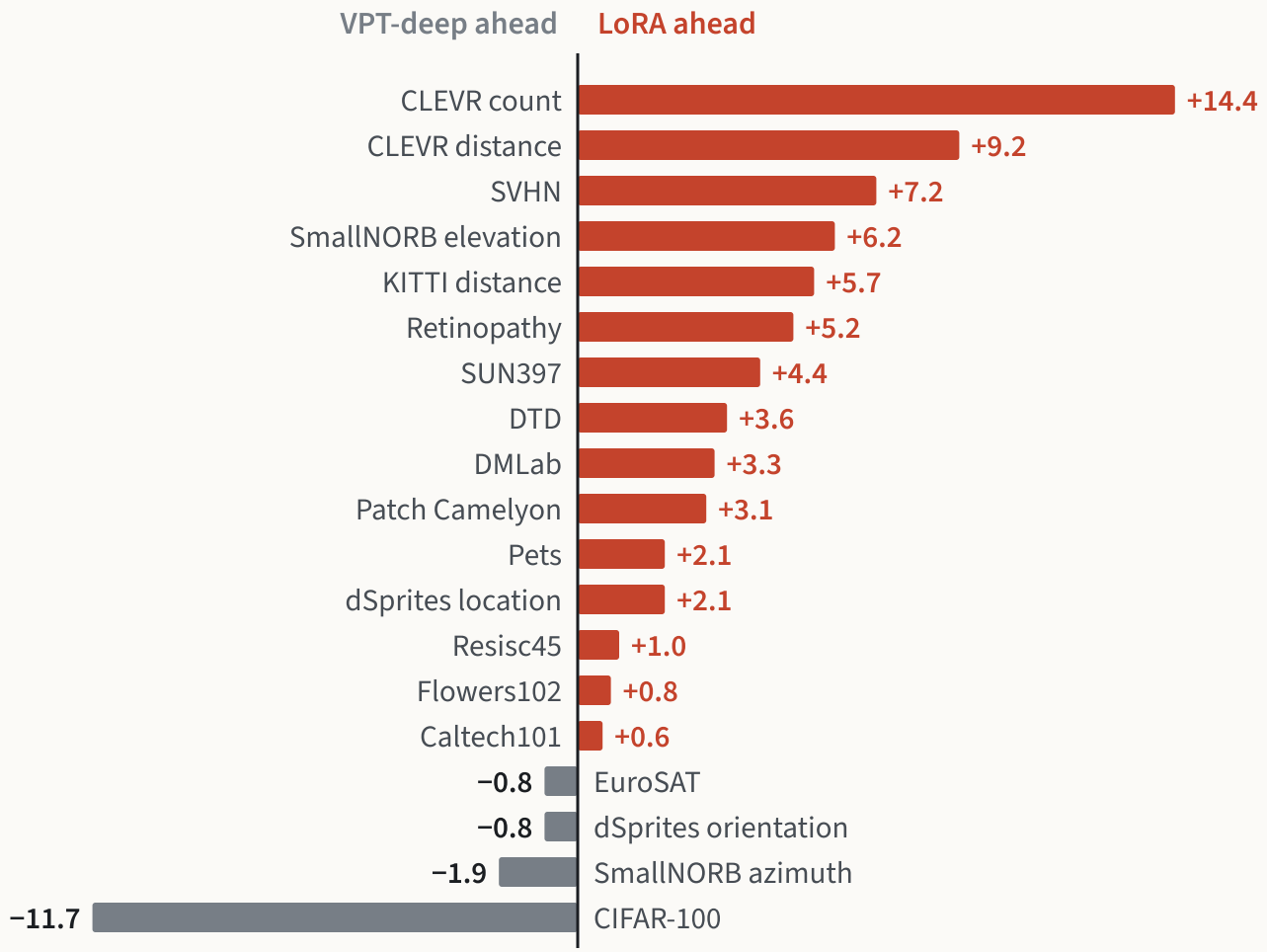


Over all three groups LoRA reaches 74.6, VPT-deep 72.0 and full fine-tuning 69.0.

LoRA trains 0.29 million weights here and VPT-deep 0.64 million on average, so LoRA is both smaller and more accurate.

Zhang et al. 2022, Table 1: ViT-B/16 pretrained on ImageNet-21k, VTAB-1k test accuracy. LoRA (rank 8) and the adapter were trained by Zhang et al.; the other rows are Jia et al. 2022, Table 1. Group means computed from the per-task values; "all 19" is the mean of the three groups.

Neither LoRA nor VPT wins every task



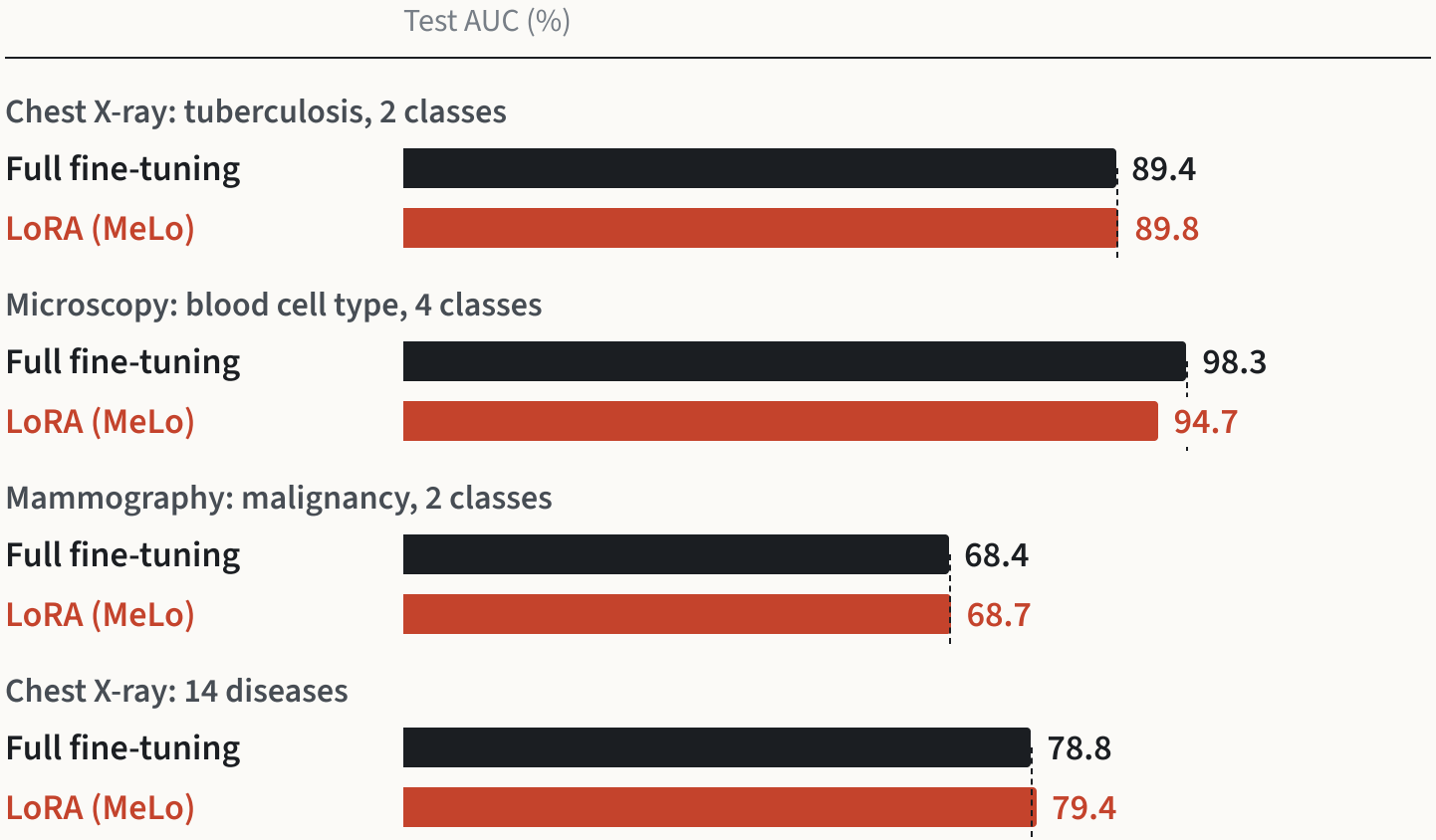
LoRA is ahead on 15 of the 19 tasks, by up to 14 points on counting objects in CLEVR. VPT-deep is ahead on 4, and on CIFAR-100 by 12 points.

Zhang et al. conclude that no single method fits every task. Their method, NOAH, searches for a mix of LoRA, adapter and prompts per task, and reaches 75.5 on average.

Zhang et al. 2022, Table 1: per-task test accuracy of LoRA minus VPT-deep on VTAB-1k, ViT-B/16. Right of the line, LoRA is ahead.

LoRA on medical images

AUC | Accuracy | Switching tasks



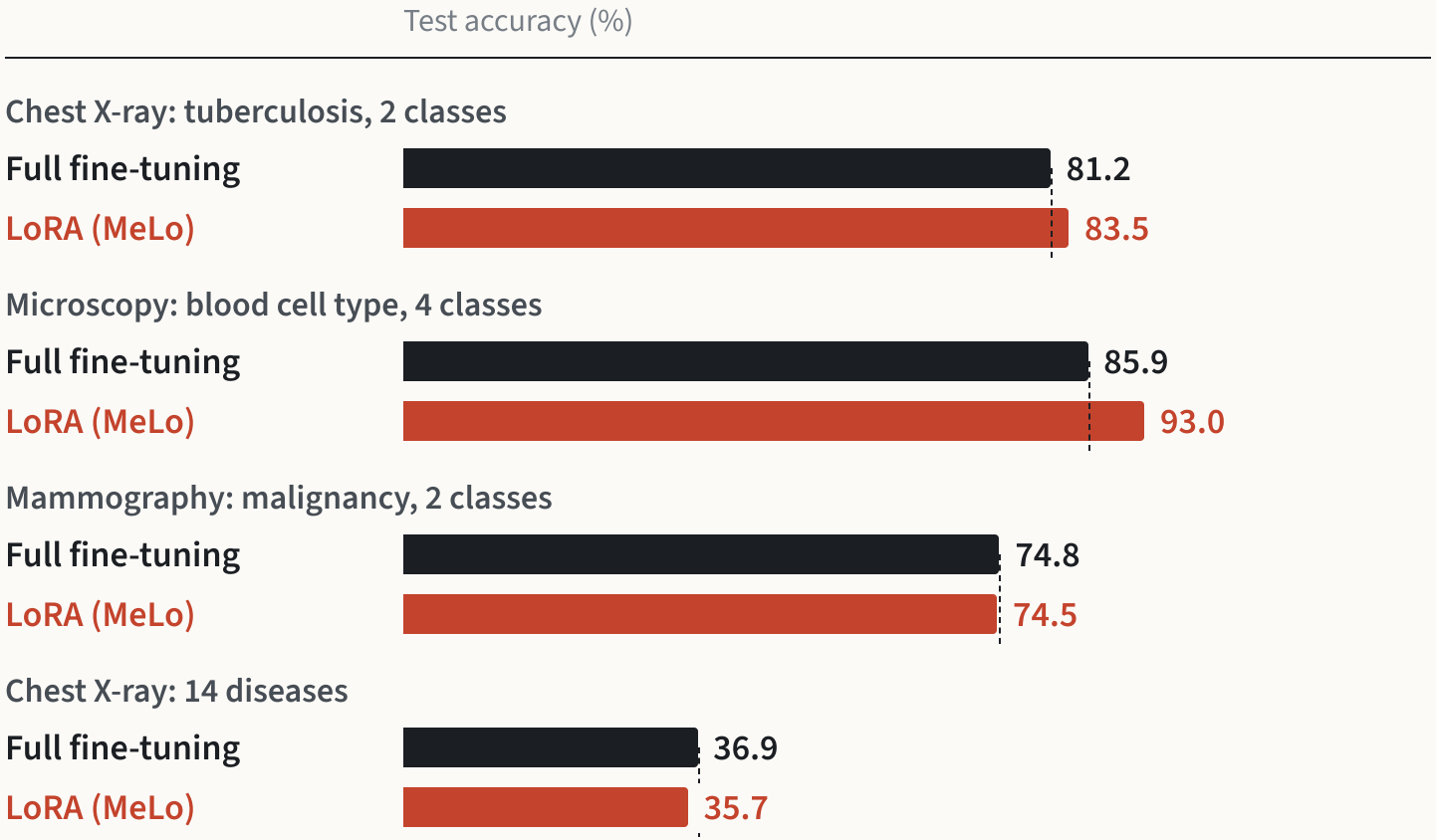
MeLo adds LoRA pairs of rank 4 to the query and value weights of an ImageNet ViT-B, training 0.14 million weights, 0.17% of the model.

Its AUC is within a point of full fine-tuning on three tasks and 3.6 points lower on blood cells.

Zhu et al. 2024 (MeLo, ISBI), Table 1: ViT-B pretrained on ImageNet, LoRA rank 4 on query and value, single runs on their own test splits. The 14-disease task is multi-label.

LoRA on medical images

AUC | **Accuracy** | Switching tasks



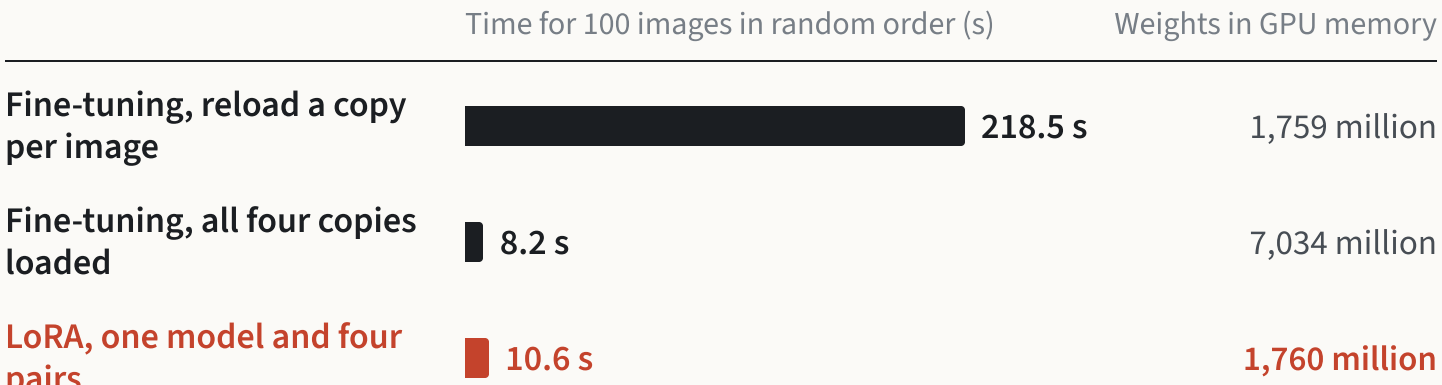
In accuracy MeLo is ahead on two tasks and slightly behind on two.

These are single runs on small test sets, about 80 mammograms for INbreast, so the fair reading of the title, "better than fine-tuning", is "about the same".

Zhu et al. 2024 (MeLo, ISBI), Table 1: ViT-B pretrained on ImageNet, LoRA rank 4 on query and value, single runs on their own test splits. The 14-disease task is multi-label.

LoRA on medical images

AUC | Accuracy | **Switching tasks**



Their practical point is switching between tasks, measured with a giant CLIP ViT on 25 images from each task in random order.

Reloading a fine-tuned copy for each image takes 218 s, and keeping all four copies loaded needs four times the memory. One frozen model with four LoRA pairs takes 10.6 s at the memory of one model.

Zhu et al. 2024, Table 2: ViT-giga pretrained with CLIP (1.76 billion weights), 100 images, 25 from each of the four datasets, in random order, on one A100 GPU.

Where LoRA falls short

The rank and the weights are chosen by hand

The paper picks the query and value weights and a small rank by trying settings, and leaves a principled choice to future work. On VTAB-1k the best method even changes from task to task.

Merged weights serve one task at a time

Once BA is added into W, every input in a batch must belong to that task. Mixing tasks in one batch means keeping the pairs separate, which brings back a small extra cost; VPT's prompts mix freely.

A small rank assumes a nearby task

The authors expect a small rank to fail when the new task is far from pretraining, such as a different language. The low-rank analysis was only done on GPT-3, not on vision models.

Training still runs through the whole model

LoRA saves the optimizer state of the frozen weights, but the gradient still passes back through every layer. On ViT-B most memory is activations: freezing the backbone for VPT only cut it from 11.7 to 10.3 GB.

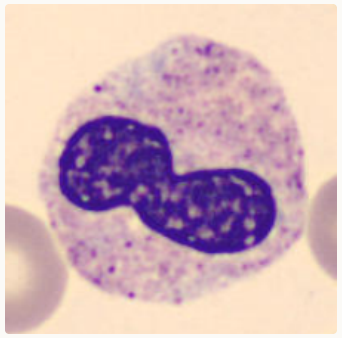
Hu et al. 2022, Sec. 4.2 (batching), footnote 6 (a different language), Sec. 7 and 8 (analysis on GPT-3, future work). Zhang et al. 2022, Fig. 1b. Jia et al. 2022, Table 12 (ViT-B/16, batch 64).

My experiments

I re-implemented VPT and LoRA and trained both on three medical imaging datasets from MedMNIST. A linear probe and full fine-tuning ran next to them, all four starting from the same ViT-B/16.

First the results, then all four models running live in this browser.

Skin lesions



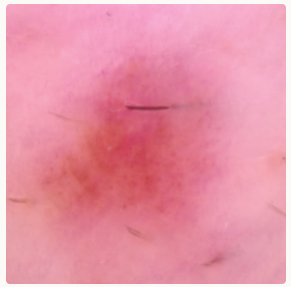
Blood cells

Chest X-rays

I tested VPT and LoRA on three medical datasets

Each method was trained twice: on 1,000 images, as in VTAB-1k, and on the full training set.

melanoma

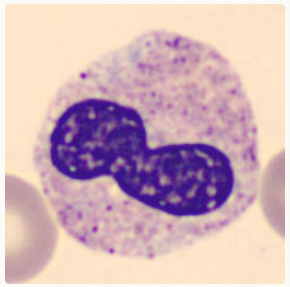


nevus

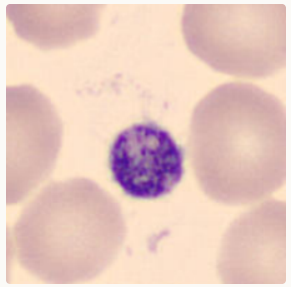
Skin lesions

Dermoscopy, 7 classes

1,000 or all **7,007** training images



neutrophil



platelet

Blood cells

Microscopy, 8 cell types

1,000 or all **11,959** training images

normal

pneumonia

Chest X-rays

Radiographs, pneumonia or normal

1,000 or all **4,708** training images

MedMNIST+ at 224 × 224 (Yang et al. 2023); skin lesions from HAM10000 (Tschandl et al. 2018). RTX 5070 Ti laptop GPU. SGD for linear probe and VPT, AdamW for LoRA and full fine-tuning; learning rate picked on validation.

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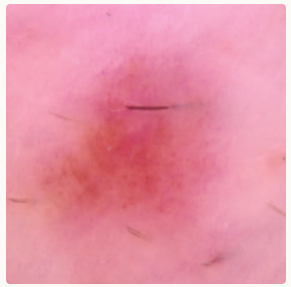
melanoma



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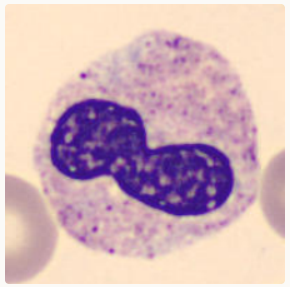
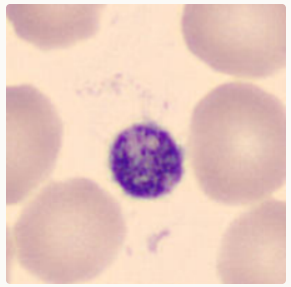


neutrophil



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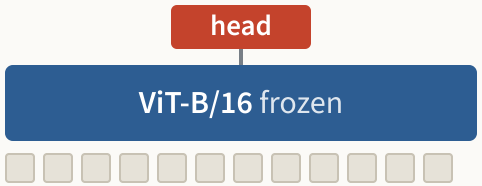
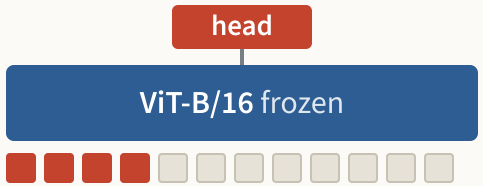


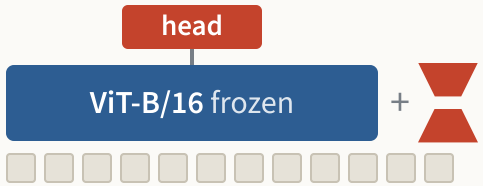
Linear probe

Trains only a new head.



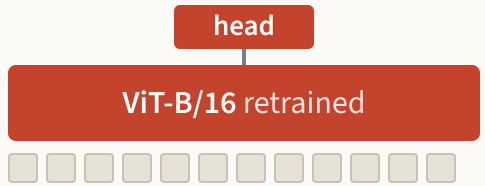
VPT-deep

Trains a head and 50 prompts per layer.



LoRA

Trains a head and rank-8 pairs beside W_q and W_v.



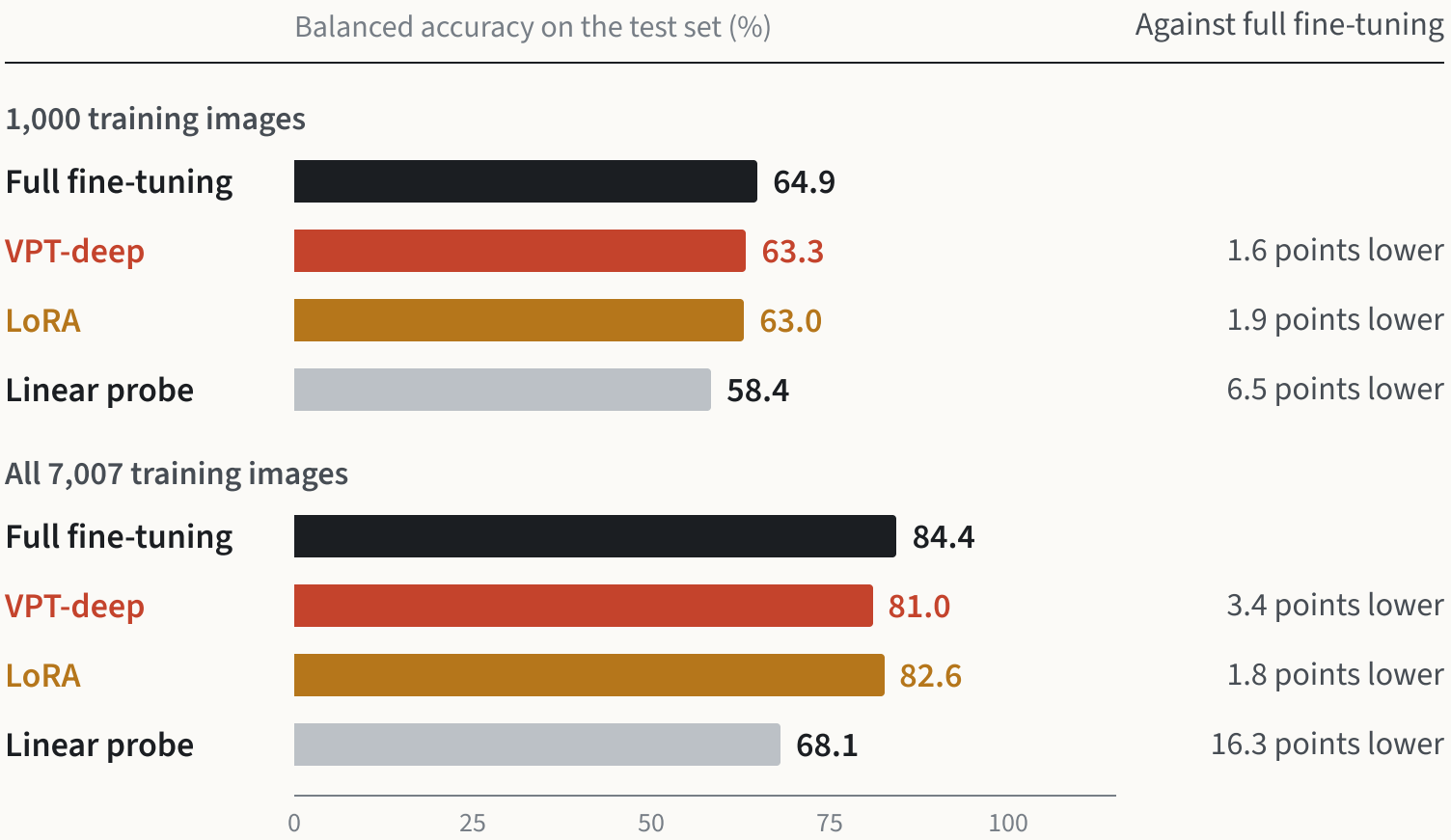
Full fine-tuning

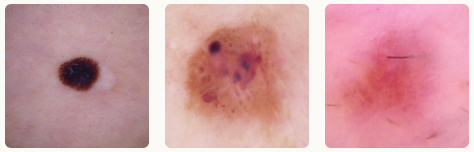
Retrains all 85.8 million weights.

MedMNIST+ at 224 × 224 (Yang et al. 2023); skin lesions from HAM10000 (Tschandl et al. 2018). RTX 5070 Ti laptop GPU. SGD for linear probe and VPT, AdamW for LoRA and full fine-tuning; learning rate picked on validation.

VPT and LoRA on my three datasets

Skin lesions | Blood cells | Chest X-rays





Seven kinds of skin lesion, and two thirds of the images show harmless moles. This is the hardest of the three tasks.

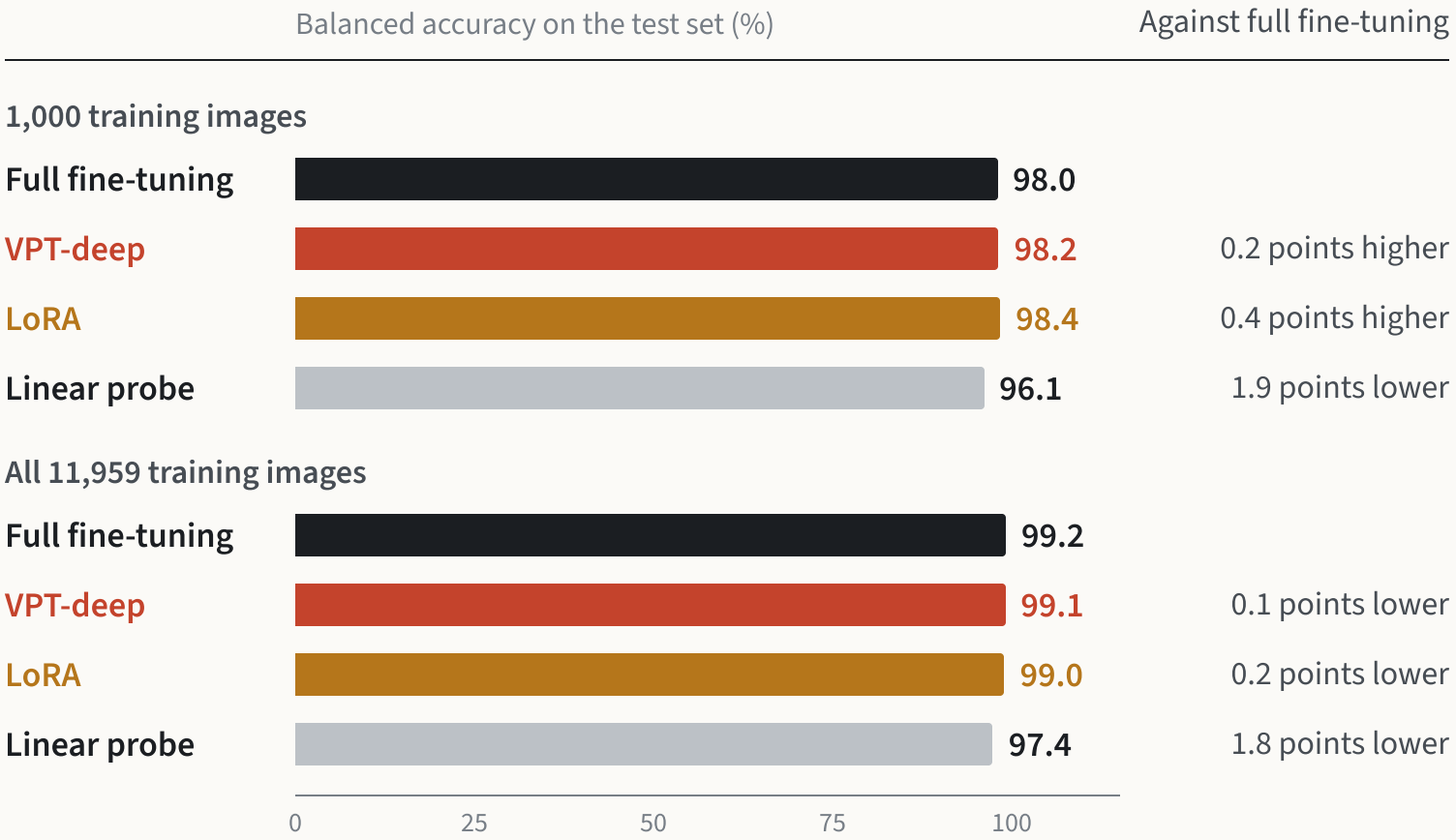
With all 7,007 images, LoRA is 1.6 points ahead of VPT-deep and 1.8 points behind full fine-tuning.

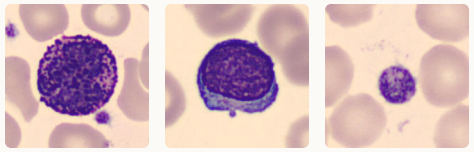
With 1,000 images, LoRA is 0.3 points behind VPT-deep and 1.9 points behind full fine-tuning.

My runs: ViT-B/16 (ImageNet-21k), MedMNIST+ 224 × 224, balanced accuracy on the test set (the mean per-class recall), bars start at 0. Each method was trained once with 1,000 images and once with the full training set; there are no points in between.

VPT and LoRA on my three datasets

Skin lesions | **Blood cells** | Chest X-rays





Eight kinds of blood cell under a microscope. Every method does well here, so the bars are almost the same length.

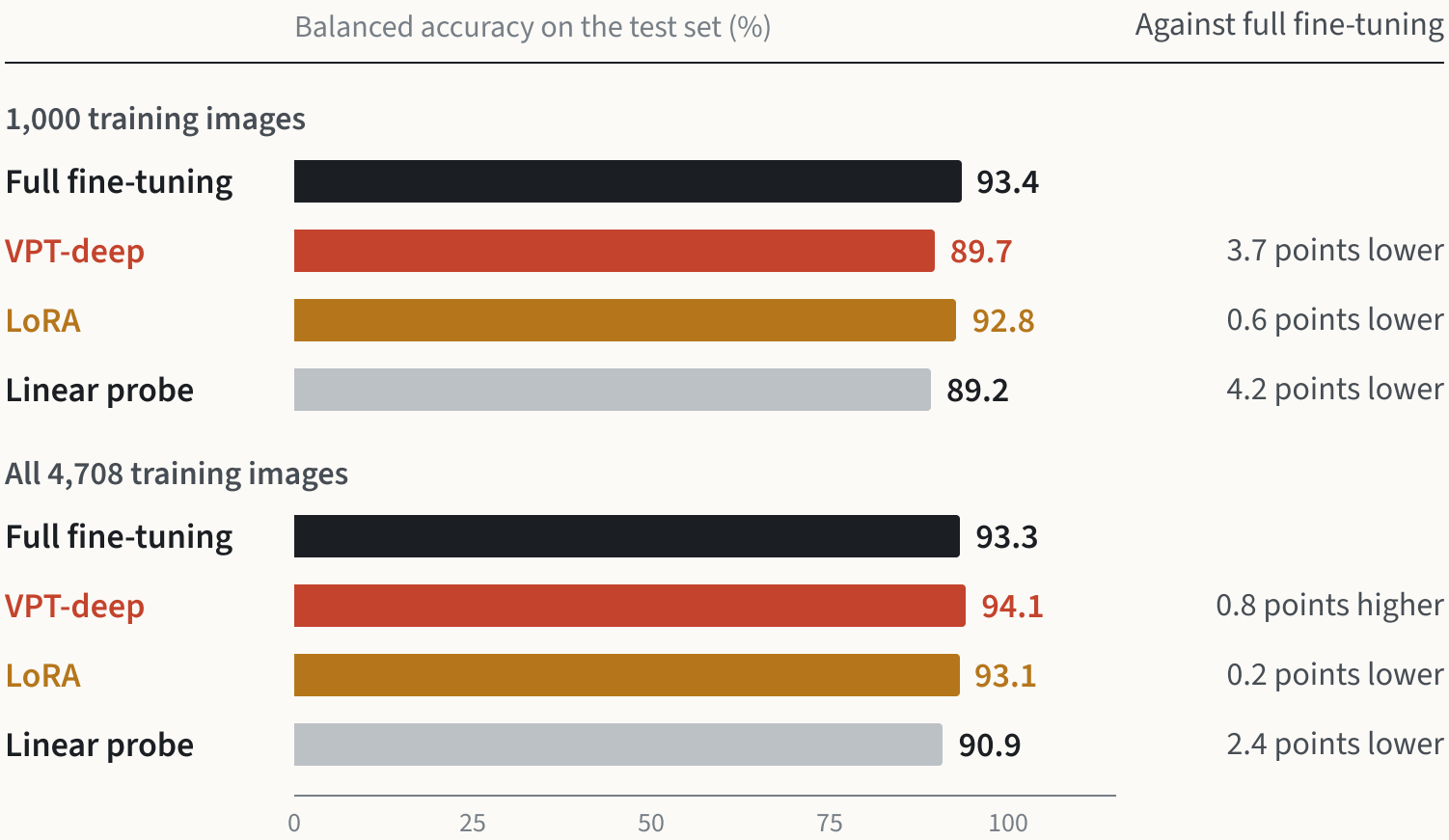
With all 11,959 images, LoRA is 0.1 points behind VPT-deep and 0.2 points behind full fine-tuning.

With 1,000 images, LoRA is 0.2 points ahead of VPT-deep and 0.4 points ahead of full fine-tuning.

My runs: ViT-B/16 (ImageNet-21k), MedMNIST+ 224 × 224, balanced accuracy on the test set (the mean per-class recall), bars start at 0. Each method was trained once with 1,000 images and once with the full training set; there are no points in between.

VPT and LoRA on my three datasets

Skin lesions | Blood cells | **Chest X-rays**



Chest X-rays of children, labelled normal or pneumonia.

With all 4,708 images, LoRA is 1.0 points behind VPT-deep and 0.2 points behind full fine-tuning.

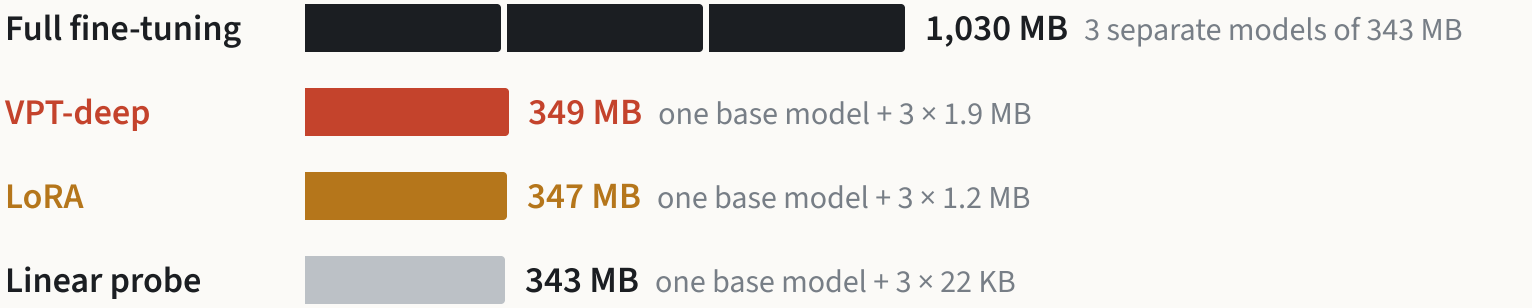
With 1,000 images, LoRA is 3.1 points ahead of VPT-deep and 0.6 points behind full fine-tuning.

My runs: ViT-B/16 (ImageNet-21k), MedMNIST+ 224 × 224, balanced accuracy on the test set (the mean per-class recall), bars start at 0. Each method was trained once with 1,000 images and once with the full training set; there are no points in between.

What each method costs on skin lesions

Storage and accuracy | Training | Inference

Stored for all three tasks



Balanced accuracy on the test set

LoRA stores 1.2 MB per task and VPT-deep 1.9 MB, against 343 MB for a fine-tuned copy. Both share one frozen base model across all tasks.

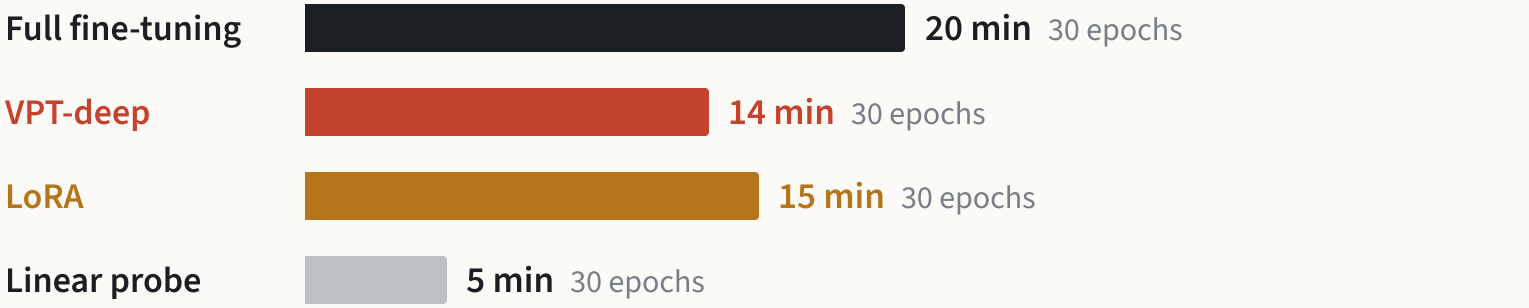
Here LoRA reaches 82.6% and VPT-deep 81.0%, against 84.4% for full fine-tuning and 68.1% for the linear probe.

Skin lesions (DermaMNIST), all 7,007 training images. RTX 5070 Ti Laptop GPU, batch 32; CPU inference on 2 threads; 32-bit weights; LoRA rank 8 on the query and value weights.

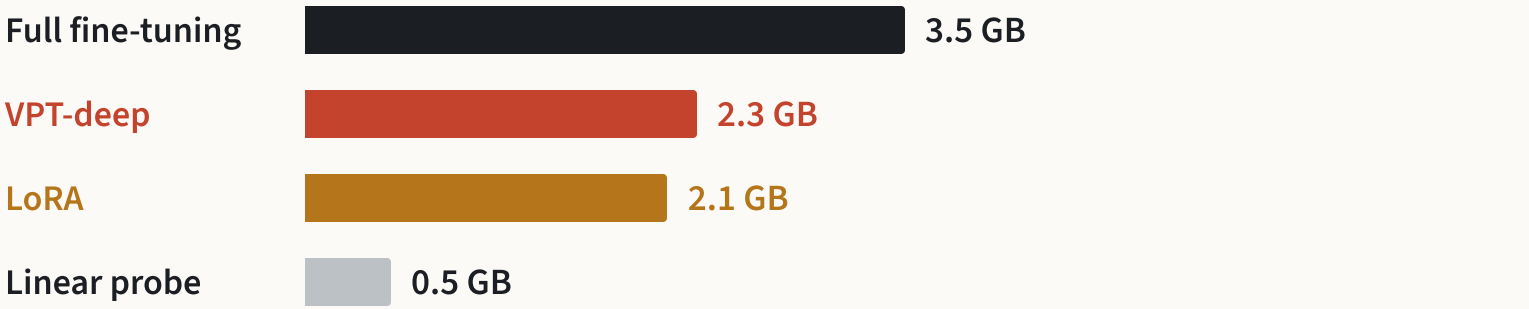
What each method costs on skin lesions

Storage and accuracy | **Training** | Inference

Training time on all 7,007 images



Peak GPU memory during training, batch 32



LoRA needed 2.1 GB of GPU memory and VPT-deep 2.3 GB, against 3.5 GB for full fine-tuning.

Both still pass the error signal back through all twelve frozen layers, so only the linear probe is really cheap to train.

Skin lesions (DermaMNIST), all 7,007 training images. RTX 5070 Ti Laptop GPU, batch 32; CPU inference on 2 threads; 32-bit weights; LoRA rank 8 on the query and value weights.

What each method costs on skin lesions

Storage and accuracy | Training | **Inference**

Time per image on the CPU, 2 threads

Tokens that every layer processes

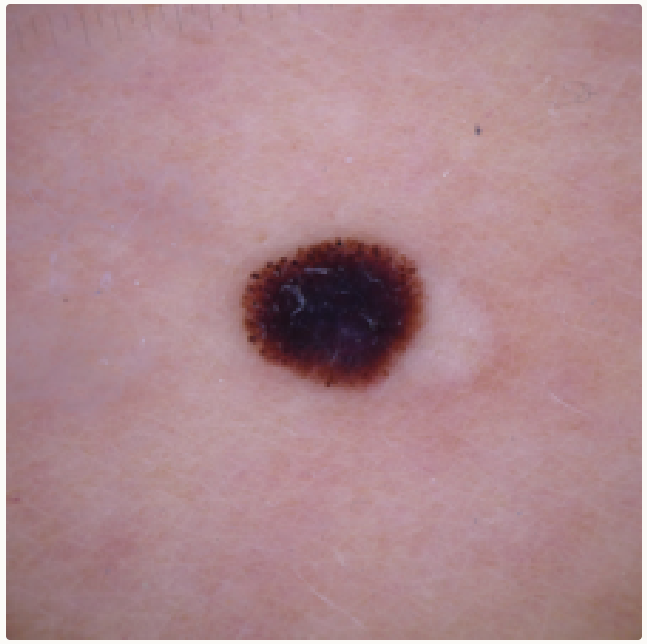
VPT-deep needs 250 ms per image, because every layer processes 50 extra tokens. LoRA with separate pairs needs 190 ms.

Merged into the weights, LoRA takes 188 ms, the time of the unchanged model (189 ms).

Skin lesions (DermaMNIST), all 7,007 training images. RTX 5070 Ti Laptop GPU, batch 32; CPU inference on 2 threads; 32-bit weights; LoRA rank 8 on the query and value weights.

Live demo: one image, four models

Skin lesion, one test image, models trained on **all training images**.



True label: **melanoma**

Linear probe

shared model + head

68.5% test accuracy

melanocytic n…	**73.7%**
melanoma	**23.8%**
benign keratosi…	**2.1%**

Wrong

178 ms per image

VPT-deep

shared model + prompts

81.0% test accuracy

melanoma	**99.5%**
benign keratosi…	**0.3%**
melanocytic nevi	**0.1%**

Correct

204 ms per image

LoRA

shared model + pairs

82.5% test accuracy

melanocytic n…	**92.8%**
melanoma	**7.1%**
benign keratosi…	**0.0%**

Wrong

173 ms per image

Full fine-tuning

own retrained model

84.4% test accuracy

melanoma	**97.7%**
melanocytic nevi	**2.2%**
benign keratosi…	**0.0%**

Correct

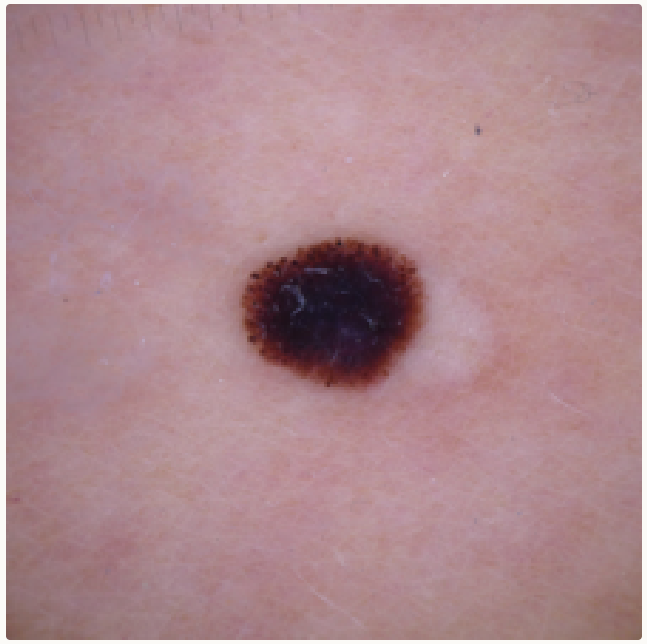
163 ms per image

Test accuracy is balanced accuracy on the whole test set, for exactly these models. Changing the training set swaps one 1.2 MB file for LoRA, but a whole second 172 MB model for full fine-tuning.

A recorded run. In the HTML version of this deck the four models run live in the browser.

Live demo: one image, four models

Skin lesion, one test image, models trained on **1,000 training images**.



True label: **melanoma**

Linear probe

shared model + head

58.1% test accuracy

melanocytic n…	**84.5%**
benign keratosi…	**7.9%**
melanoma	**7.5%**

Wrong

142 ms per image

VPT-deep

shared model + prompts

62.5% test accuracy

melanoma	**69.2%**
melanocytic n…	**30.8%**
dermatofibroma	**0.0%**

Correct

218 ms per image

LoRA

shared model + pairs

63.0% test accuracy

melanocytic n…	**86.6%**
melanoma	**12.8%**
benign keratosi…	**0.4%**

Wrong

166 ms per image

Full fine-tuning

own retrained model

64.8% test accuracy

melanocytic n…	**95.1%**
melanoma	**4.2%**
benign keratosi…	**0.2%**

Wrong

169 ms per image

Test accuracy is balanced accuracy on the whole test set, for exactly these models. Changing the training set swaps one 1.2 MB file for LoRA, but a whole second 172 MB model for full fine-tuning.

A recorded run. In the HTML version of this deck the four models run live in the browser.

Summary

20 of 24	tasks where VPT-deep beat full fine-tuning, training 0.6% of a ViT-B pretrained with labels.
99.4%	less storage per task than a fully fine-tuned copy: 2.2 MB for VPT instead of 343 MB, and 1.5 MB for LoRA.
1.00×	the test-time cost of the original model with LoRA, because its change is added into the weights. VPT's prompts cost 1.27×.
74.6	mean VTAB-1k accuracy of LoRA on ViT-B/16, against 72.0 for VPT-deep and 69.0 for full fine-tuning.
88.1%	average balanced accuracy of LoRA on my three medical datasets, against 87.6% for VPT-deep, 88.8% for full fine-tuning and 83.3% for the linear probe.

Jia et al. 2022, Table 1 (24 tasks, ViT-B/16) and Table 12; 0.6% is VPT-deep with 50 prompts per layer and a 102-class head, LoRA is rank 8 on the query and value weights with the same head. Zhang et al. 2022, Table 1, mean of the three VTAB-1k groups. Medical datasets: my runs, balanced accuracy on the test set.

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